



SECJ2203: Software Engineering

System Documentation (SD)

Project Title

Version 2.0

Date

Faculty of Computing

Prepared by: CO-T

Revision Page

a. Overview

In this System Documentation (SD), a detailed documentation of the automated cooperative system for Koperasi KADA will be provided. The documentation consists of the introduction that comprises the purpose and scope of the system. The documentation will also provide the specific requirements that consist of user persona, user story and system feature. The documentation will be supported with different use case diagrams. The documentation also covers the non-functional and the design constraints imposed by Koperasi KADA.

b. Target Audience

The target audience for this system is the non-member, member, staff and Ahli Lembaga Koperasi (ALK) of Koperasi KADA.

c. Project Team Members

Table 01 below shows the task distribution for each member in completion of SD. The table shows the roles and status for each assigned task for each member by sections for this SD version (complete, partially complete, incomplete).

Table 1 : Task distribution

Member Name	Role	Task	Status
Nurul Ika Syafiny Binti Azhar	Team Leader	Project planning	Complete
		Project management	Incomplete
		Develop system	Incomplete
Nuraisyah Binti Mohd Zikre	Technical Writer	Gather user requirements	Complete
		Create System Requirement Specification	Complete
		Create system documentation	Incomplete
Harini A/P Sangaran	System Analyst	Create system architecture	Complete
		Design system functional diagram	Complete
		System Implementation	Complete

Damiya Aina Binti Basir Abd Shammad	System Designer	System interface design	Complete
		System testing	Incomplete

d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Nurul Ika Syafiny Binti Azhar	Completed Chapter 1, Section 1.1 Chapter 2, Section 2.1.4, 2.2.1, 2.4.4, 2.6. Chapter 3, Section 3.1. Chapter 4, Section 4.1, 4.5.	22/1/2025
1.0	Nuraisyah Binti Mohd Zikre	Completed Revision Page, Chapter 1 (Introduction), Section 1.3, Section 2.1.1, Section 2.2.2, Section 2.4.1, Section 2.6, Section 4.4, Chapter 5	28/1/2025
1.0	Harini A/P Sangaran	Completed section 1.2, section 2.1.2, section 2.2.2, section 2.4.2, and section 2.5, section 4.3, Chapter 6	2/2/2025
1.0	Damiya Aina Binti Basir Abd Shammad	Complete section 2.1.3, section 2.3, section 2.5 section 4.2, Chapter 7	22/1/2025

Table of Contents

1	Introduction		4-6
	1.1	Purpose	4
	1.2	Scope	4-5
	1.3	Definitions, Acronyms and Abbreviations	5
	1.4	References	5
	1.5	Overview	5-6
2	Specific Requirements		7-63
	2.1	Persona	7-19
		2.1.1 Persona 1 (Member/Actor 1)	7-10
		2.1.2 Persona 2 (Admin/Actor 2)	10-15
		2.1.3 Persona 3 (ALK/Actor 3)	15-17
		2.1.4 Persona 4 (Non-Member/Actor 4)	17-19
	2.2	System Features	19-26
	2.3	Launch Phase	26-29
	2.4	User Story Details	29-61
		2.4.1 US001: User Story <User Story 1> User Story description of US001 Activity Diagram of US001 System Sequence Diagram of US001	29-37
		2.4.2 US002: User Story <User Story 2> User Story description of US002 Activity Diagram of US002 System Sequence Diagram of US002	37-52
		2.4.3 US003: User Story <User Story 3> User Story description of US003	53-59

		Activity Diagram of US003 System Sequence Diagram of US003 2.4.4 US004: User Story <User Story 4> User Story description of US004 Activity Diagram of US004 System Sequence Diagram of US004	59-61
	2.5	Performance and Other Requirements	61-62
	2.6	Design Constraints	63
3		Architectural Rationale	64
	3.1	Architectural Style and Rationale	64
4		Architectural Views	65-82
	4.1	Use Case View	67
	4.2	Implementation View	67-72
	4.3	Logical View	73-75
	4.4	Process View	76-81
	4.5	Physical/ Deployment View	82
5		Data Design	83-87
	5.1	Data Dictionary	85-87
		5.1.1 Entity: <ALK>	85
		5.1.2 Entity: <Invoice>	85
		5.1.3 Entity: <Koperasi Member>	85
		5.1.4 Entity: <Loan Application>	85
		5.1.5 Entity: <Loan Balance>	85-86
		5.1.6 Entity: <Member Application>	86
		5.1.7 Entity: <Non-Member>	86
		5.1.8 Entity: <Savings>	86-87
		5.1.9 Entity: <Admin>	87

6	User Interface Design	88-115
	6.1 Screen Images	92-115
7	Traceability	116

Introduction

In this System Documentation (SD), detailed information of the system that has been prepared by the team member after the discussion will be included. In this documentation, it consists of the details of Software Requirements Specification (SRS), Software Design Description (SDD) and Software Test Documentation (STD). Each documentation of SRS, SDD and STD will provide a better understanding in the development of the system and the implementation of the system to achieve a system that satisfies the target user.

1.1 Purpose

This System Documentation (SD) describes the system's users, functions and features in detail to improve better understanding of the system. This is to help the developer grasp clearly the structure of the system and system flow. This SD includes system details such as Use Case Diagram, Class Diagram and State Diagram further help designer and developer get the clear picture of the system through diagram. Moreover, this SD will also include the functional and non-functional requirements so that the designers can design a system that fulfills them all and the tester can check if all the requirements are satisfied. This is also for client reference so that they can add up the missing functions and give feedback about the planned system before moving further to the implementation phase. Lastly, this SD will also include the design constraints so that the team can focus on it and work together to find a solution.

1.2 Scope

The software product that is used in the proposed system is phpMyAdmin MySQL. MySQL is used to store the databases that are used in the proposed system. For example, the proposed system for Koperasi KADA has multiple databases to store information for non-members, members, staff and Ahli Lembaga Koperasi (ALK). Storing the data in databases is crucial so that the information can be used for verification and can be stored for the future purposes of staff and ALK. Using databases in the proposed system is efficient because the user details are interrelated. For example, the membership application list will be used both by the staff and ALK. Hence, the retrieval of user information will be easier, time efficient and any modification made to the user information has to be made once only at the database.

Besides that, XAMPP Control Panel is the software used as the web server for the proposed system. XAMPP Has multiple modules under its software, such as Apache, MySQL, FileZilla, Mercury and Tomcat. For our proposed system, only Apache and MySQL will be used. Apache functions as the web server, enabling the execution of PHP scripts, which are developed using Visual Studio Code.

1.3 Definitions, Acronyms and Abbreviation

This subsection will provide the definitions of all terms, acronyms, and abbreviations that will be used in this documentation.

- ALK: The board members that were responsible for approving the application applied.
- Non-member: A person who is not a member of Koperasi KADA
- Member: A person is a member of Koperasi KADA
- Staff: An employee in Koperasi KADA who will review the application and send it to ALK
- SD: System Documentation

1.4 References

[1] UXPin. (2023, January 23). *7 Constraints in Design and How to Overcome Them*. Studio by UXPin. <https://www.uxpin.com/studio/blog/constraints-in-design/>

1.5 Overview

The entire process of designing and developing the proposed Koperasi KADA system is described in this System Documentation (SD). It outlines the team's approach for resolving the inefficiencies of the current manual procedures by using a web-based, digital platform that is customized to meet the requirements of the cooperative. The goal, requirements, design, and testing techniques of the system are all included in the multiple parts that make up the SD. Each part verifies that the system's usability, functionality, and operational effectiveness are thoroughly documented for easy setup and maintenance.

The System Documentation is organized to provide a clear and logical progression through the system development process. It begins with the System Introduction, which outlines the objectives and rationale for developing the Koperasi KADA system, focusing on addressing

the inefficiencies of the current manual processes. The system's functional and non-functional requirements are then described in the Specific Requirements part, assisting in defining the system's goals and parameters. To make sure that the system design meets the demands of the target users, this part contains a Persona subsection that characterizes them.

The main features of the system, including online membership registration, loan applications and real-time financial tracking, are then covered in detail in the System Features section. The Launch Phase then outlines the deployment plan, including how to roll out the system, onboard users, and replace the manual processes that are currently in place. Additionally, use case descriptions that show how users will interact with the system in different scenarios are included in the User Story Details section. These explanations help to make sure that every feature successfully satisfies the user's wants and is in line with the system's overall objectives. To guarantee that the platform operates at its best in a variety of circumstances, the Performance and Other Requirements section also concentrates on technical and operational factors like system scalability, dependability, and response time.

Finally, the Design Constraints section highlights the limitations and specific boundaries within which the system must operate, ensuring the design remains feasible and realistic. By organizing the document into these sections, the SD provides a clear and structured roadmap for the system's development, ensuring that all stakeholders can contribute effectively to achieving a successful implementation.

2. Specific Requirements

This section of SD contains all the details of software requirements such as Persona and Use Case Diagram of the system to enable designers to design the system that satisfies all the requirements. Throughout this section, every stated requirement is explained clearly in many kinds of forms and diagrams to enhance the understanding of users, operators and other external systems. These requirements also include a description of every input into the system, every output from the system, and all functions performed by the system in response to an input or in support of an output.

2.1 Persona

In this subtopic, we will describe the intended users of the system and their characteristics to provide the developer with a clear understanding of whom the system is designed for and helps to ensure that the system is developed to meet the needs of its intended user. This system has four types of users which are non-members, members, staff and Ahli Lembaga Koperasi. The details of these actors will be explained using persona as in below.

2.1.1 Persona 1 (Member/Actor 1)

User Need

- A member needs a way to login to the system to apply for a loan, view loan application approval, loan balance, savings, financial statement and apply for membership termination.

User Stories

- As a member, I want to login to the system so that I can apply for a loan or membership termination, view my loan application result whether it is approved or not, view the balance of loan that needs to be paid, view the savings from the deduction of the membership fee every month and check the payment statement.

A member apply for loan
To apply for a loan you must first open the website of Koperasi KADA and login into the system. If you did not log in into the system, you cannot apply for a loan. After successfully logging into the system, you can apply for a loan after clicking on the “Mohon Pinjaman”. You are required to fill in the required details in the form in order to submit the application. If you do not fill in the required details, you are unable to submit the form. After successfully submitting the form, the application will be processed. In all cases, you need to log in into the system in order to apply for a loan.

Table 2: A member user story 1

A member view loan application approval
If you apply for a loan, you can check your loan application approval by clicking on the “Semak Kelulusan Pinjaman”. If your application is approved, the details of the application will be displayed. If you did not apply for a loan, the system will show that you did not apply for a loan. In all cases, you need to log in into the system in order to view loan application approval.

Table 3: A member user story 2

A member view loan balance
If you apply for a loan and the application is approved, you can check your loan balance by clicking on the “Baki Pinjaman”. The balance of the loan that needs to be paid will be displayed by the system. If you did not apply for a loan, the system will show that you did not balance that need to be paid. In all cases, you need to log in into the system in order to view loan balance.

Table 4: A member user story 3

A member view savings
If you want to view savings, you can click on the “Semak Simpanan”. The details of the savings will be shown by system. In all cases, you need to log in into the system in order to view savings.

Table 5: A member user story 4

A member view financial statement
If you want to view the financial statement, you can click on the “Semak Penyata”. The system will display the details of every transaction of money for every month. In all cases, you need to log in into the system in order to view savings.

Table 6: A member user story 5

A member apply for membership termination
If you want to apply for membership discontinuation, you can click on “Mohon Berhenti Ahli”. You are required to fill in the required details in the form in order to submit the application. If you do not fill in the required details, you are unable to submit the form. After successfully submitting the form, the application will be processed. In all cases, you need to log in into the system in order to apply for a membership termination.

Table 7: A member user story 6

2.1.2 Persona 2 (Admin/Actor 2)

User Need

- A user who is a staff of Koperasi KADA needs to be able to view the list of applicants who applied for Koperasi KADA membership, list of Koperasi KADA members who applied for a loan with Koperasi KADA, list of Koperasi KADA members who applied for membership termination, validate the verification documents of users who applied for membership, loan and approve membership discontinuation, view Koperasi KADA report and add new admin and ALK to the system.

User Stories

- As an admin, I want to validate the verification documents of non-members who applied for a membership, validate the verification documents of members who applied for a loan and approve membership termination applications. I also want to view the Koperasi KADA report and add new admin and ALK to the system.

An admin viewing list of non-members who applied for Koperasi KADA membership
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership, list of members who applied for loans, list of members who applied for membership termination, view Koperasi KADA report and add new admin or ALK to the system. If you click to view the list of applicants who applied for Koperasi KADA membership, a list of applicants along with their identification card number and verification documents will be displayed. You can click on the verification documents to check the documents.

Table 8: An admin user story 1

An admin viewing list of Koperasi KADA members who applied for loan
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership, list of members who applied for loans, list of members who applied for membership termination, view Koperasi KADA report and add new admin or ALK to the system. If you click to view the list of Koperasi KADA members who have applied for a loan with Koperasi KADA, a list of members along with their identification card number, the type of loan they applied for and the verification documents will be displayed. You can click on the verification documents to check the documents.

Table 9: An admin user story 2

An admin viewing list of Koperasi KADA members who applied for membership termination
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership, list of members who applied for loans, list of members who applied for membership termination, view Koperasi KADA report and add new admin or ALK to the system. If you click to view the list of Koperasi KADA members who have applied for membership discontinuation, a list of members along with their ID number, identification card number and the verification documents will be displayed. You can click on the verification documents to check the documents.

Table 10: An admin user story 3

An admin validating the verification documents of non-members who applied for membership
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership, list of members who applied for loans, list of members who applied for membership termination, view Koperasi KADA report and add new admin or ALK to the system. If you click to view the list of non-members who applied for membership, a list of applicants with their identification card number and verification documents will be displayed. You can click on the verification documents to check the documents. If the documents are as per requirements, you can validate the verification documents. Once you have validated the verification documents, the validation status of the documents will be updated. You can also choose to update the validity of the verification documents later on if there's any changes.

Table 11: An admin user story 4

An admin validating the verification documents of members who applied for loan
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership, list of members who applied for loans, list of members who applied for membership termination, view Koperasi KADA report and add new admin or ALK to the system. If you click to view the list of members who applied for a loan, a list of members with their identification card number, the type of loan they applied for and the verification documents will be displayed. You can click on the verification documents to check the documents. If the documents are as per requirements, you can validate the verification documents. Once you have validated the verification documents, the validation status of the documents will be updated. You can also choose to update the validity of the verification documents later on if there's any changes.

Table 12: An admin user story 5

An admin approving the membership termination application
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership, list of members who applied for loans, list of members who applied for membership termination, view Koperasi KADA report and add new admin or ALK to the system. If you click to view the list of Koperasi KADA members who have applied for membership discontinuation, a list of members along with their ID number, identification card number and the verification documents will be displayed. You can click on the verification documents to check the documents. If the documents are as per requirements, you can approve the membership discontinuation application. Once you have approved the application, the application status will be updated. You can also choose to update the application status later on if there's any changes.

Table 13: An admin user story 6

An admin viewing Koperasi KADA report
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership, list of members who applied for loans, list of members who applied for membership termination, view Koperasi KADA report and add new admin or ALK to the system. If you click to view the Koperasi KADA report, a calendar will be displayed along with the monthly report and annual report for employer and employee share, membership applications, loan applications and membership discontinuation applications. To view up to a certain date's report only, you have to click on the certain date on the calendar so that the system filters the report up to that specific date only for all the reports. To view the monthly report, click on the specific month. To view the annual report, click on the latest date of the year.

Table 14: An admin user story 7

An admin adding new admin to the system
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership, list of members who applied for loans, list of members who applied for membership termination, view Koperasi KADA report and add new admin or ALK to the system. If you click to add a new admin to the system, you have to key in the new admin's identification card number and click submit. Upon successful submission, an appropriate message will be displayed.

Table 15: An admin user story 8

An admin adding new ALK to the system
Once you have logged into the system, you will be able to view a homepage with features to view the list of Koperasi KADA members, list of Koperasi KADA members who have a loan with Koperasi KADA, list of non-members who applied for membership, list of members who applied for loans, list of members who applied for membership termination, view Koperasi KADA report and add new admin or ALK to the system. If you click to add a new ALK to the system, you have to key in the new ALK's identification card number and click submit. Upon successful submission, an appropriate message will be displayed.

Table 16: An admin user story 9

2.1.3 Persona 3 (ALK/Actor 3)

User Need

- A user who is on the high committee in Koperasi KADA needs a way to verify the membership applications by a non-member and the loan applications by a member of Koperasi KADA and view the Koperasi KADA report.

User Stories

- As a high committee member, I want to verify and approve the membership and loan applications and view the Koperasi KADA report.

An ALK viewing the validated membership applications
Once logged into the system, you will be able to view the homepage with features to view membership applications from non-members, loan applications from members and Koperasi KADA report.

If you click to view the validated membership applications, a list of applicants with valid verification documents for membership application will be displayed. You can click on the verification documents to check the documents.

Table 17: An ALK user story 1

An ALK viewing the validated loan applications
Once logged into the system, you will be able to view the homepage with features to view membership applications from non-members, loan applications from members and Koperasi KADA report. If you click to view the validated loan applications, a list of members with valid verification documents for loan applications will be displayed. You can click on the verification documents to check the documents.

Table 18: An ALK user story 2

An ALK approving the membership application
Once logged into the system, you will be able to view the homepage with features to view membership applications from non-members, loan applications from members and Koperasi KADA report. If you click to view the validated membership applications, a list of applicants with valid verification documents for membership application will be displayed. You can click on the verification documents to check the documents. If the documents are as per requirements, you can approve the application. Once you have approved the application, the application status will be updated. You can also choose to update the application status later on if there's any changes.

Table 20: An ALK user story 3

An ALK approving the loan application
Once logged into the system, you will be able to view the homepage with features to view membership applications from non-members, loan applications from members and Koperasi KADA report. If you click to view the validated loan applications, a list of members with valid verification documents for loan application will be displayed. You can click on the verification documents to check the documents. If the documents are as per requirements, you can approve the application. Once you have approved the application, the application status will be updated. You can also choose to update the application status later on if there's any changes.

Table 21: An ALK user story 4

An ALK viewing Koperasi KADA report
If you click on the list of system reports, a calendar will be displayed along with the monthly and annual report for employer and employee shares, membership applications, loan applications and membership discontinuation applications. To view the reports up to a certain date, the specific date has to be clicked on the calendar. Thus, the report will update accordingly.

Table 22: An ALK user story 5

2.1.3 Persona 4(Non-Member /Actor 4)

User Need

- A user who is a non-member of Koperasi KADA needs a way to become a member of Koperasi KADA.

User Stories

- As a non-member user, I want to become a member of Koperasi KADA so that I can log into the KADA Corporation system.

A non-member apply to be a member
You must first open the website of Koperasi KADA to see the application for membership function. Click on the apply membership at the website to be directed to the application forms. Once you see the application forms, you need to complete all details in their forms and upload the mandatory documents to the system. You are required to make a mandatory payment to apply as a member for the first time. Once the payment is made and the receipt is uploaded, you can submit your application to the system. Once you click 'Submit Application', your form will be recorded in the system for further processing by Koperasi KADA staff and ALK to approve your application . In seven working days, you will get an email about their membership approval status either success or fail. If your application is approved, you will get your member number and password to log in into the system. If you fail, you will be notified through email too and asked to resubmit your application if you want.

Table 23: A non-member user story 1

A non-member who just become a member want to login for the first time
Once you get login details through email, press the 'Daftar Masuk' button at the homescreen to be directed to the login page of the system. Choose 'Ahli' at the button below the 'Log Masuk' button, then insert the login and password details given in the approval email. If login success, you will be directed to the main tab where all the member functions will be in display and available to use. You are now a Koperasi member. If you fail, try login again or contact the staff for further assistance.

Table 24: A non-member user story 2

2.2 System Features

The KADA Corporation System is a software system that operates on a web site. The system provides a means for Koperasi KADA members to get updates fast, apply to member's loan easier and get the details for transactions made. It also helps the staff to manage corporation work, like the application forms easier and faster than the manual method.

2.2.1 Use Case Diagram

The system features are illustrated in the use case diagram at Figure 1 below. The detailed description of each module and functions is tabulated in Table 25.

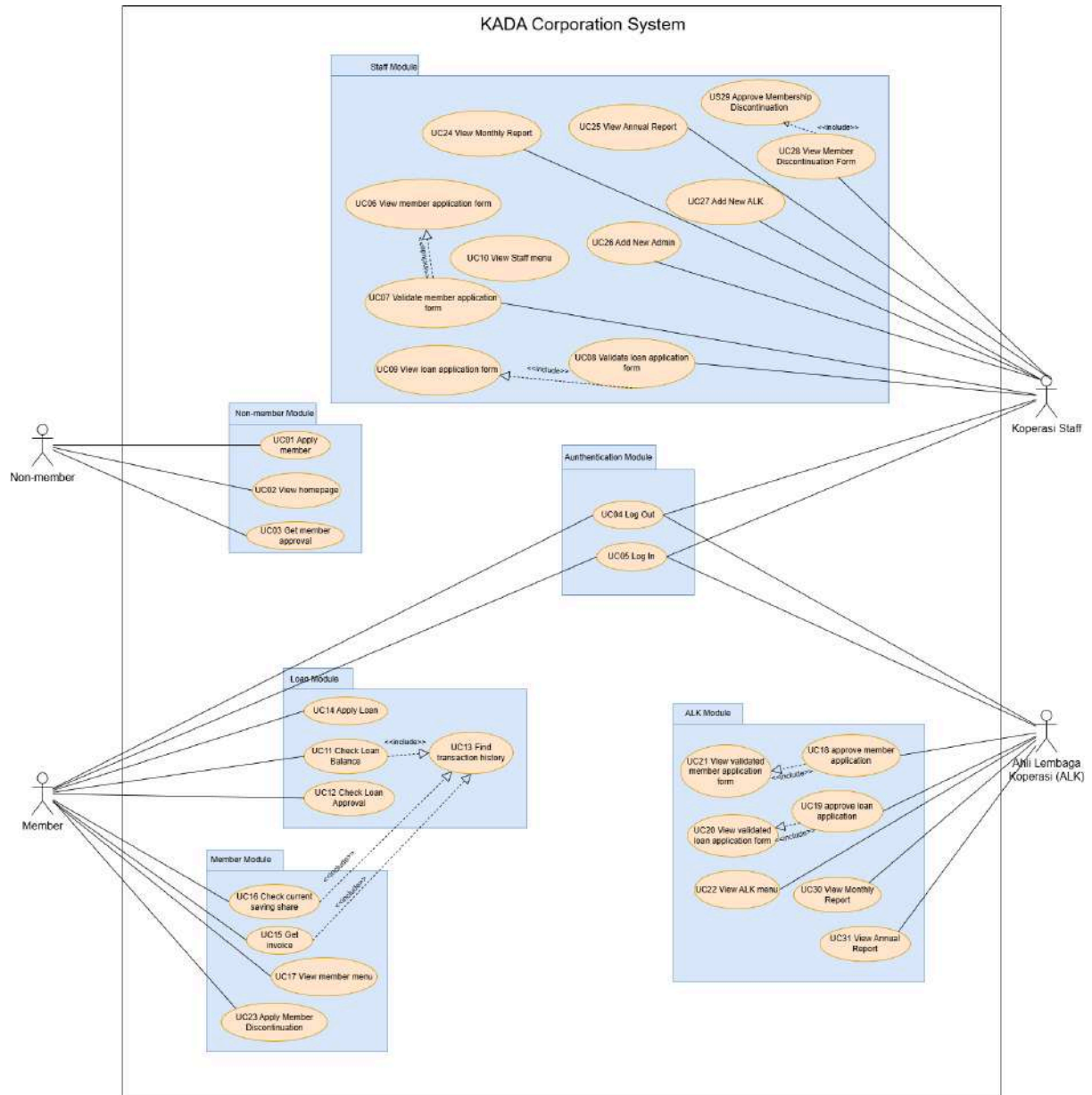


Figure 1: Use Case Diagram for Koperasi KADA System

Use case	Function	Description
UC01	Apply member	This use case allows users to apply for Koperasi KADA membership.

UC02	Get member approval	This use case will notify users when their membership application is approved and provide login details to them.
UC03	View Homepage	This use case allows all users to see the homepage of the KADA website without logging in.
UC04	Log out	This use case allows users to exit the system.
UC05	Log in	This use case allows users to access the system.
UC06	View member application form	This use case lets users see the submitted membership application form.
UC07	Validate member application form	This use case is for users to validate each submitted membership application form and make sure the details are true before being sent to the ALK.
UC08	Validate loan application form	This use case is for users to validate each submitted loan application form and make sure the details are true before being sent to the ALK.
UC09	View loan application form	This use case lets users see the submitted loan application form.
UC10	View staff menu	This use case allows staff to see the menu of the system where all their functions are available there.
UC11	Check loan balance	This use case lets users check their loan balance.
UC12	Check loan approval	This use case notifies users about the status of their loan applications approval.
UC13	Find transaction history	This use case sends all the transaction history to users.
UC14	Apply loan	This use case allows users to send their loan application form for approval.

UC15	Get invoice	This use case prints out the invoice of every transaction made to users.
UC16	Check current saving share	This use case allows users to see their current share and savings.
UC17	View member menu	This use case allows members to see the menu of the system where all their functions are available there.
UC18	Approve member application	This use case allows users to approve the membership application and trigger application status.
UC19	Approve loan application	This use case allows users to approve the loan application and trigger application status.
UC20	View validated loan application form	This use case lets users see the submitted and validated loan application form.
UC21	View validated member application form	This use case lets users see the submitted and validated membership application form.
UC22	View ALK menu	This use case allows ALK to see the menu of the system where all their functions are available there.
UC23	Apply member termination	This use case allows users to send their membership termination form for approval.
UC24	View monthly report	This use case allows users to see the monthly report for membership and loan application graphically.
UC25	View annual report	This use case allows users to see the annual report for membership and loan application graphically.
UC26	Add new admin	This use case allows users to add new admin for this system.
UC27	Add new ALK	This use case allows users to add new ALK for this system.

UC28	View membership termination form	This use case allows users to see the submitted membership termination application form.
UC29	Approve membership termination	This use case allows users to approve the membership termination application and trigger application status.
UC30	View monthly report	This use case allows users to see the monthly report for membership and loan application graphically
UC31	View annual report	This use case allows users to see the annual report for membership and loan application graphically

Table 25: Description of module and functions for Koperasi KADA System

2.2.2 Domain Model: Class Diagram

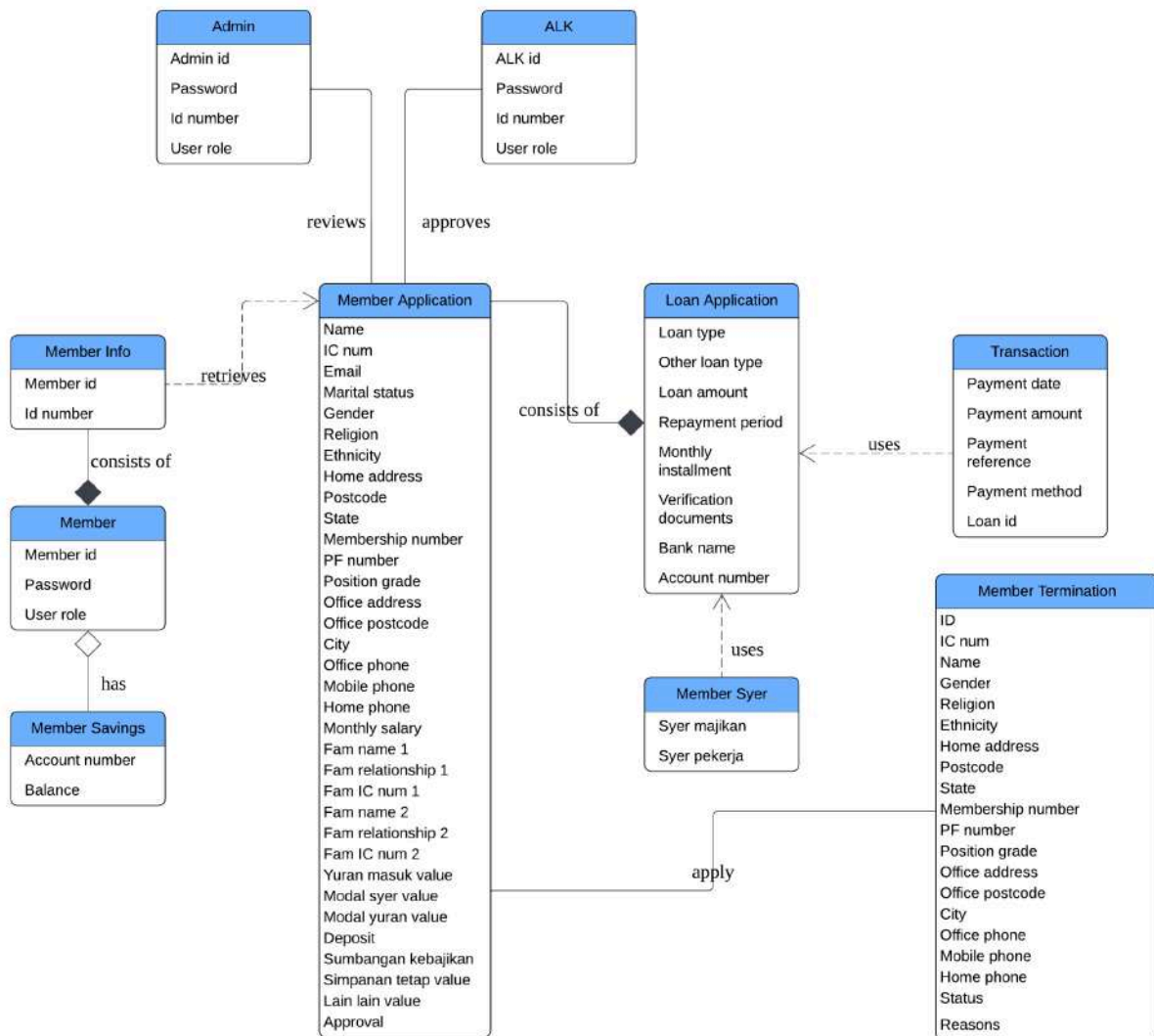


Figure 2: Domain Model for Koperasi KADA system

The classes used in the Koperasi KADA system are illustrated in the domain model in Figure 2 along with the relationship for each class. Each member consists of a Member Info table which has member id and id number. Member Info table retrieves member information from Member Application table using id number. A member can have savings which the information will be stored in the Member Savings table. A member can apply for loans in which the information will be stored in the Loan Application table. The Loan Application table consists of member information retrieved from the Member Application table. If a member has any loans, then Member Syer and Transaction depends on the loan. Admin and Ahli Lembaga

Koperasi (ALK) review and approve the member application respectively. Member can also apply for membership termination if they wish to.

2.2.3 State Diagram

Figure 3 below shows the main flow of the system where a non-member of Koperasi KADA wants to apply for a membership. The flow starts with a non-member to fill in the form provided and after the submission of the form, it will be reviewed by the staff of Koperasi KADA. If the submitted form has been reviewed and approved by staff, the process will continue by the ALK to review the form. If ALK approves the application, non-members will change their status to member of Koperasi KADA upon the membership is approved by ALK. For a disapproved application, non-members can apply for a membership again by filling in the form again and going through the same process again. Same cycle happens for Figure xxx for a member of Koperasi KADA to apply for a loan where the difference is the loan application can only be applied for members of Koperasi KADA only.

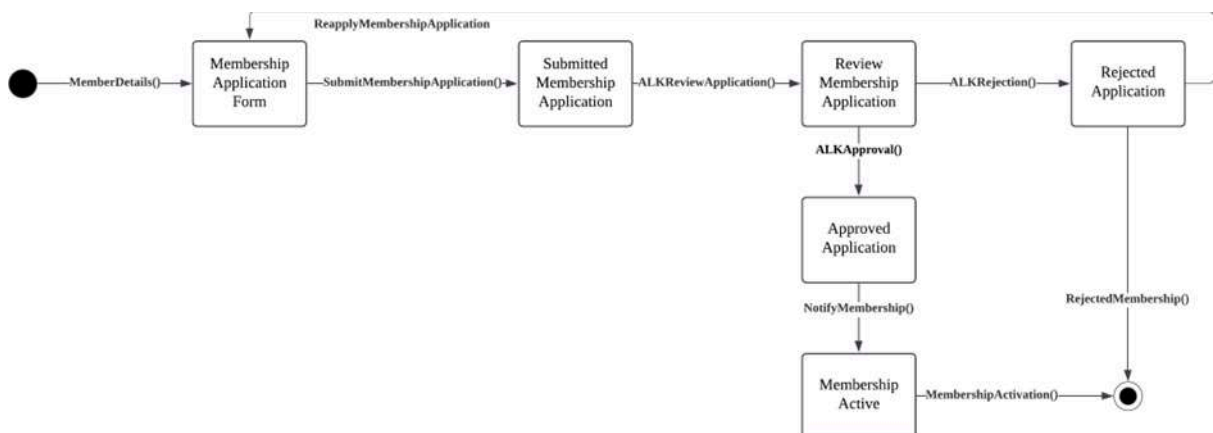


Figure 3: State Machine Diagram for <Membership Application System>

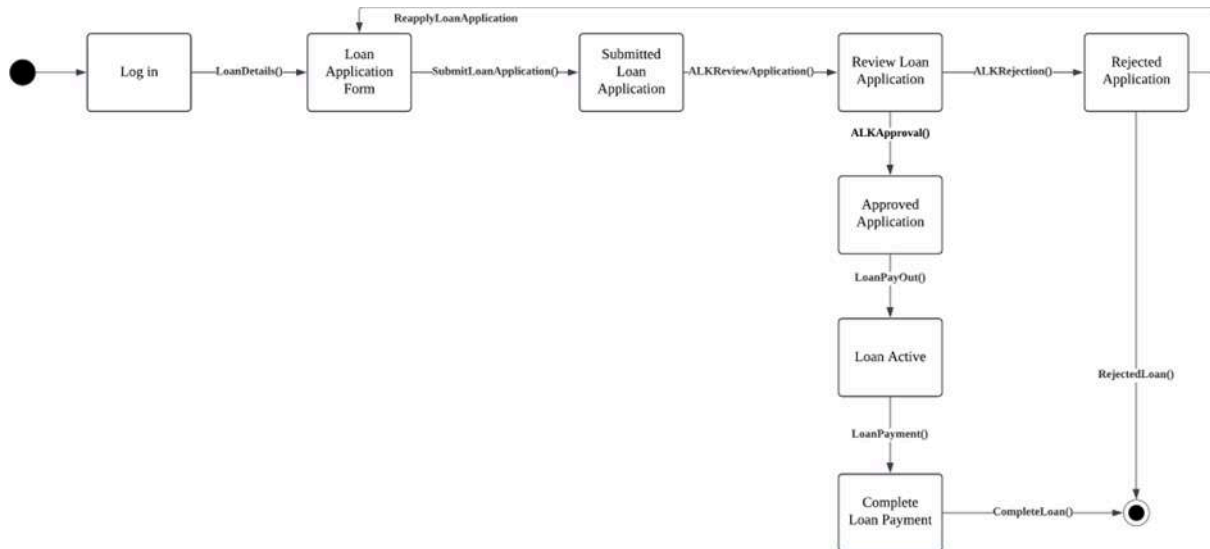


Figure 4: State Machine Diagram for <Loan Application System>

2.3 Launch Phase

Backlog Items:

1. Develop a non-member module to allow non-members to apply for membership and track their application status.
2. Develop a member registration and login module to enable new users to register and access the system securely.
3. Implement a loan application and approval module to allow members to submit loan applications and check approval statuses.
4. Create a staff dashboard that facilitates membership and loan validation processes efficiently.
5. Build an ALK dashboard to enable high committee members to review and approve membership and loan applications.
6. Incorporate security and encryption measures to protect sensitive data and ensure compliance with privacy standards.
7. Optimize system performance to handle simultaneous user requests and maintain quick response times.
8. Provide user training and documentation to ensure smooth adoption and usability of the system.

Sprint	Team members assigned
Sprint #1: Non-Member Module Development • User Stories Covered: - US004: Non-member applies for membership. • Tasks: - Develop membership application forms for non-members. - Implement functionality to upload required verification documents. - Enable non-members to track application status and receive notifications. - Test non-member functionalities to ensure smooth operations.	Nurul Ika Syafiny binti Azhar
Sprint #2: Member Module Development • User Stories Covered: - US001: Member login and registration. - US001: Member loan application and approval. • Tasks: - Implement login and registration features for members. - Develop loan application forms and validation processes. - Create views for loan approvals and balance. - Test functionalities to ensure smooth operations.	Nuraisyah Binti Mohd Zikre

Sprint #3: Admin Module Development • User Stories Covered: - US002: Admin viewing and validating membership applications. - US002: Admin validating loan applications. • Tasks: - Create an admin dashboard for membership and loan validation. - Develop forms for document validation. - Provide download features for lists of members and loans. - Conduct unit testing for admin features.	Harini A/P Sangaran
Sprint #4: ALK Module Development • User Stories Covered: - US003: ALK verifying membership applications. - US003: ALK validating loan applications. • Tasks: - Build an ALK dashboard for reviewing and approving applications. - Enable document verification workflows. - Test ALK functionalities for accuracy.	Damiya Aina binti Basir Abd Shammad

Sprint #5: Final Testing and Deployment • Tasks: - Conduct integration testing for all modules. - Optimize system performance. - Prepare deployment scripts. - Provide user training and manuals.	All members
--	-------------

Table 26: Sprint table for each member

2.4 User Story Details

This section is a detailed explanation of the user stories described in Persona's user stories. The User Story ID for members is US001 whereas the User Story ID for staff is US002. User Story ID for ALK is US003 whereas the User Story ID for non-members US004. US001 is the first User Story, followed by US002, US003 and US004 respectively.

2.4.1 US001: User Story <User Story 1>

User story: Member
ID: US001
User Story Description : As a member of Koperasi KADA, I want to log in into the system so that I can apply for a loan, view my loan application approval, view loan balance, view savings and financial statements.

Flow of events:

1. The member logs into the system.
2. If the member wants to apply for a loan,
 - 2.1 The member needs to click on the loan application button.
 - 2.2 The system will display form by the system for member to fill in the required details
 - 2.3 After filled in all the required details member need to submit the form by agreeing to all the term
3. If the member wants to view loan application approval
 - 3.1 The member needs to click on the loan application approval button
 - 3.2 The system will display the status of approval of the application whether it is approved or not
4. If the member wants to loan balance
 - 4.1 The member needs to click on the loan balance button.
 - 4.2 The system will display the details of the balance of the loan that needs to be paid by the member.
5. If the member wants to view the saving
 - 5.1 The member needs to click on the savings button
 - 5.2 The system will display the details of saving from the deduction of the monthly membership fee
6. If the member wants to view financial statement
 - 6.1 The member needs to click on the financial statement button.
 - 6.2 The system will display the details of the transaction that occur every month based on the salary deduction
7. If the member wants to apply for membership termination
 - 7.1 The member needs to click on the membership termination application button.
 - 7.2 The system will display form by the system for member to fill in the required details
 - 7.3 After filled in all the required details member need to submit the form by agreeing to all the term

Alternative flow 1:

If the member does not apply for a loan when view loan application approval

- The system will show that the member has not apply for a loan yet

Alternative flow 2:

If the member does not apply for a loan when view loan balance

- The system will show that the member has not apply for a loan yet

Acceptance Criteria

Postcondition: The user is a member of Koperasi KADA

Precondition: The loan application applied by member will be recorded in the system

Other conditions: The details filled by member will be validated during the process for approval

Table 27: User story description for US001

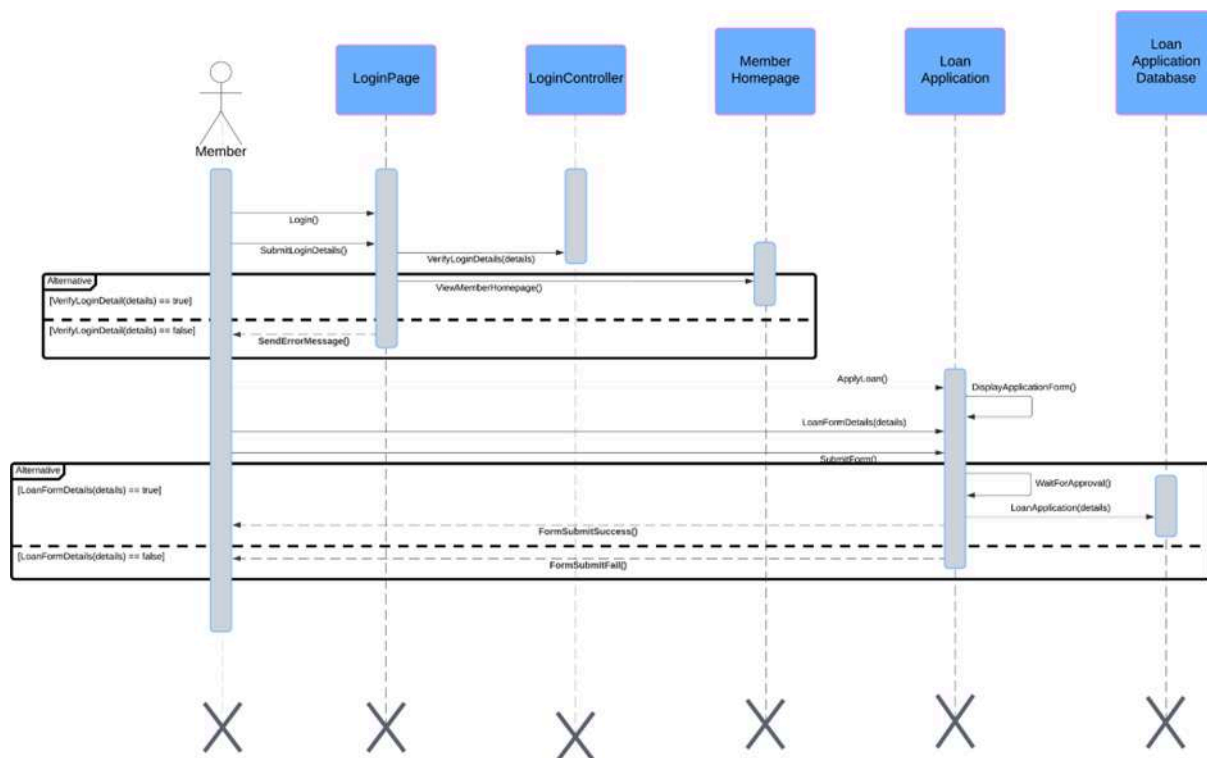


Figure 5: SD001 sequence diagram for member apply loan

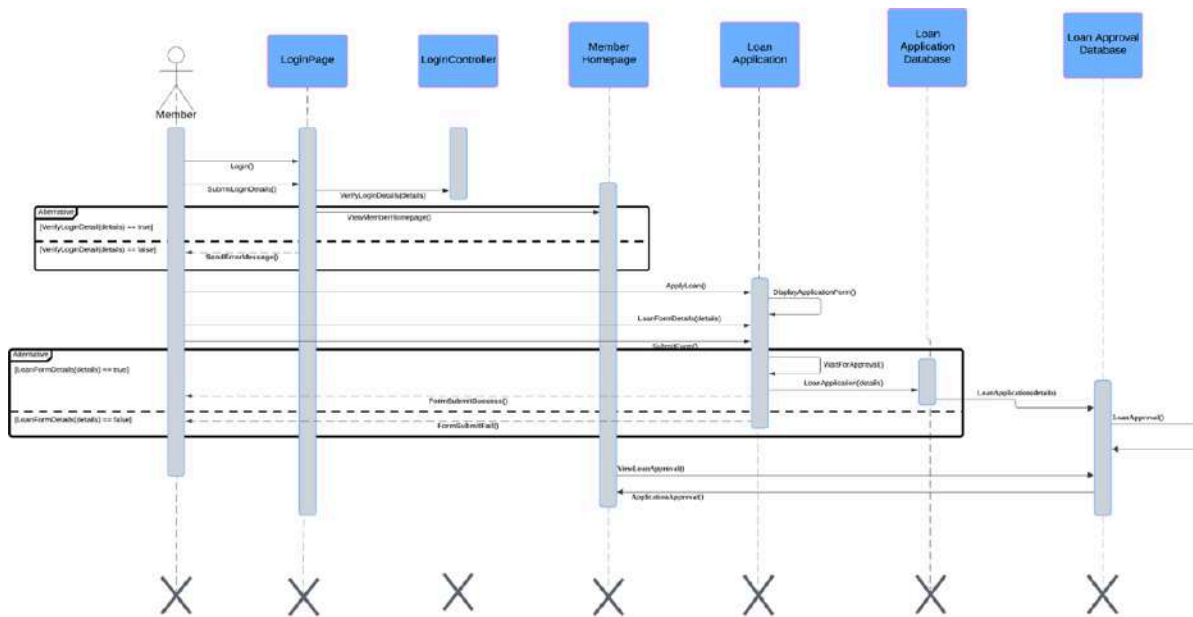


Figure 6: SD002 sequence diagram for member view loan application form

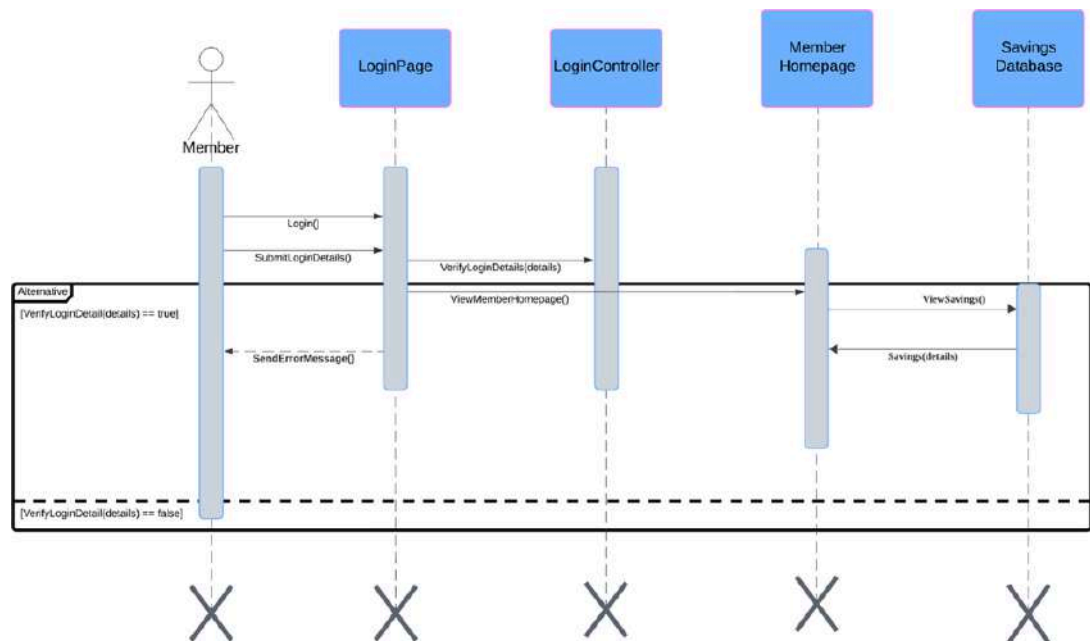


Figure 7: SD003 sequence diagram for member check current saving

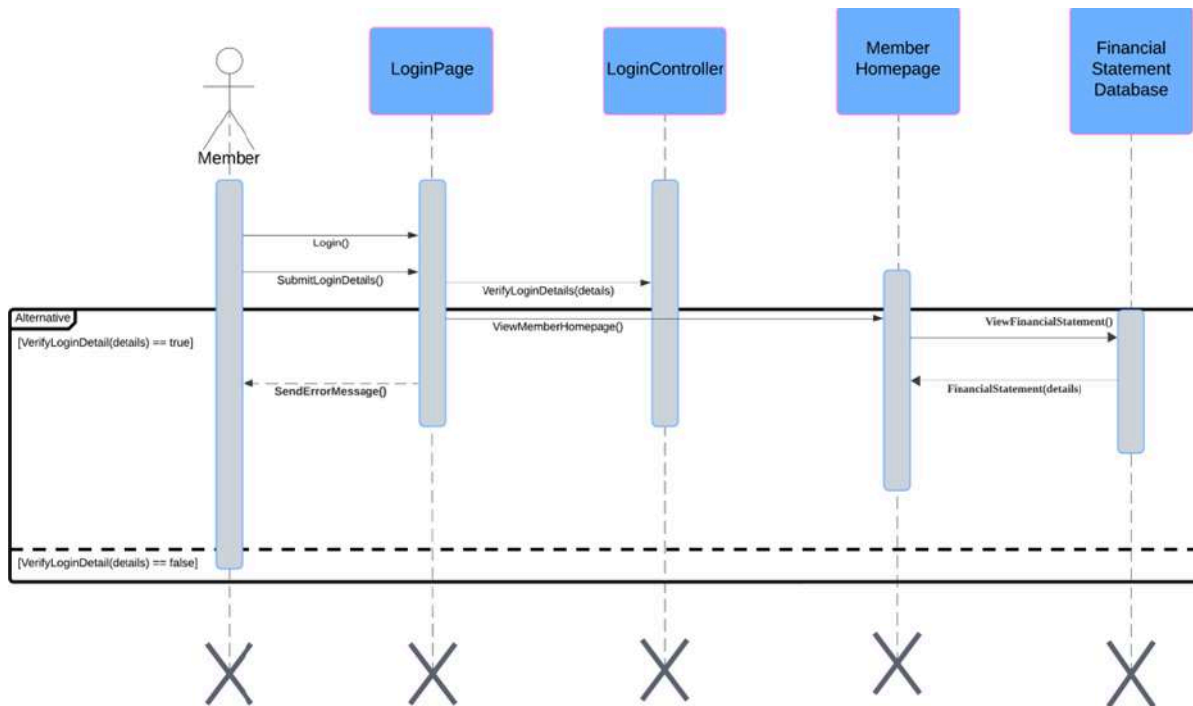


Figure 8: SD004 sequence diagram for member find transaction history

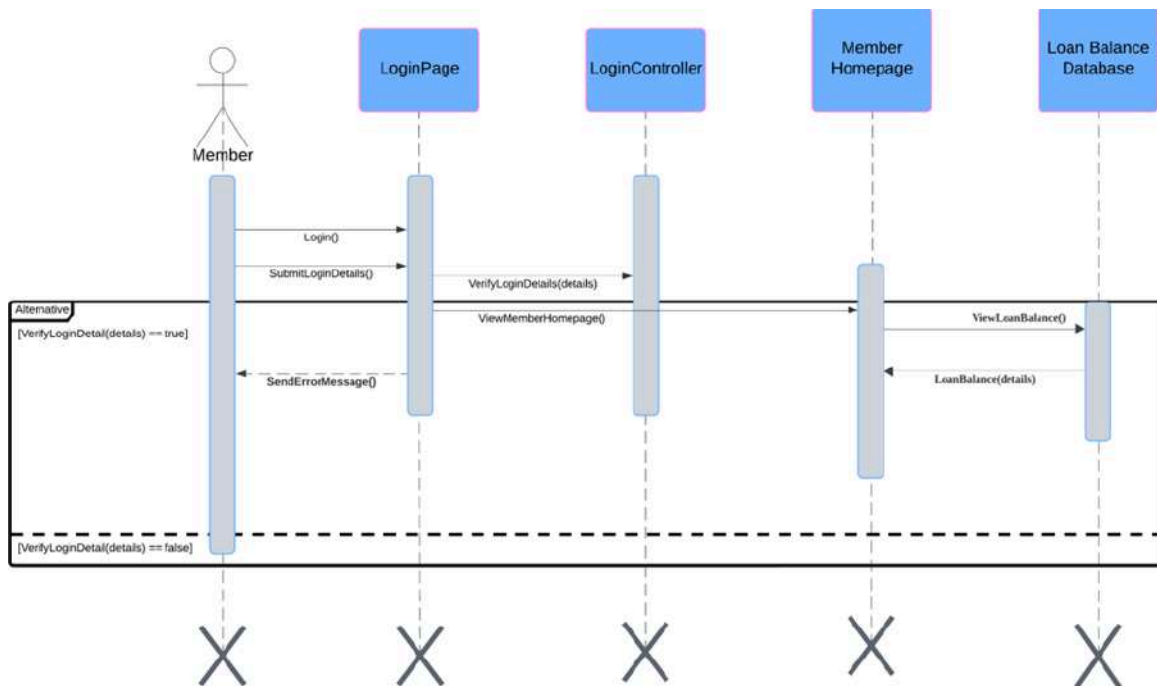


Figure 9: SD005 sequence diagram for check loan balance

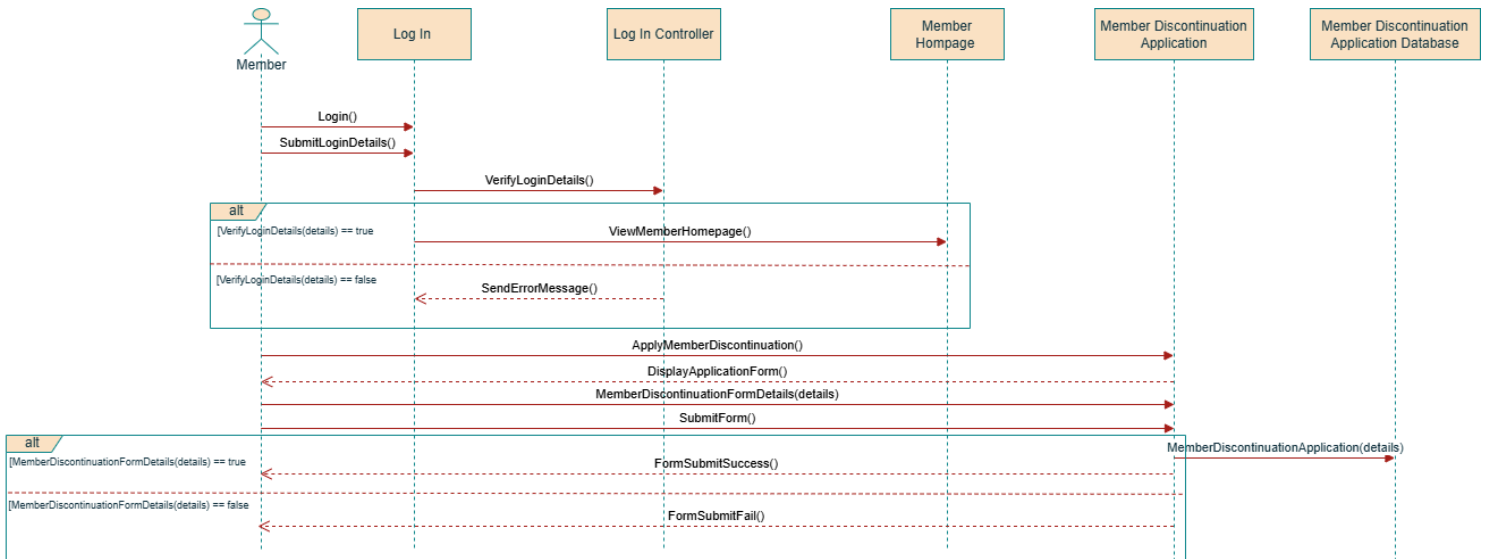


Figure 10: SD006 sequence diagram for apply membership termination

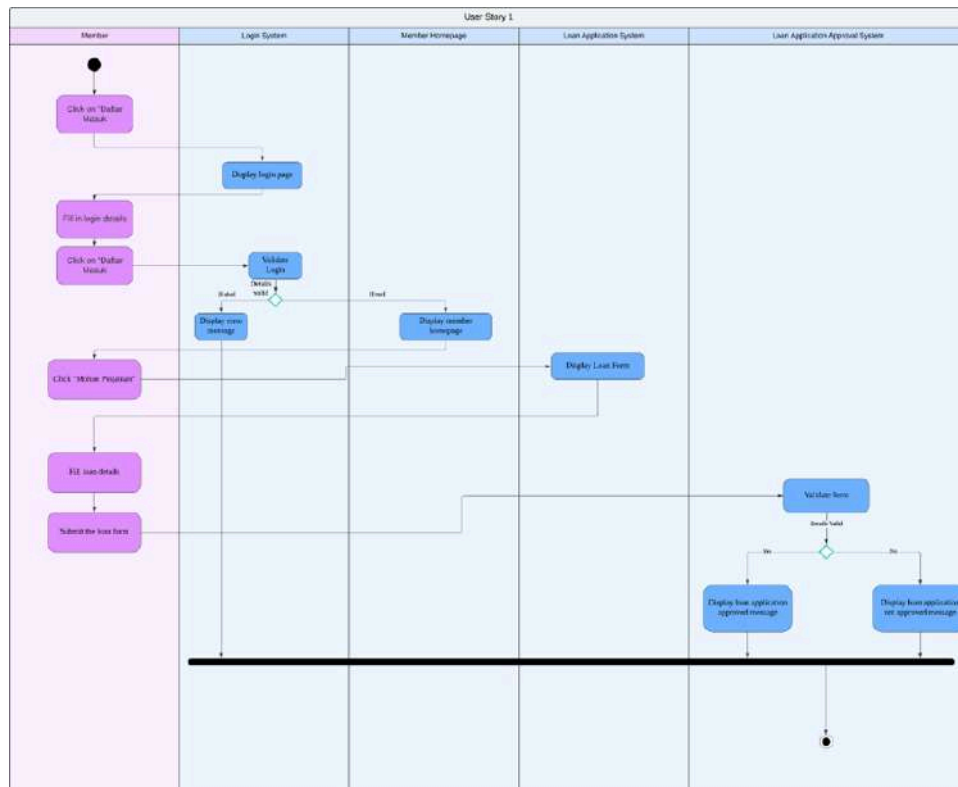


Figure 11: Activity diagram for member apply loan application

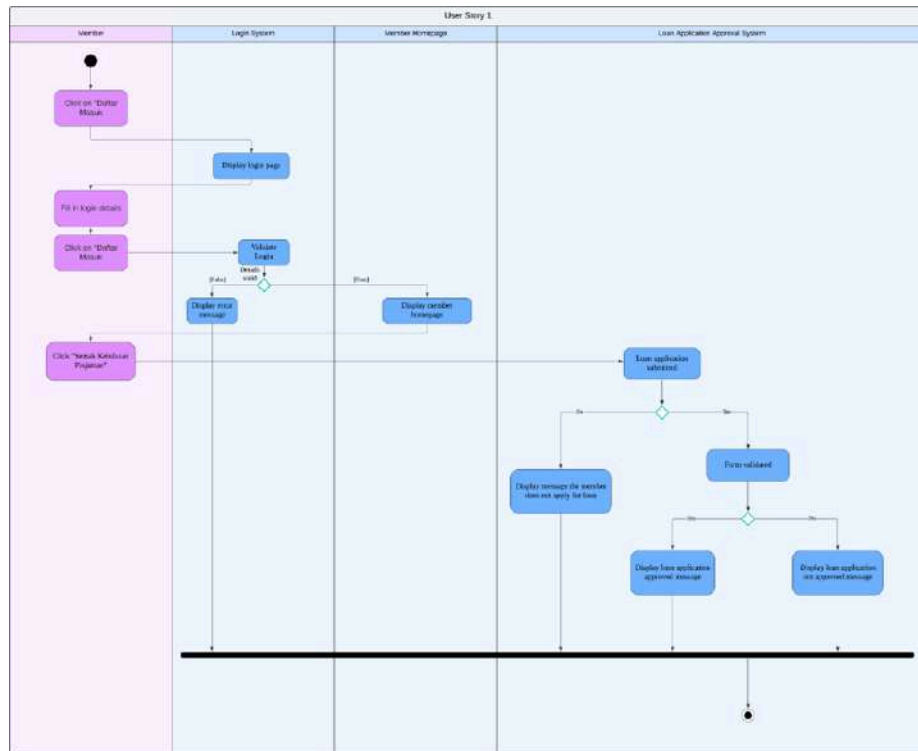


Figure 12: Activity diagram for member view approved loan application

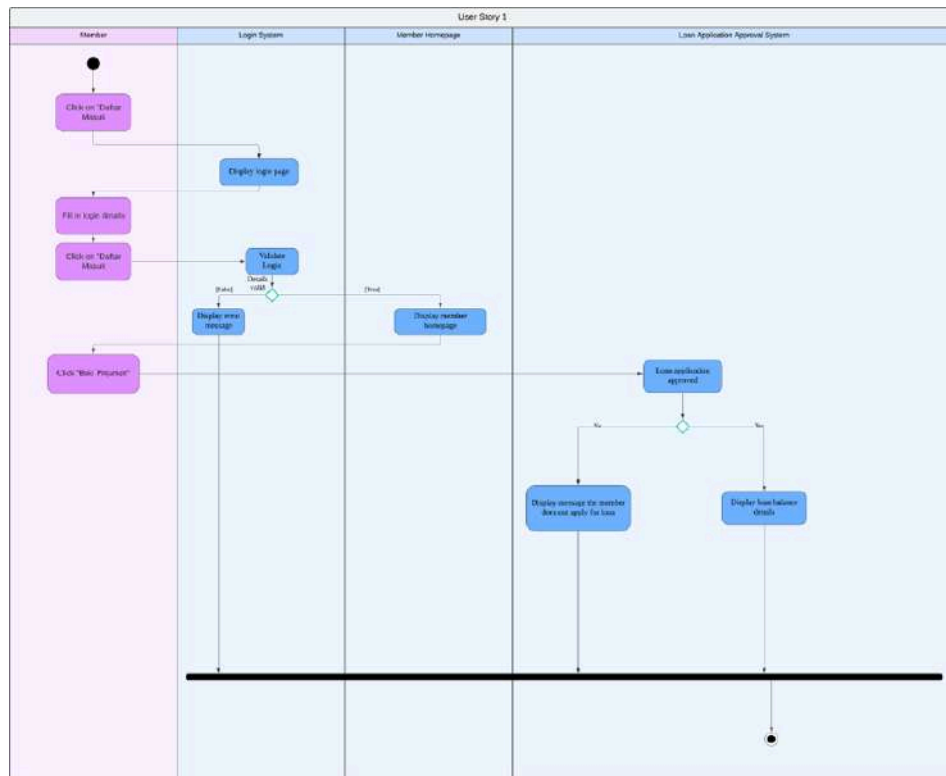


Figure 13: Activity diagram for member check the loan balance

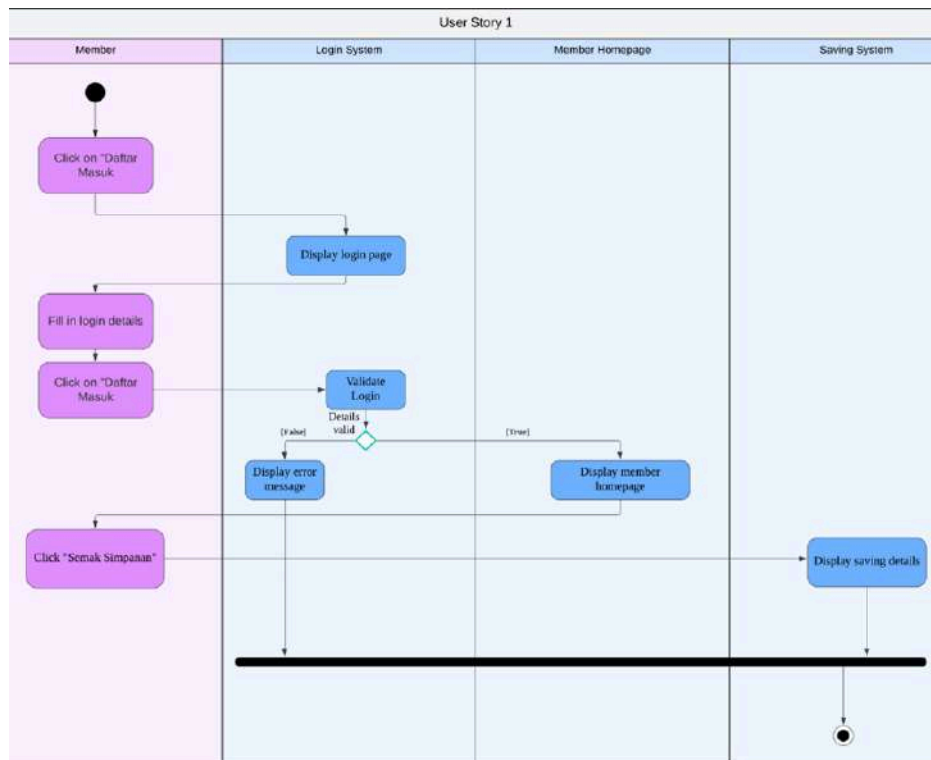


Figure 14: Activity diagram for member check current saving

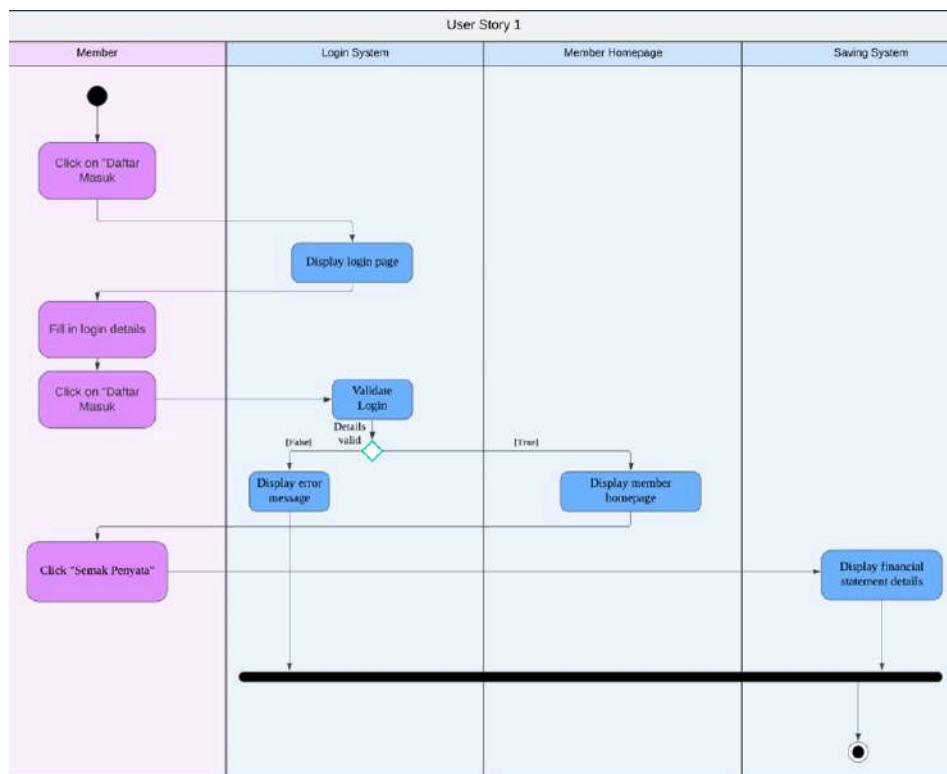


Figure 15: Activity diagram for member find transaction history

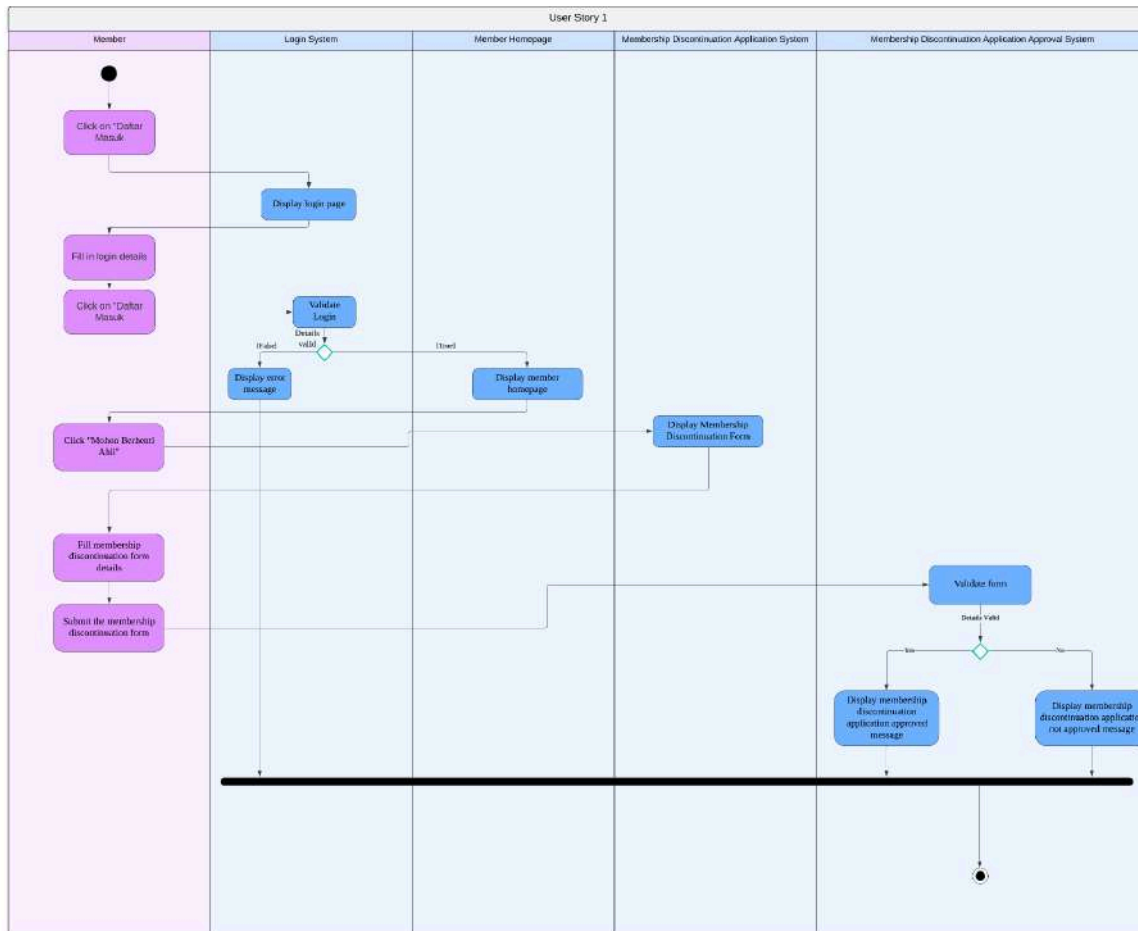


Figure 16: Activity diagram for member apply membership discontinuation

2.4.2 US002: User Story <User Story 2>

User story: Admin
ID: US002
User Story Description : As an admin of Koperasi KADA I want to log in into the system so that I can view the list of members and list of members with loan and validate the verification documents of non-members who applied for membership and members who applied for a loan.
Flow of events: 1. The admin logs into the system. 2. If the admin wants to view the list of applicants who applied for membership, 2.1 The admin needs to click on the view membership application list button.

2.2 The system will display a list with applicants' names, identification number and verification documents.

3. If the admin wants to view the list of members who applied for loan

3.1 The admin needs to click on the view loan application list button.

3.2 The system will display the members' names, identification card numbers, the type of loan they applied for and the verification documents.

4. If the admin wants to view the list of members who applied for membership termination

4.1 The admin needs to click on the view membership termination application button.

4.2 The system will display the members' names, identification card numbers and the verification documents.

5. If the admin wants to validate the verification documents for membership application

5.1 The admin needs to click on the membership application list button.

5.2 The system will display a list with applicants' names, identification number and verification documents and the status validity status of the verification documents.

5.3 If the admin wants to validate the verification documents

5.3.1 The admin must click on the verification documents to check the documents.

5.3.2 If the documents are as per requirements

5.3.2.1 The admin validates the verification documents.

5.3.3 Else

5.3.3.1 The admin does not approve the verification documents.

5.3.4 The validity status of the verification documents is updated.

5.4 If the admin wants to update the validity of the documents

5.4.1 The admin must click on the validity status of the verification documents.

5.4.2 If the admin wants to validate the verification documents

5.4.2.1 The admin must click on the validate button.

5.4.2.2 The admin must submit the form.

5.4.2.3 The validity status of the verification documents is updated.

5.4.3 Else

5.4.3.1 The admin must click on the disapprove documents button.

5.4.3.2 The admin must submit the form.

5.4.3.3 The validity status of the verification documents is updated.

6. If the admin wants to validate the verification documents for loan application

6.1 The admin needs to click on the loan application list button.

6.2 The system will display the members' names, identification card numbers, their verification documents and the validity status of the verification documents.

6.3 If the admin wants to validate the verification documents

6.3.1 The admin must click on the verification documents to check the documents.

6.3.2 If the documents are as per requirements

6.3.2.1 The admin validates the verification documents.

6.3.3 Else

6.3.3.1 The admin does not approve the verification documents.

6.3.4 The validity status of the verification documents is updated.

6.4 If the admin wants to update the validity of the documents

6.4.1 The admin must click on the validity status of the verification documents.

6.4.2 If the admin wants to validate the verification documents

6.4.2.1 The admin has to click on the validate button.

6.4.2.2 The admin must submit the form.

6.4.2.3 The validity status of the verification documents is updated.

6.4.3 Else

6.4.3.1 The admin must click on the disapprove documents button.

6.4.3.2 The admin must submit the form.

6.4.3.3 The validity status of the verification documents is updated.

7. If the admin wants to approve the membership termination applications

7.1 The admin needs to click on the membership termination application list button.

7.2 The system will display the members' names, identification card numbers, their verification documents and the application status.

7.3 If the admin wants to approve the application

7.3.1 The admin must click on the verification documents to check the documents.

7.3.2 If the documents are as per requirements

7.3.2.1 The admin approves the application.

7.3.3 Else

7.3.3.1 The admin does not approve the application.

7.3.4 The application status is updated.

7.4 If the admin wants to update the application status

7.4.1 The admin must click on the application status.

7.4.2 If the admin wants to approve the application

7.4.2.1 The admin has to click on the approve button.

7.4.2.2 The admin must submit the form.

7.4.2.3 The application status is updated.

7.4.3 Else

7.4.3.1 The admin must click on the disapprove button.

7.4.3.2 The admin must submit the form.

7.4.3.3 The application status is updated.

8. If the admin wants to view the system report 8.1 The admin must click on the system report button. 8.2 The system displays the monthly and annual report for employer and employee shares, membership applications, loan applications and membership termination applications. 8.3 If the admin wants to view system report up to a certain date only 8.3.1 The admin must click on the specific date on the calendar displayed by the system. 8.3.2 The system report will update up to the specific date. 9. If the admin wants to add a new admin to the system 9.1 The admin must click on the add new admin button. 9.2 The admin needs to key in the new admin's identification card number. 9.3 The admin needs to submit the form. 9.4 New admin is added to the system successfully. 10. If the admin wants to add a new ALK to the system. 10.1 The admin must click on the add new ALK button. 10.2 The admin needs to key in the new ALK's identification card number. 10.3 The admin needs to submit the form. 10.4 New ALK is added to the system successfully.
Alternative flow 1: If the admin fails to login to the system • The system will show that login is unsuccessful
Acceptance Criteria Precondition: The user is an admin of Koperasi KADA

Table 28: User story description for US002

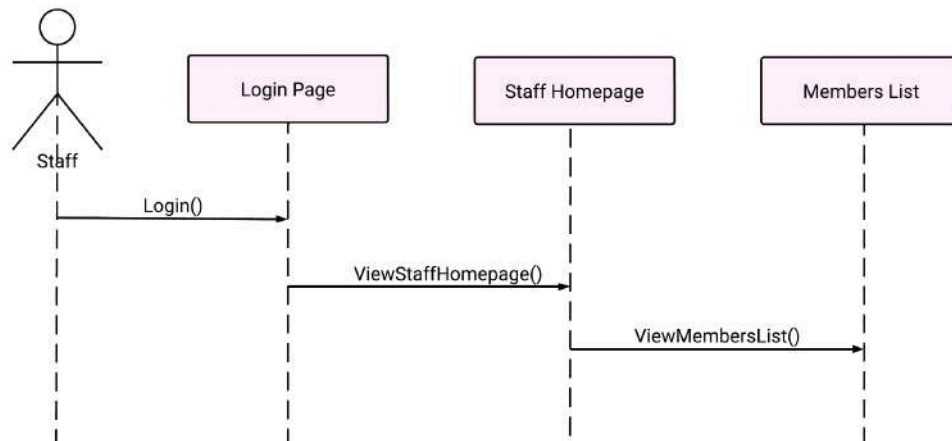


Figure 17: SD007 sequence diagram for view member application form

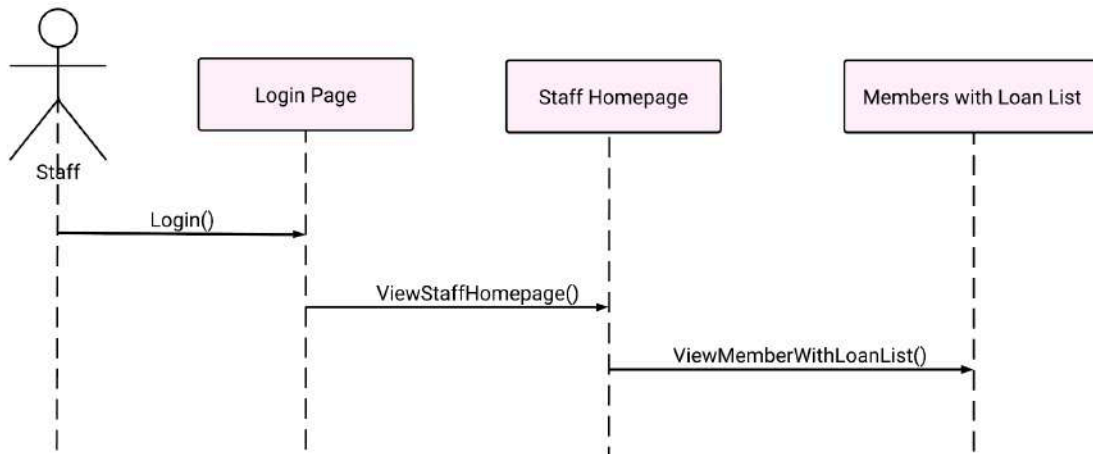


Figure 18: SD008 sequence diagram for view loan application form

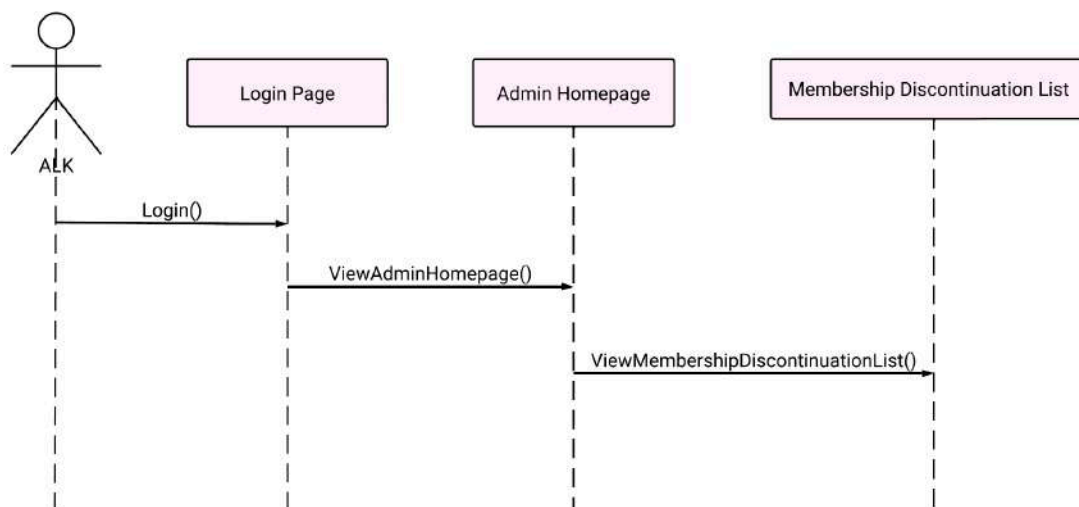


Figure 19: SD009 sequence diagram for view membership termination application form

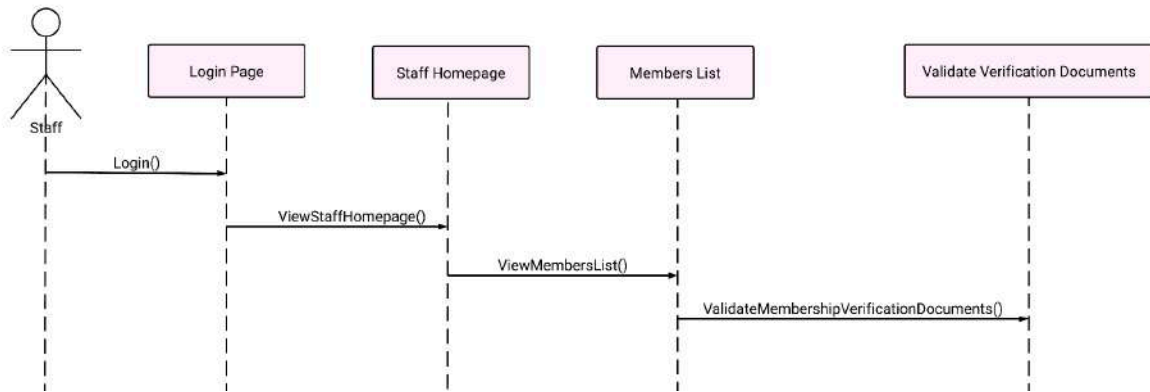


Figure 20: SD010 sequence diagram for validate membership application form

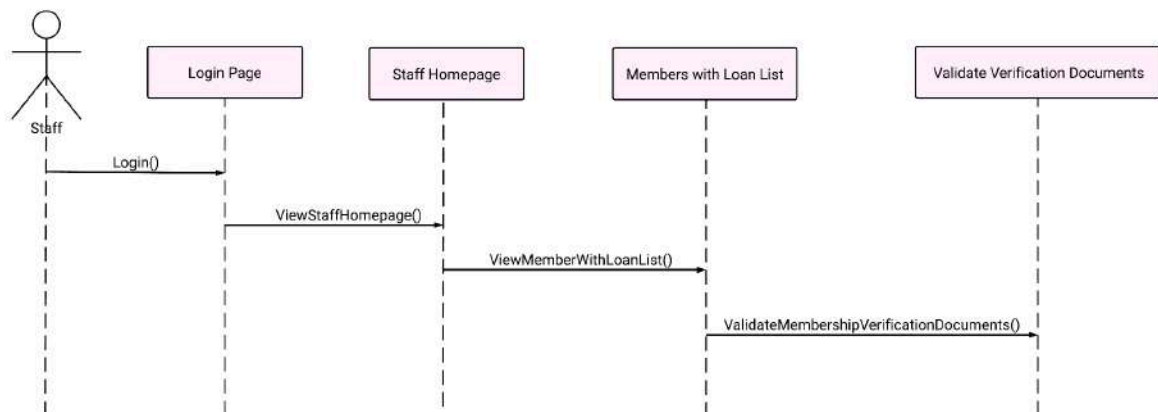


Figure 21: SD011 sequence diagram for validate loan application form

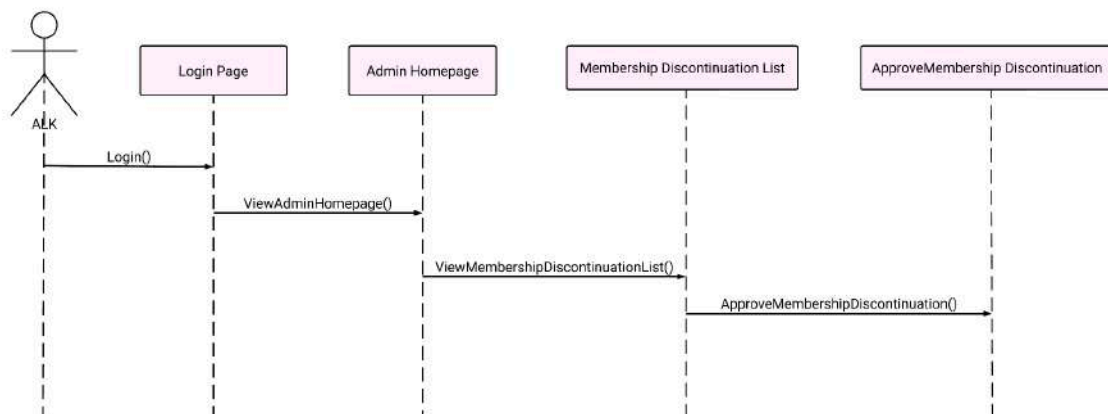


Figure 22: SD012 sequence diagram for approve membership termination application form

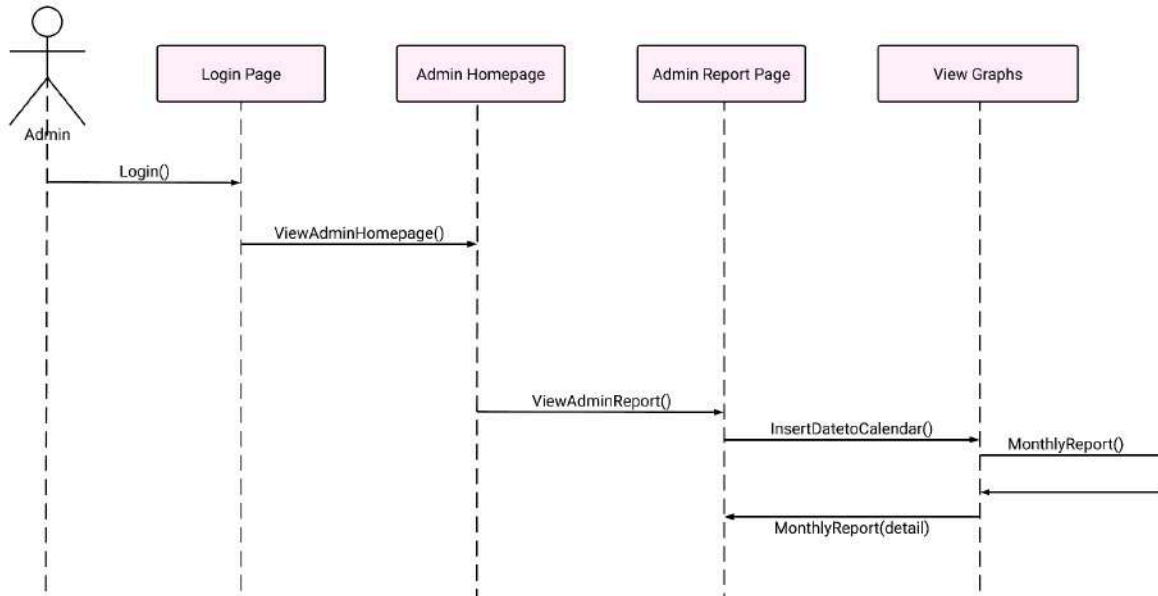


Figure 23: SD013 sequence diagram for view monthly report

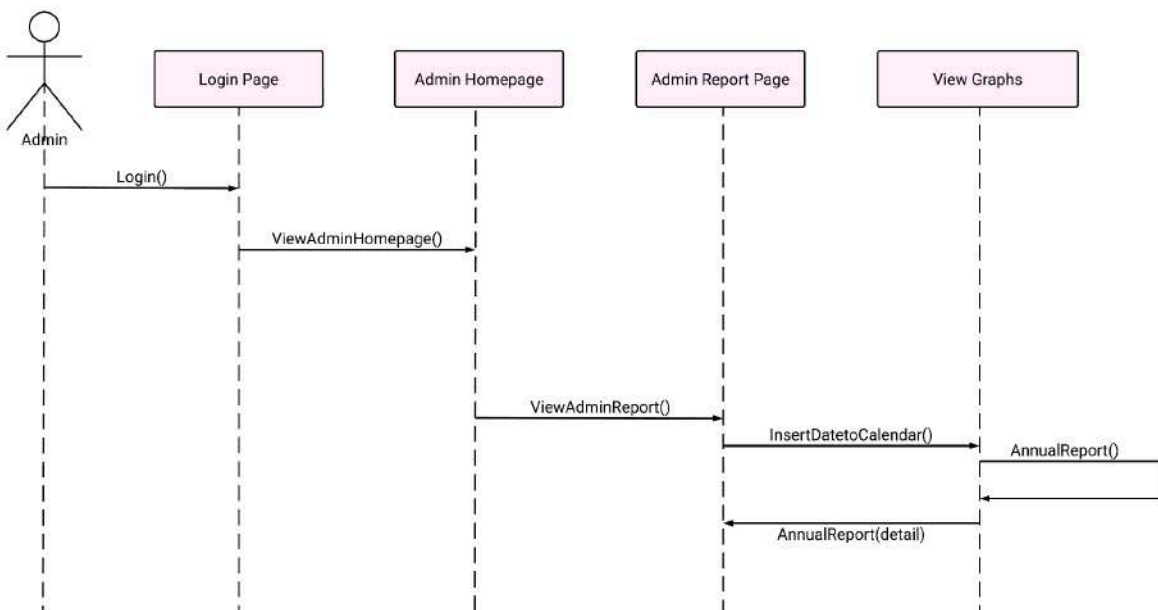


Figure 24: SD014 sequence diagram for view annual report

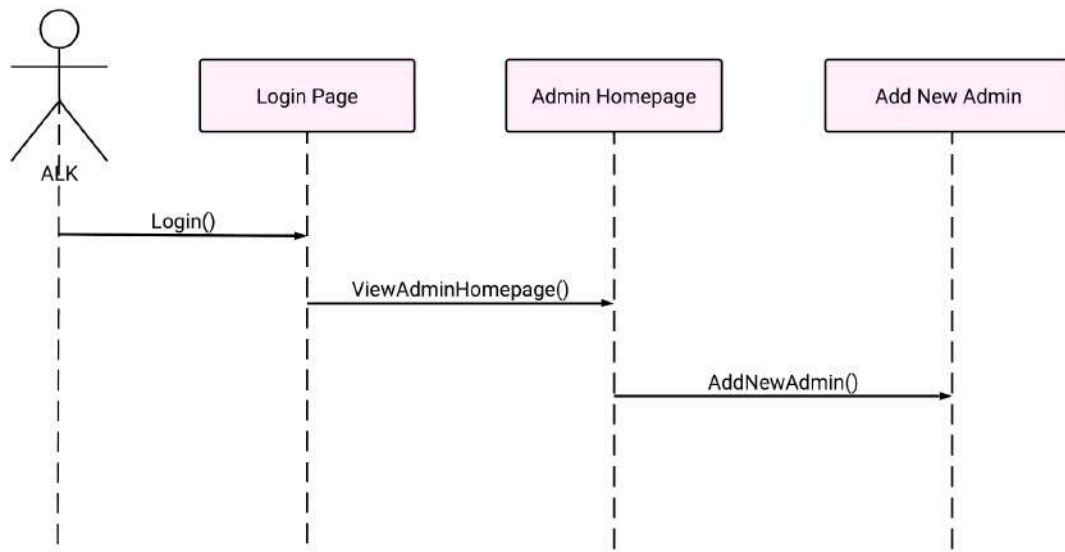


Figure 25: SD015 sequence diagram for add new admin

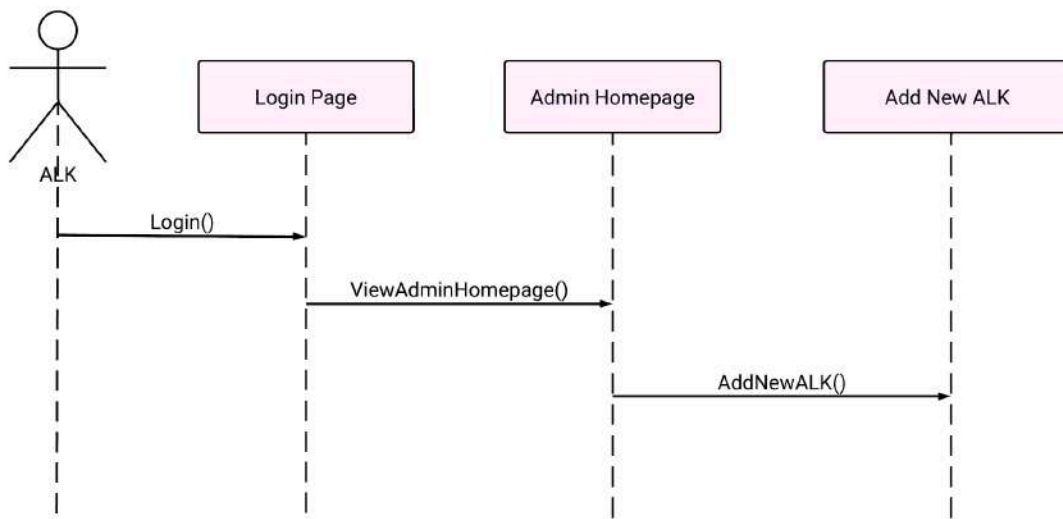


Figure 26: SD016 sequence diagram for add new ALK

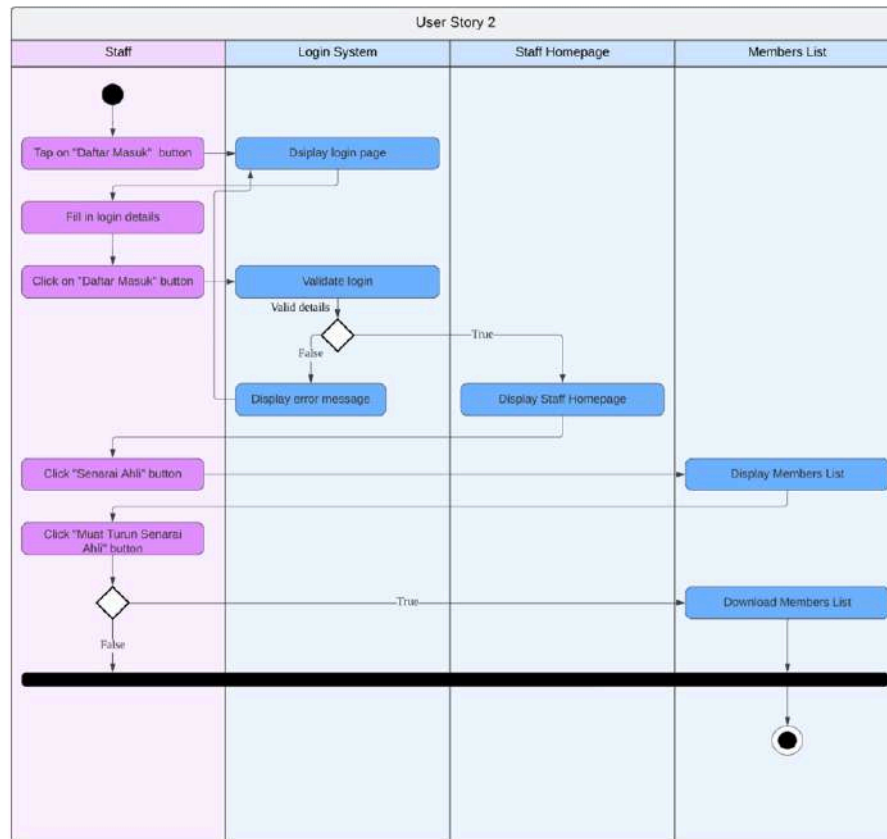


Figure 27: Activity diagram for when admin chooses to view members list

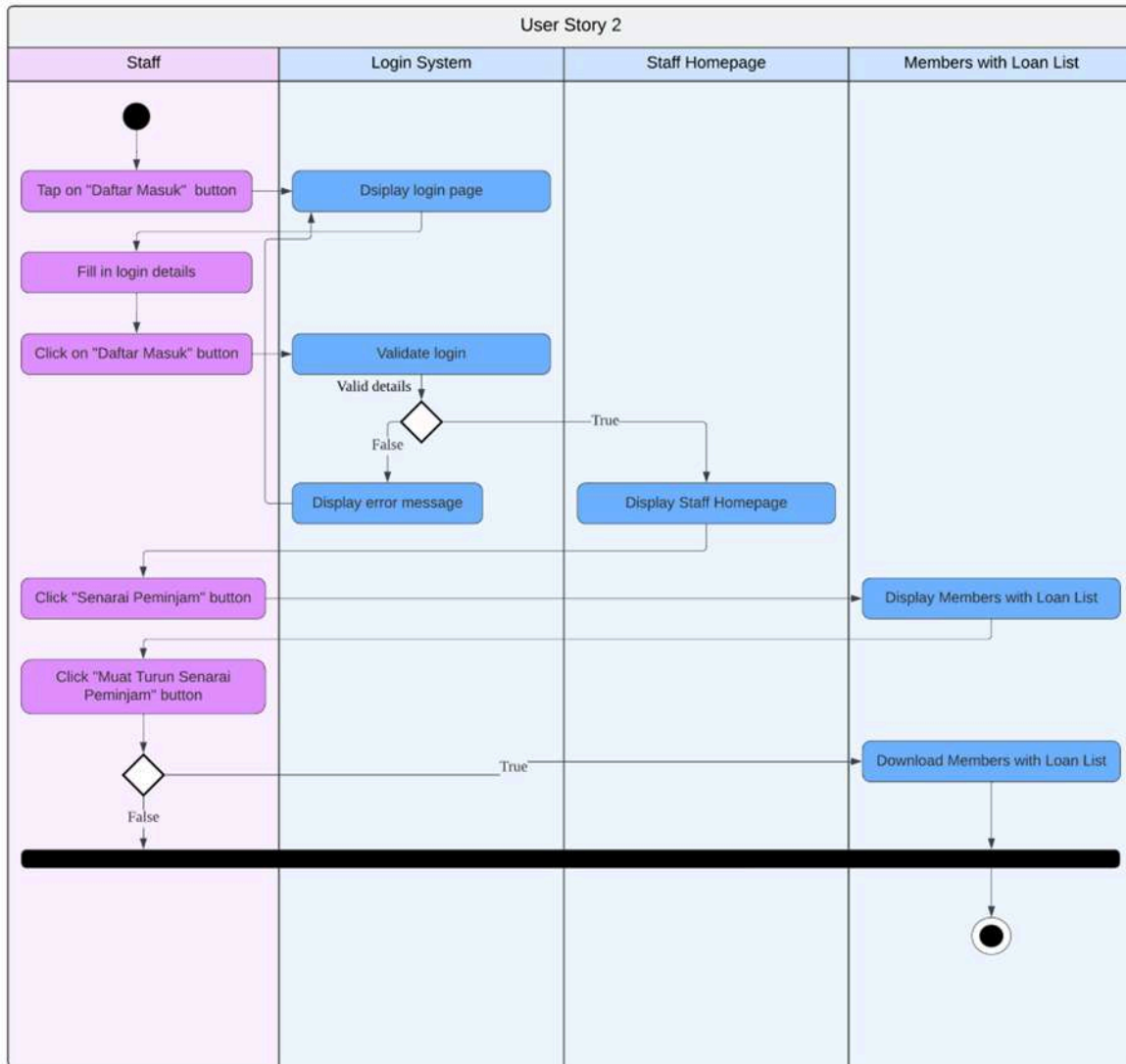


Figure 28: Activity diagram for when admin chooses to view members with loan list

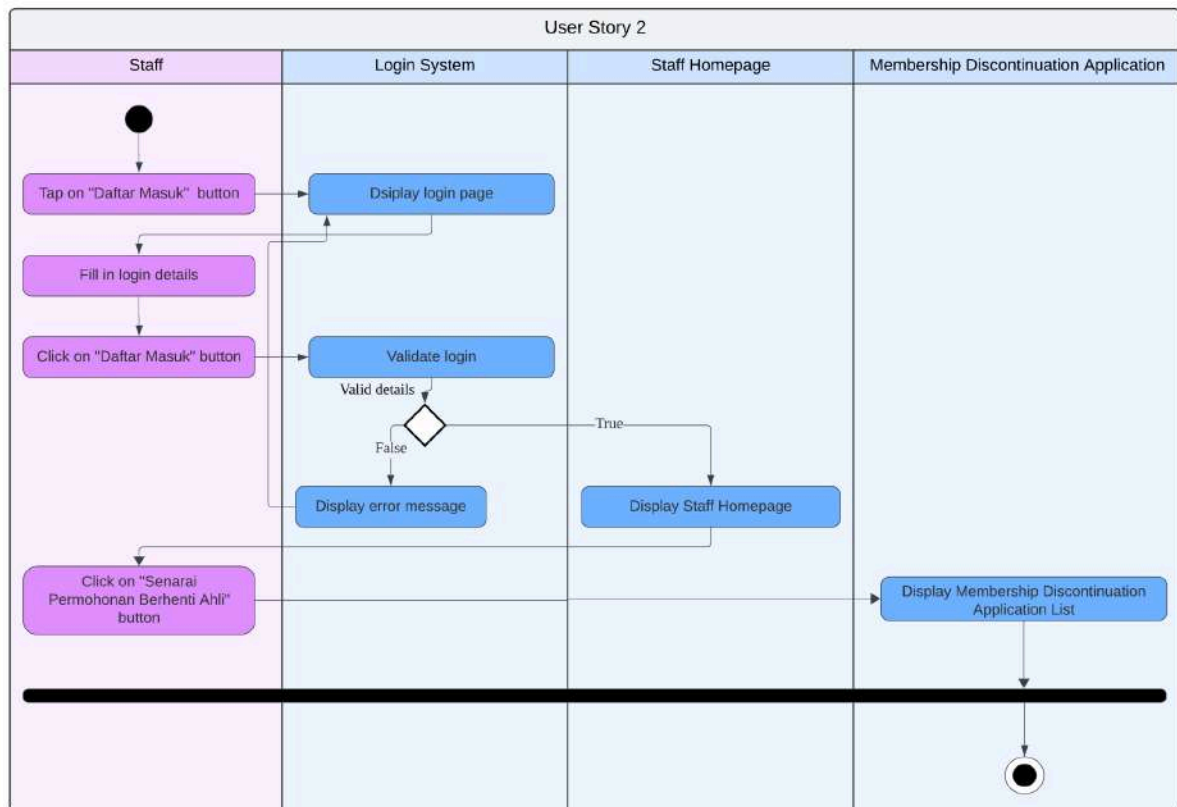


Figure 29: Activity diagram for when admin chooses to view membership termination application list

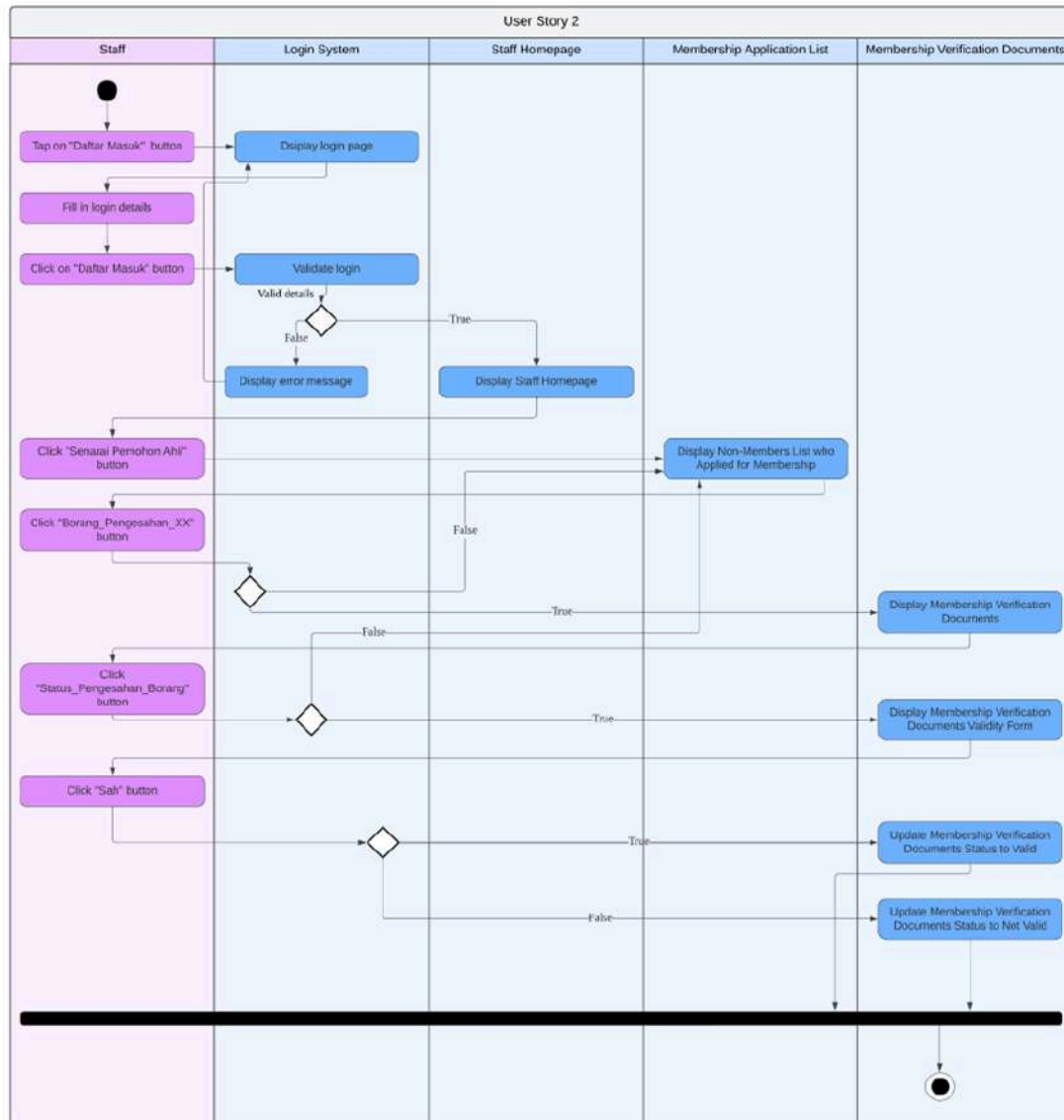


Figure 30: Activity diagram for when admin chooses to validate membership application

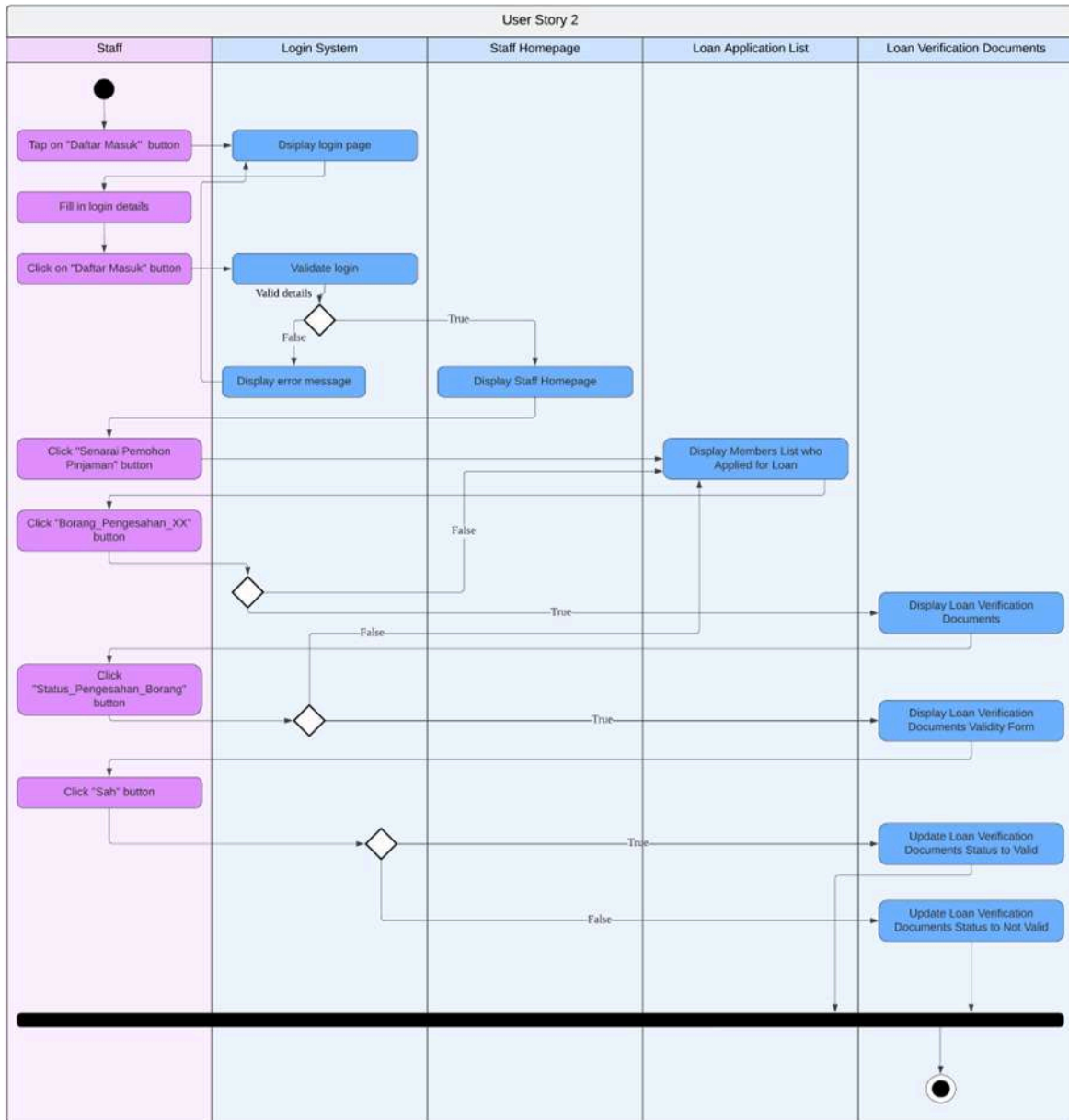


Figure 31: Activity diagram for when admin chooses to validate loan application

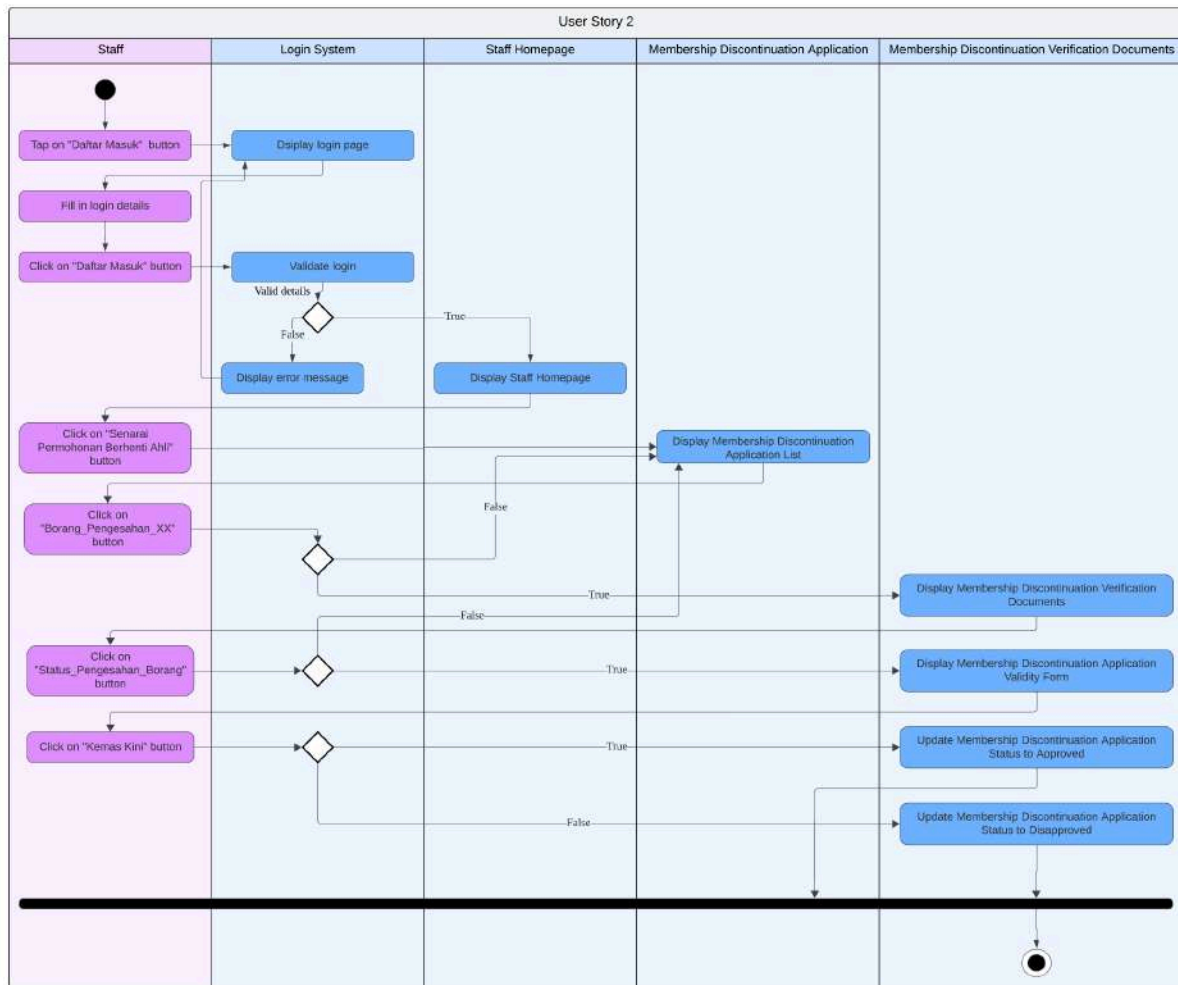


Figure 32: Activity diagram for when admin chooses to approve membership termination application

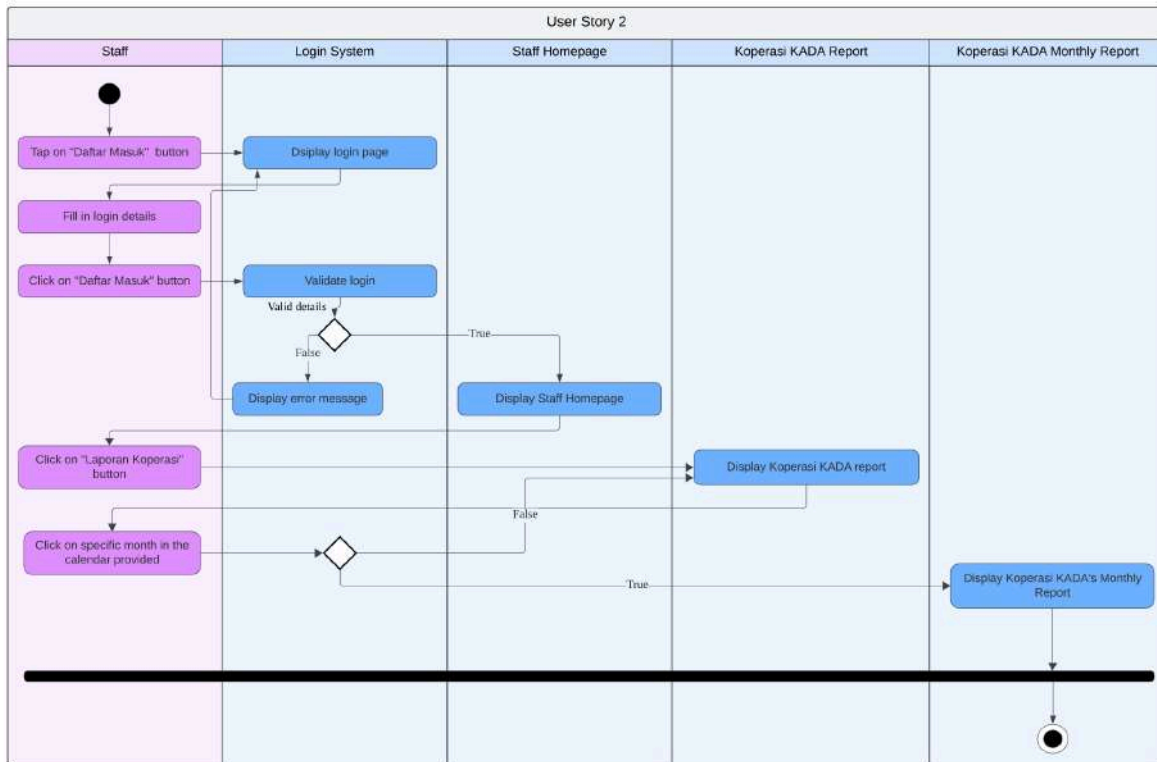


Figure 33: Activity diagram for when admin chooses to view monthly report

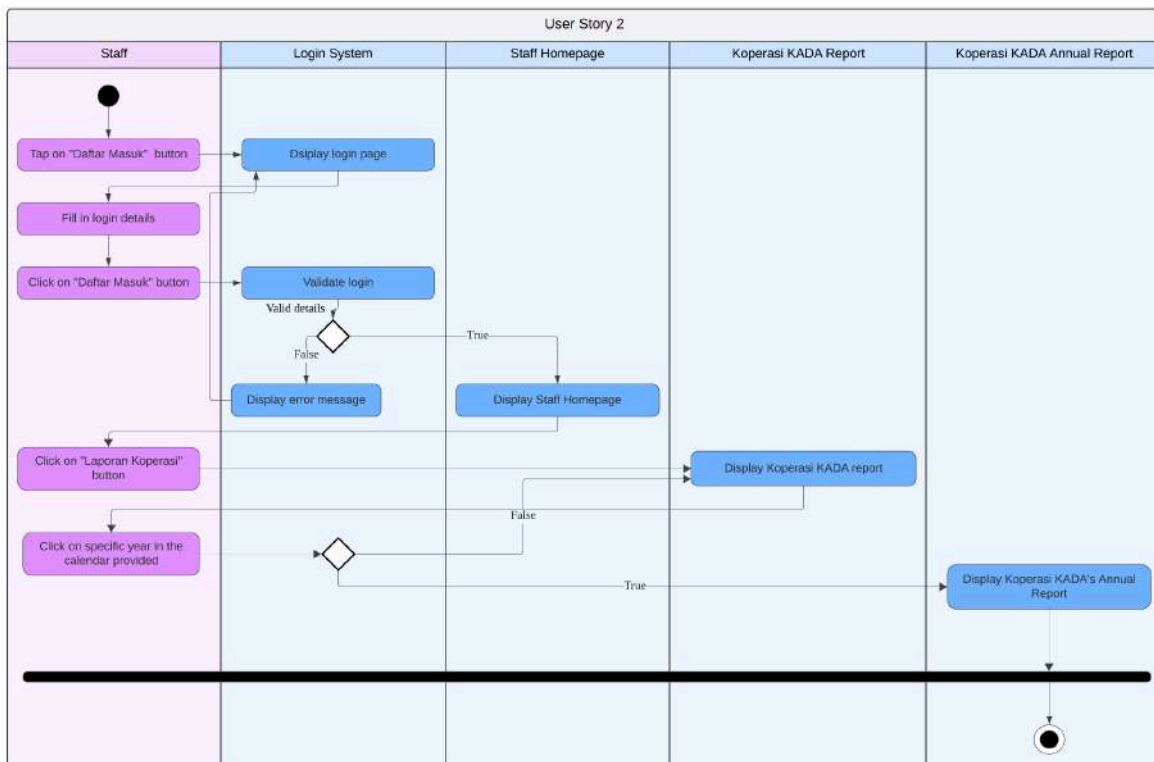


Figure 34: Activity diagram for when admin chooses to view annual report

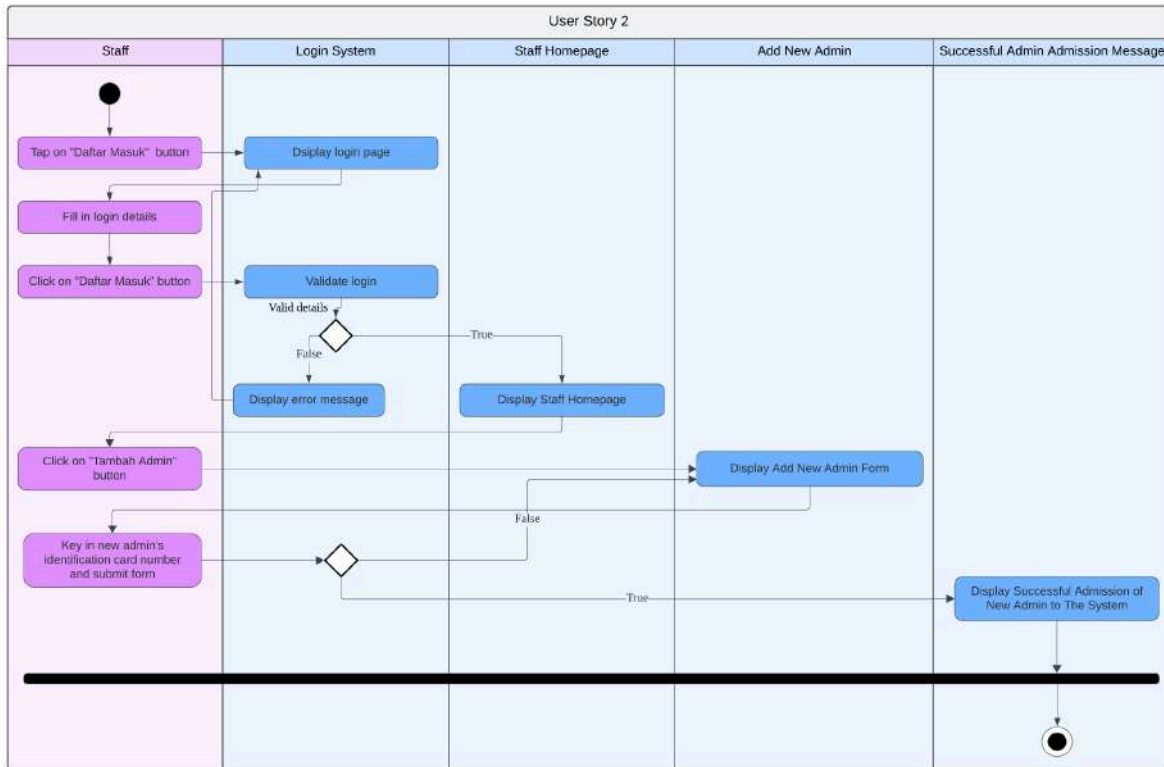


Figure 35: Activity diagram for when admin chooses to add new admin to the system

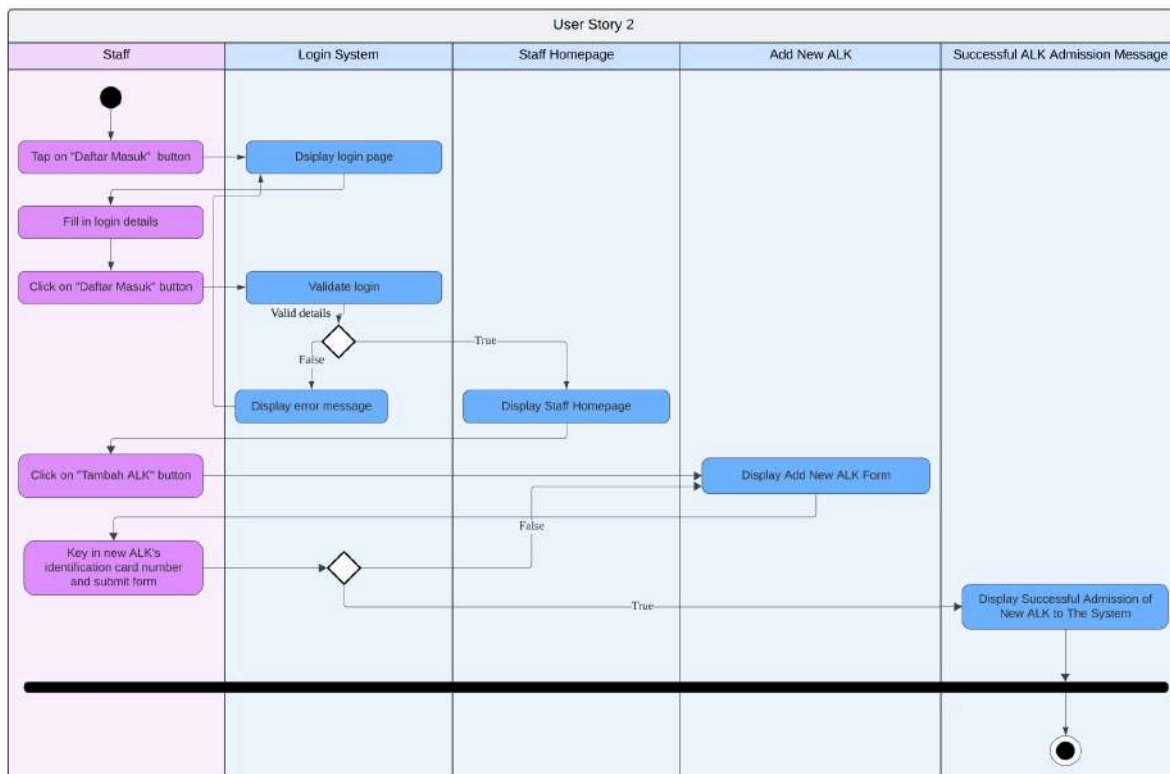


Figure 36: Activity diagram for when admin chooses to add new ALK to the system

2.4.3 US003: User Story <User Story 3>

User story: ALK module
ID: US003
User Story Description : As a high committee member (ALK) of Koperasi KADA, I want to verify membership and loan applications, so that I can ensure the eligibility of applicants and approve valid requests.
Flow of events: 1. The high committee member logs into the Koperasi KADA system by entering their ID and password. 2. The high committee member accesses the homepage with features to view: - Membership applications from non-members. - Loan applications from members. - Koperasi KADA report. 3. The high committee member clicks on the list of non-members applying for membership to view their application details. 4. The high committee member reviews the verification documents and approves the membership application. 5. The high committee member clicks on the list of members applying for loans to view their application details. 6. The high committee member reviews the loan type, verification documents, and other details to approve the loan application. 7. The system updates the status of the application (approved/rejected). 8. The high committee member may update the status later if needed. 9. The high committee clicks on the system report to view the monthly and annual report of employer and employee shares, membership applications, loan applications and membership termination applications.
Alternative flow: 1. Invalid Application Documents: The system prompts the high committee member to mark the application as rejected and notifies the applicant of the rejection. 2. Incomplete Review: The high committee member exits the application review and marks it as pending. The system saves the review progress.
Acceptance Criteria Postcondition: The status of the membership or loan application is updated (approved/rejected/pending). Precondition: The high committee member is logged into the system and has access to application details.

Table 29: User Story Description for US003

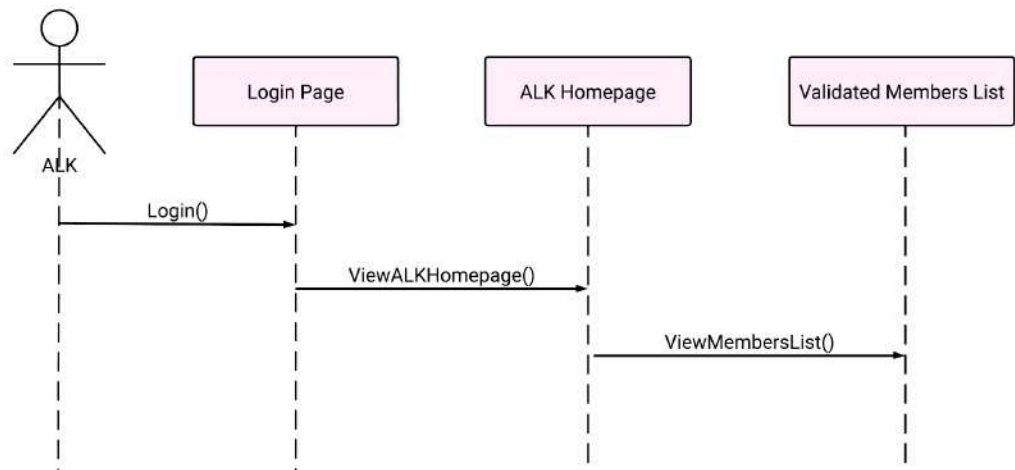


Figure 37: SD017 Sequence diagram for ALK view validated membership applications

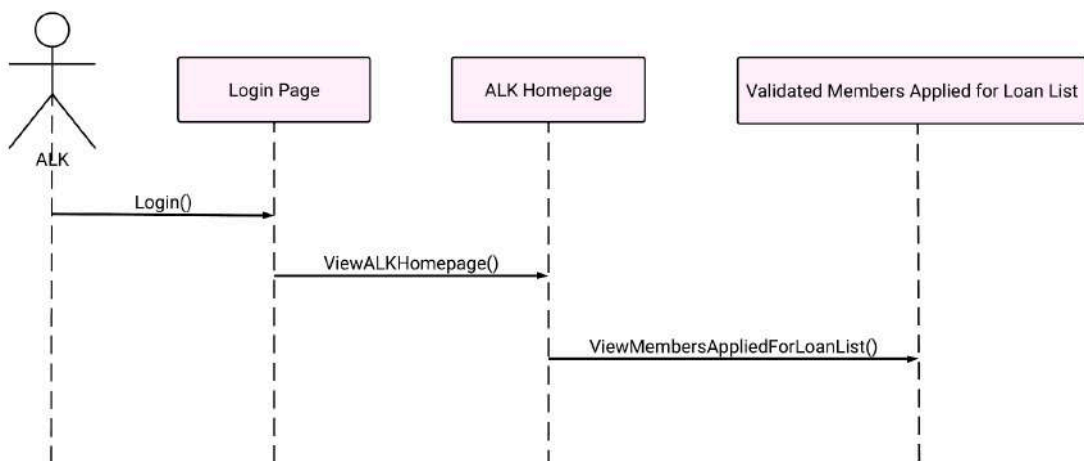


Figure 38: SD018 Sequence diagram for ALK view validated loan applications

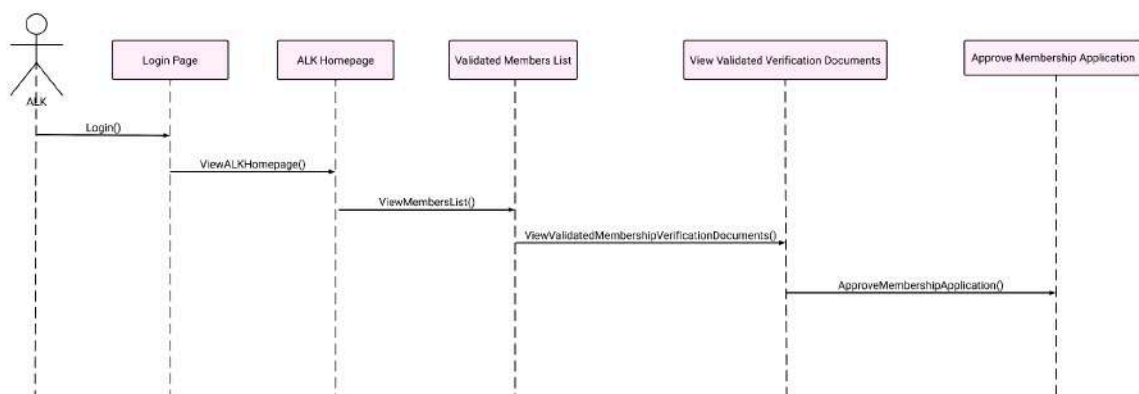


Figure 39: SD019 Sequence diagram for ALK approve membership applications

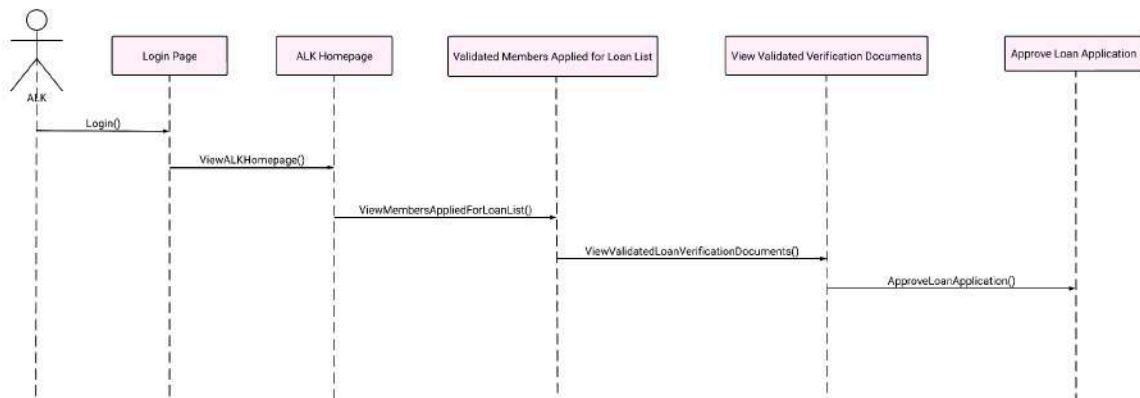


Figure 40: SD020 Sequence diagram for ALK approve loan applications

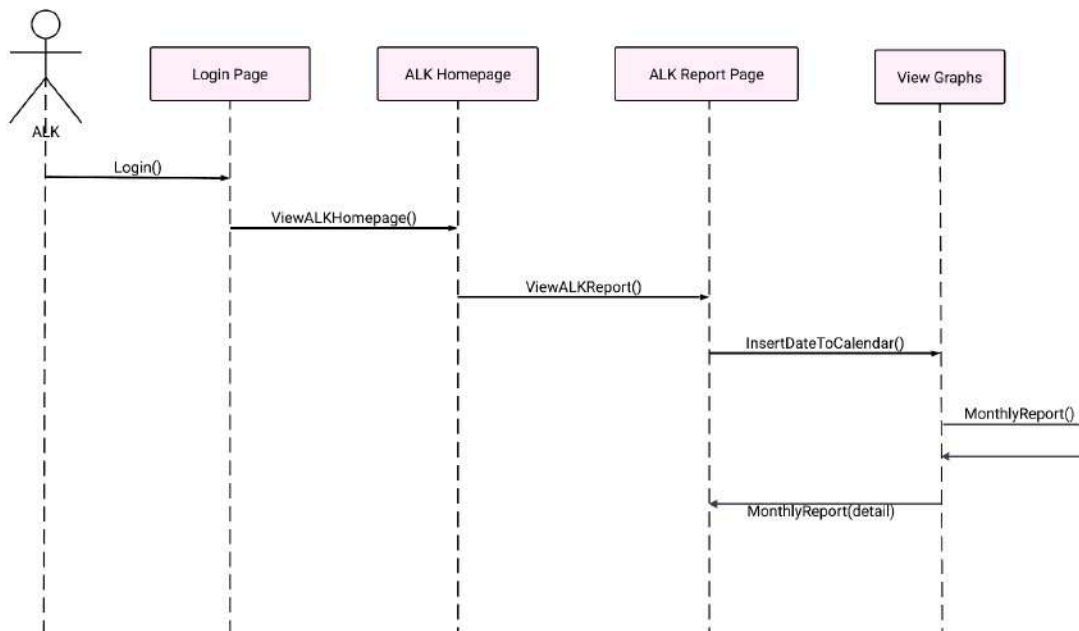


Figure 41: SD021 Sequence diagram for ALK view monthly report

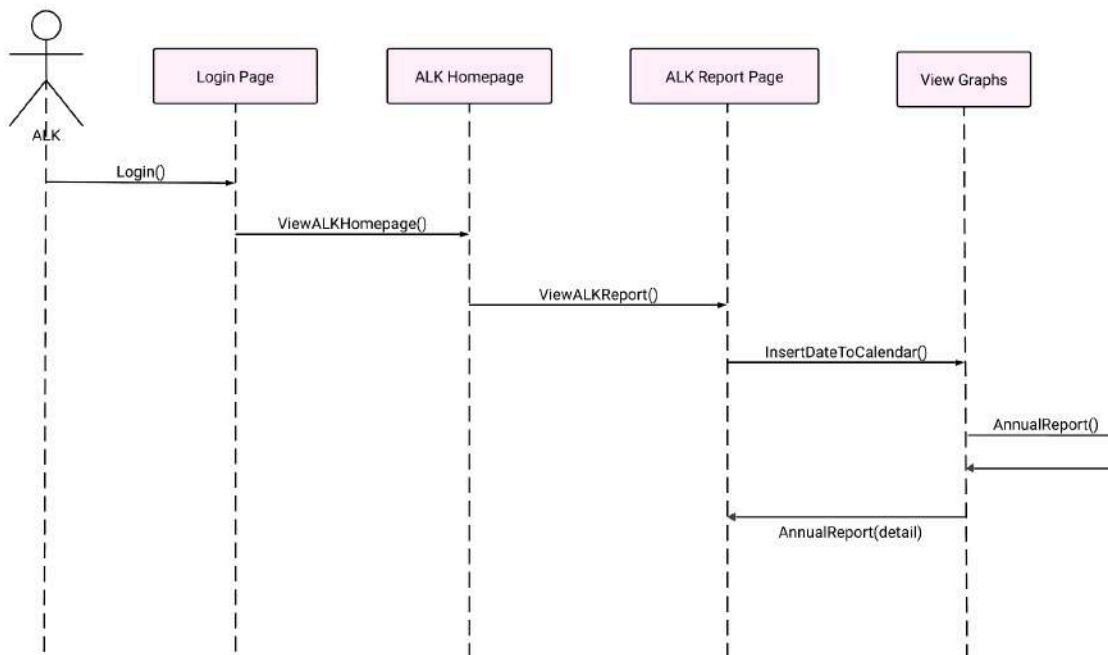


Figure 42: SD022 Sequence diagram for ALK view annual report

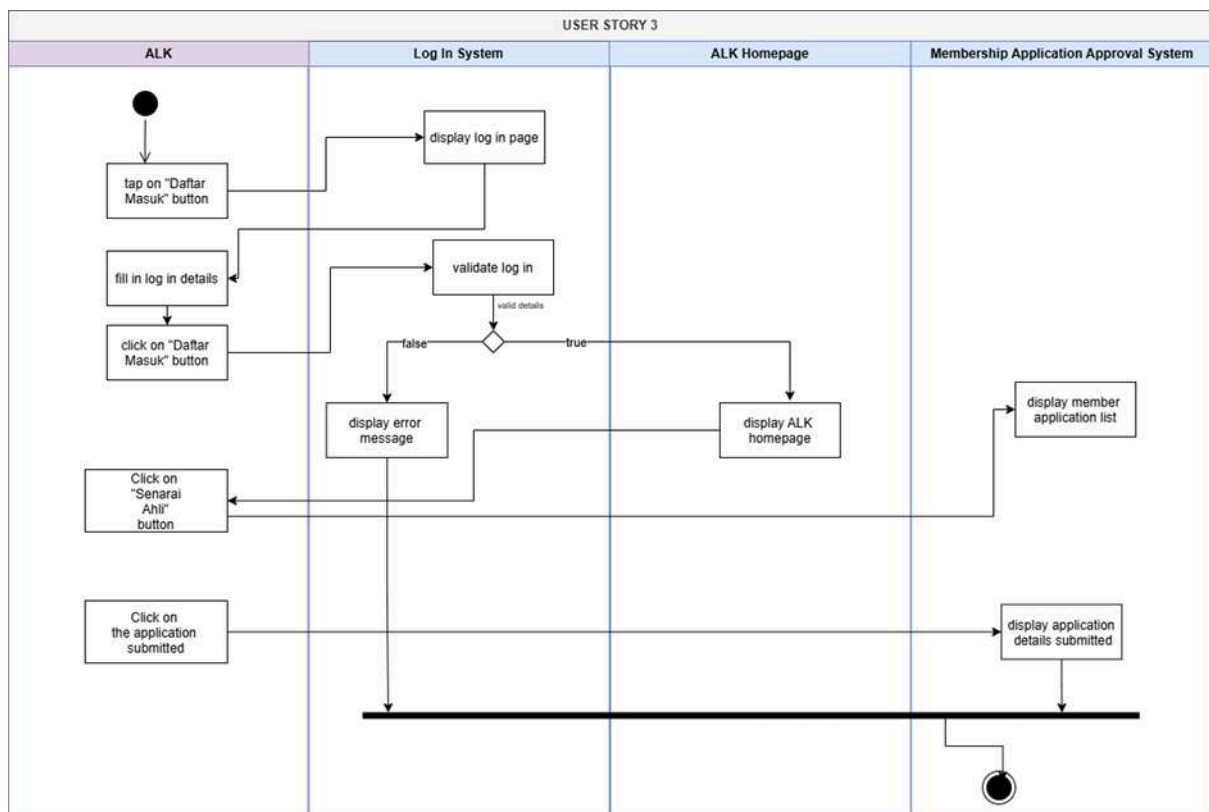


Figure 43: Activity Diagram for ALK view validated member application

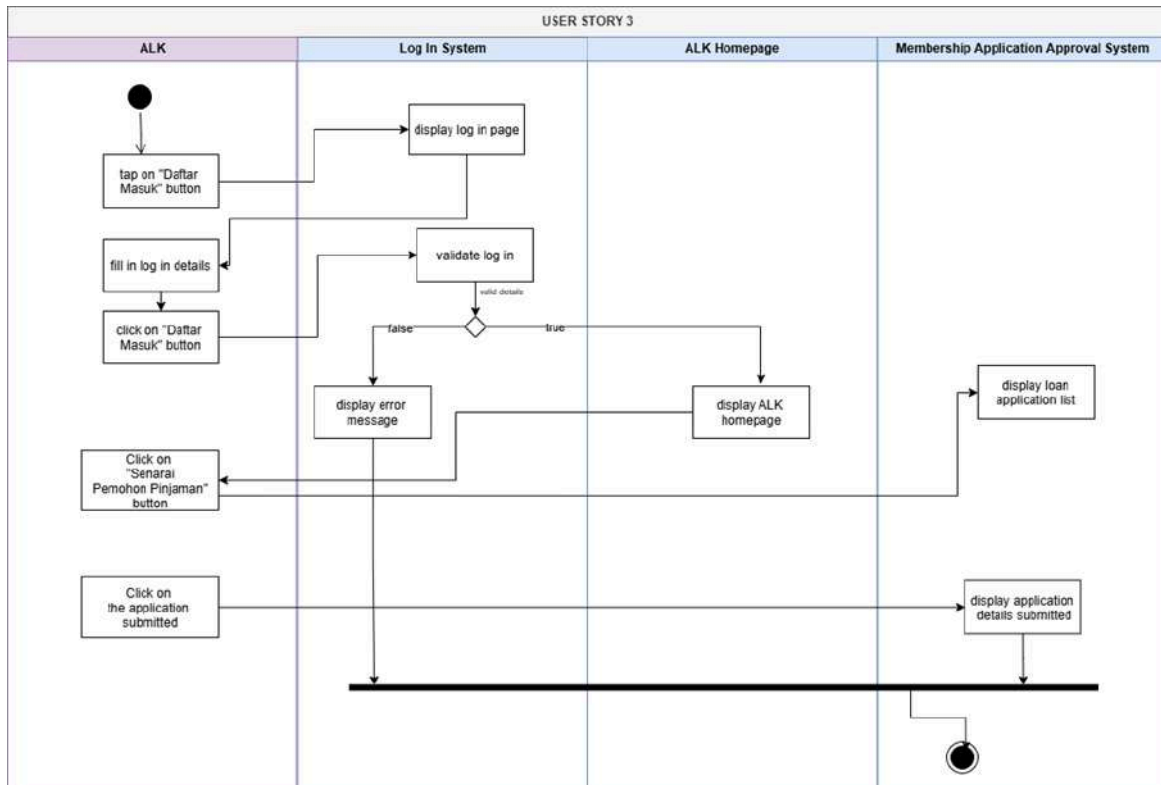


Figure 44: Activity diagram for ALK view validated loan application

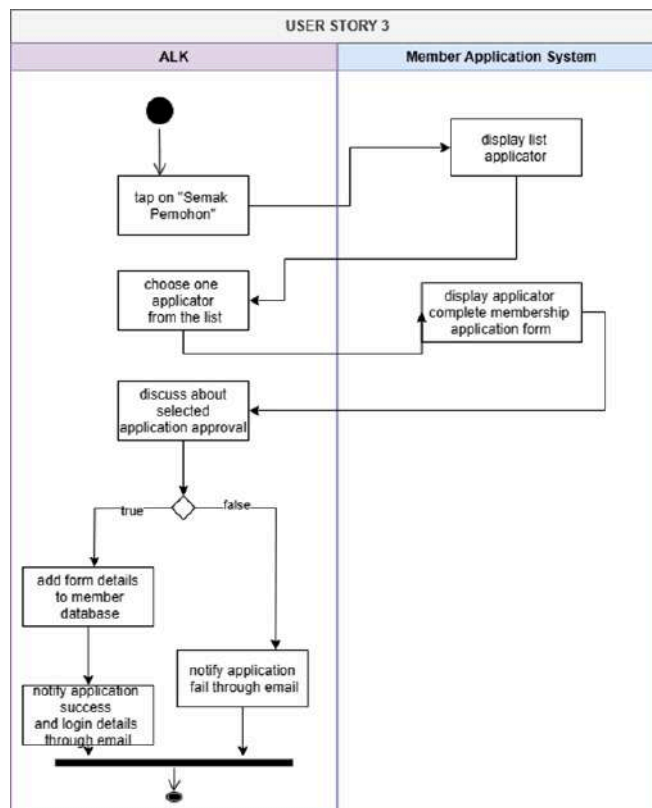


Figure 45: Activity diagram for ALK approve member application

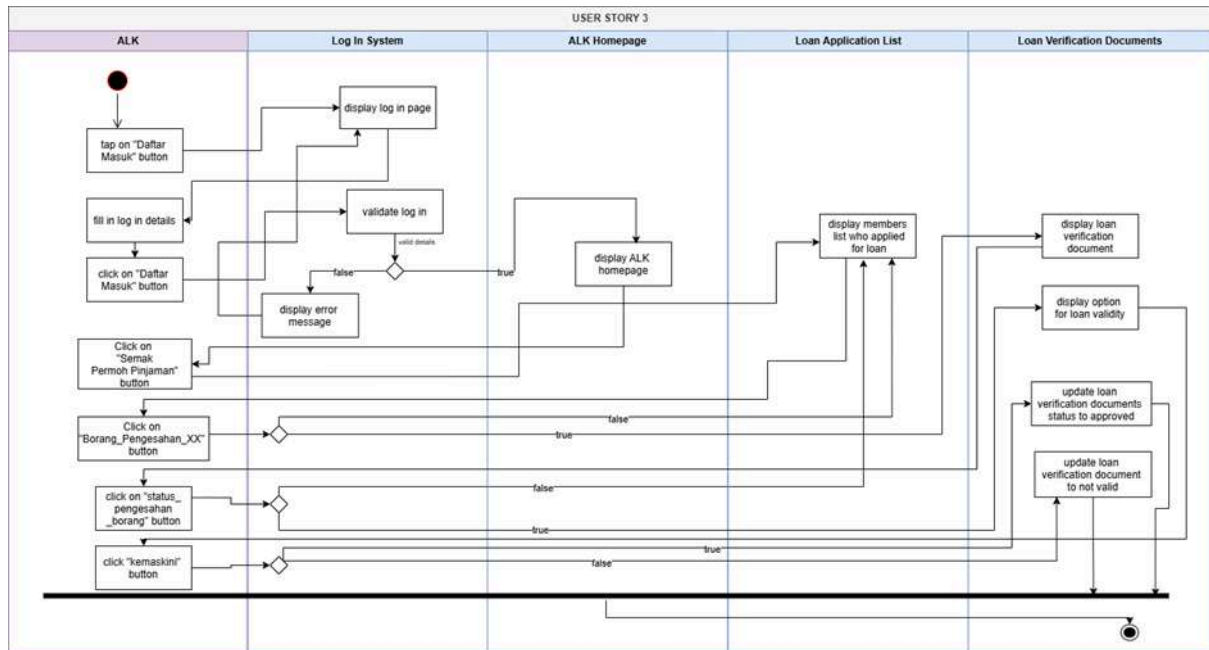


Figure 46: Activity diagram for ALK approve loan application

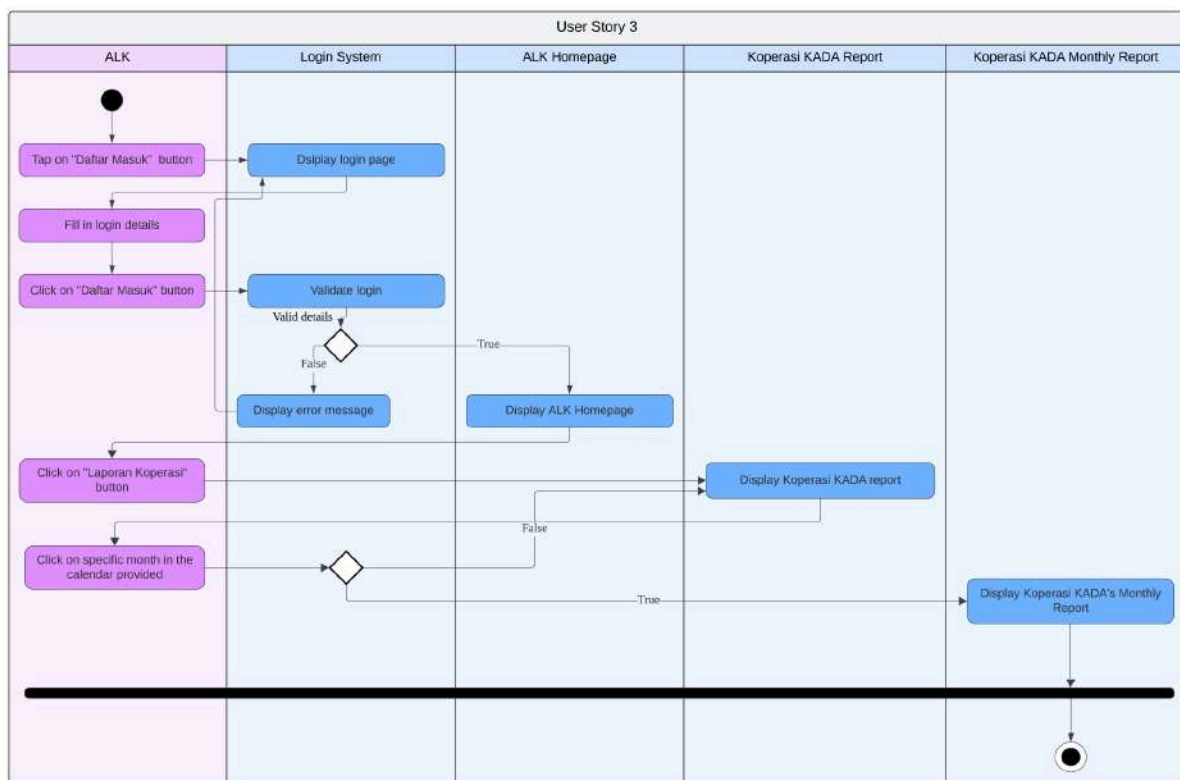


Figure 47: Activity diagram for ALK view monthly report

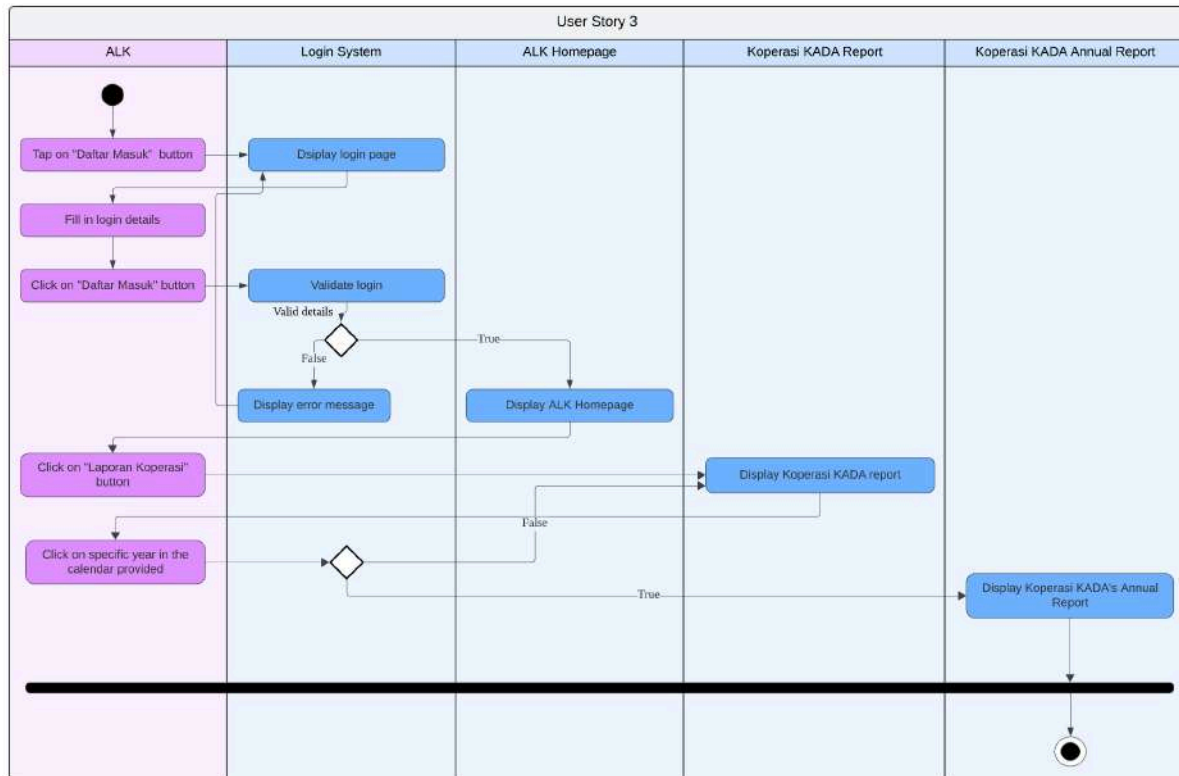


Figure 48: Activity diagram for ALK view annual report

2.4.4 US004: User Story <User Story 4>

User story: Non-member module
ID: US004
User Story Description : As a non-member of Koperasi KADA, I want to apply for membership so that I can login into the system.
Flow of events: 1. The non-member clicks on the apply membership to access the membership application form page. 2. The non-member fills in all the details needed in the form. 3. The non-member submits the form and wait for approval. 4. The non-member gets an email with login details when the application is approved. 5. The non-member accessed the login page. 6. The non-member input their login details given from email. 7. The system verified login details and showed the system homepage.

Alternative flow:

Application failed: Email sent to non-user notify them and tell them to reapply if they want.

Invalid login details: The system prompts the user to re-enter login details or contact the staff for further assistance.

Acceptance Criteria

Postcondition: The non-member is now a member of Koperasi KADA.

Precondition: The non-member is inside the Koperasi KADA homepage.

Table 29: User Story Description for US004

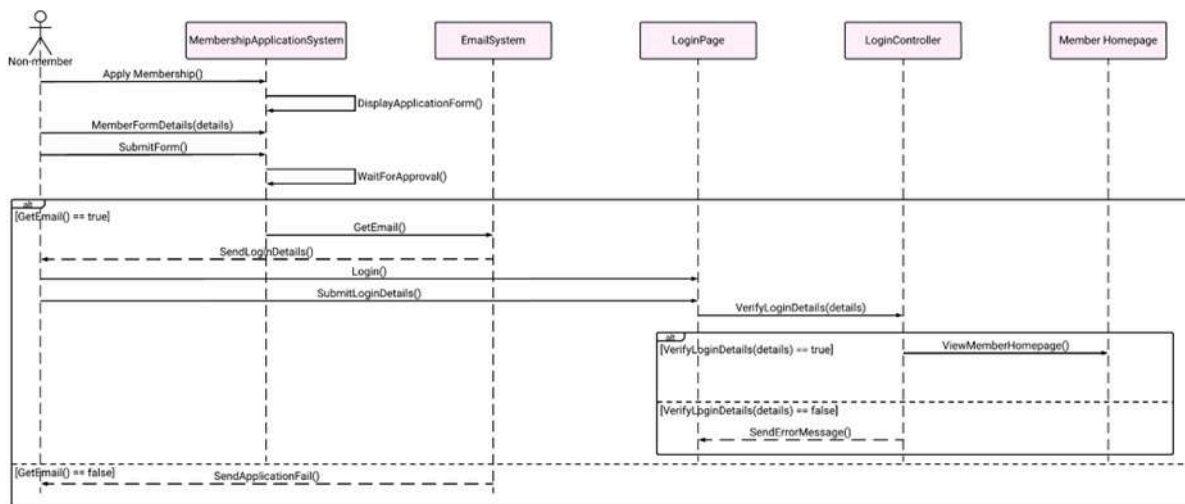


Figure 49: SD023 Sequence diagram for non-member apply for membership

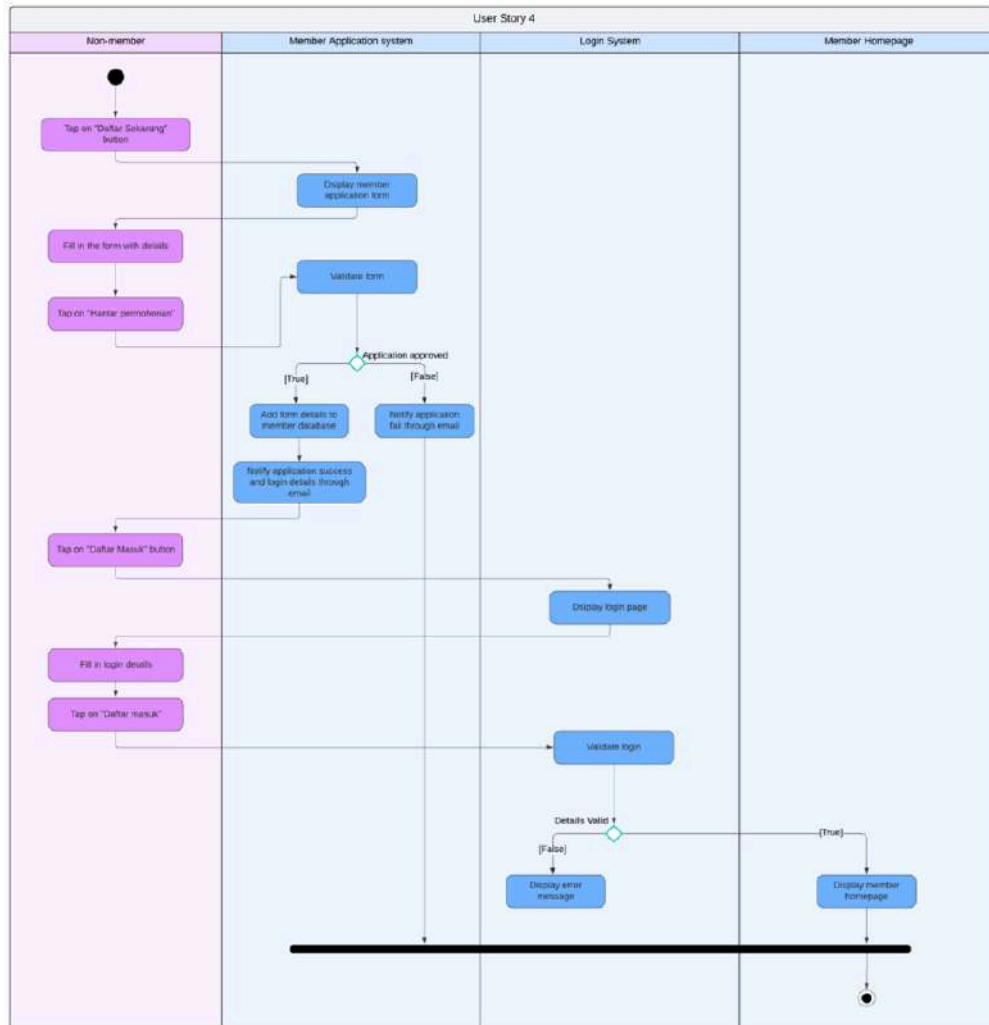


Figure 50: Activity diagram for non-member apply membership

2.5 Performance and Other Requirements

In this section, we will discuss the performance requirements and software system attributes of the proposed system for Koperasi KADA. Performance requirements include response time, throughput, capacity and availability whereas software system attributes involve usability, reliability, maintainability, portability, compatibility, security, safety, legal and regulatory, and lastly environmental impact.

Performance Requirements

- 1. Response Time:** The system is designed to respond within 3 seconds of user input, ensuring a fast and efficient experience. The response time minimizes delays and ensures smooth navigation throughout the system.
- 2. Throughput:** The system is designed to handle up to 30 requests simultaneously. This throughput ensures the proposed system can efficiently support peak usage periods or multiple simultaneous operations without experiencing slowdowns.
- 3. Capacity:** The system can manage up to 30 active users at the same time while handling a data load of approximately 10,000 entries. This capacity ensures the system can accommodate typical user activity without compromising system performance.
- 4. Availability:** The system is operational 24 hours a day, 7 days a week. Thus, users can be worry free of having a time constraint. Users can complete tasks or access information whenever needed.

Software System Attributes:

- 1. Usability:** The system must have an intuitive user interface, enabling users to navigate the platform with minimal training.
- 2. Reliability:** The system must perform consistently without errors or crashes, ensuring stable operation.
- 3. Maintainability:** The codebase should be modular and well-documented to allow easy updates and enhancements.
- 4. Portability:** The system must operate on multiple web browsers and mobile devices without compatibility issues.
- 5. Compatibility:** It should integrate seamlessly with existing KADA databases and infrastructure.
- 6. Security:** The system must employ encryption and secure authentication to protect sensitive data and ensure compliance with regulations.
- 7. Safety:** The system must operate safely, preventing any data corruption or harm to the environment.
- 8. Legal and Regulatory Compliance:** The system must comply with relevant data protection laws such as Malaysia's PDPA.
- 9. Environmental Impact:** The system should utilize optimized resources to minimize server load and energy consumption.

2.6 Design Constraints

In this section, we will point out some design constraints we found regarding the system that is about to be built, KADA Corporation System when being implemented into the organization and explain it in detail.

Compatibility constraints:

The system must seamlessly integrate with the existing workflows and infrastructure, including backend systems and databases, to maintain compatibility. The existing stored data needs to be migrated without loss or corruption.

Online accessibility constraints:

The system should support diverse devices and browser, ensuring accessibility for all users within the organization

Real-time features constraints:

The system has real-time capabilities which is crucial for providing instant updates on loan statuses, membership applications, and savings or loan balances. It should ensure the response time is less than 2 seconds and actions under normal server load conditions

Security constraints:

The system must have robust security measures, such as encryption and user authentication, must be implemented to protect sensitive financial and personal data, while ensuring compliance with data privacy regulations like PDPA (Personal Data Protection Act) in Malaysia. A role-based access control (RBAC) system needs to be implemented to prevent unauthorized access.

Scalability constraints:

The system must be scalable to accommodate a projected 15% growth in user load, reflecting the organization's expected expansion in operations.

Hardware constraints:

The system will require sufficient storage of cloud-based hosting to handle every user request. The system must be compatible with various devices with varying specifications. A compatible peripheral device is needed for administration to expand its capabilities.

3. Architectural Rationale

In this section, we will describe the architectural style chosen for this project and the justification of our selection.

3.1 Architecture Style and Rationale

For this project, we use Model-View-Controller (MVC) architecture pattern as it separates presentation and interaction from the system data making the organization of codes smoother. The system is structured into three logical components that interact with each other. The Model component manages the system data and associated operations on the data. This ensures security for the database since users will not interact with it directly. Then, the View component defines and manages how the data is presented to the user. This helps us to focus our UI design only on the files contained in the View component and enhance user experience with a good UI design. Last but not least, the Controller component manages user interaction and passes these interactions to the View and the Model. This helps us organize all our system functions in the Controller. Below is a diagram describing the organization of MVC for our system.

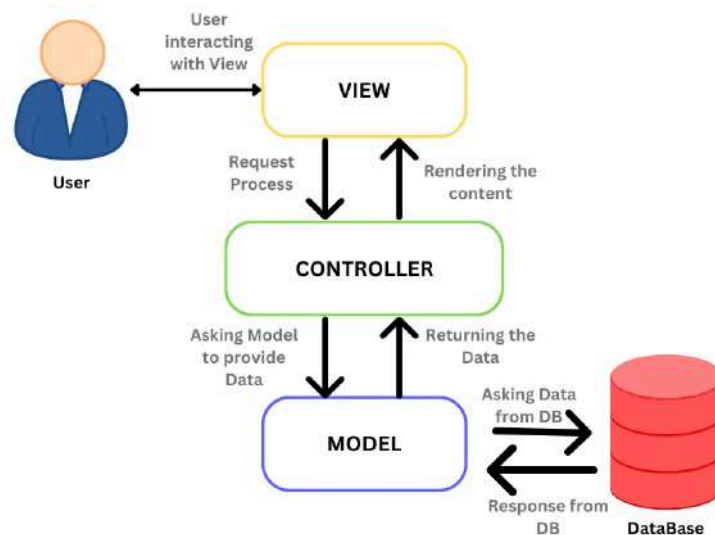


Figure 51: MVC organization

MVC is the most suitable architecture for our system and will be used throughout the project.

4. Architectural Views

The views consist of scenario view, logical view, process view, development view and physical (deployment) view as shown in Figure 52. Each view has its own usage and models or diagram to represent it as shown in Table 30.

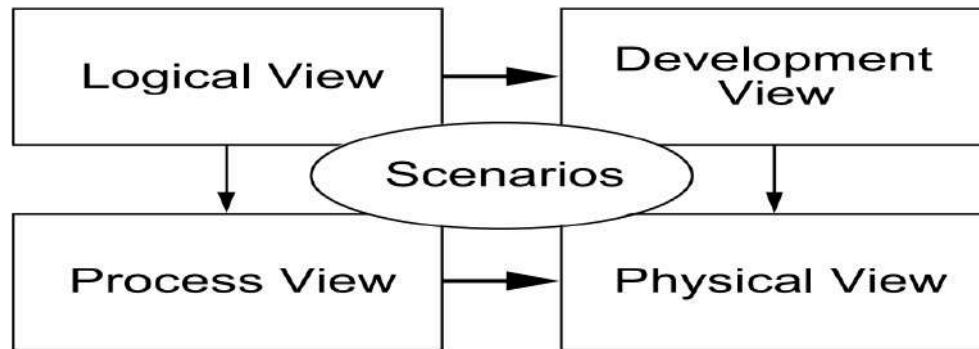


Figure 52: Architectural Views

Architectural Views	Description	Diagrams used
Use Case View	The use case view will show how the element in the four views will integrate seamlessly according to the module that has been grouped into important scenarios for every actor.	Use Case Diagram
Logical View	shows the key abstractions in the system as objects or object classes	Class Diagram
Process	The process view will explain the detailed process for each use case module that has been described in the use case view.	Sequence Diagram
Development (or Implementation) View	the implementation view shows how the software is	Component Diagram Package Diagram

	decomposed for development	
Physical (or Deployment) View	shows the system hardware and how software components are distributed across the processors in the system	Deployment Diagram

Table 30: Architectural Views

4.1 Use Case View

This section includes the use case diagram for our system.

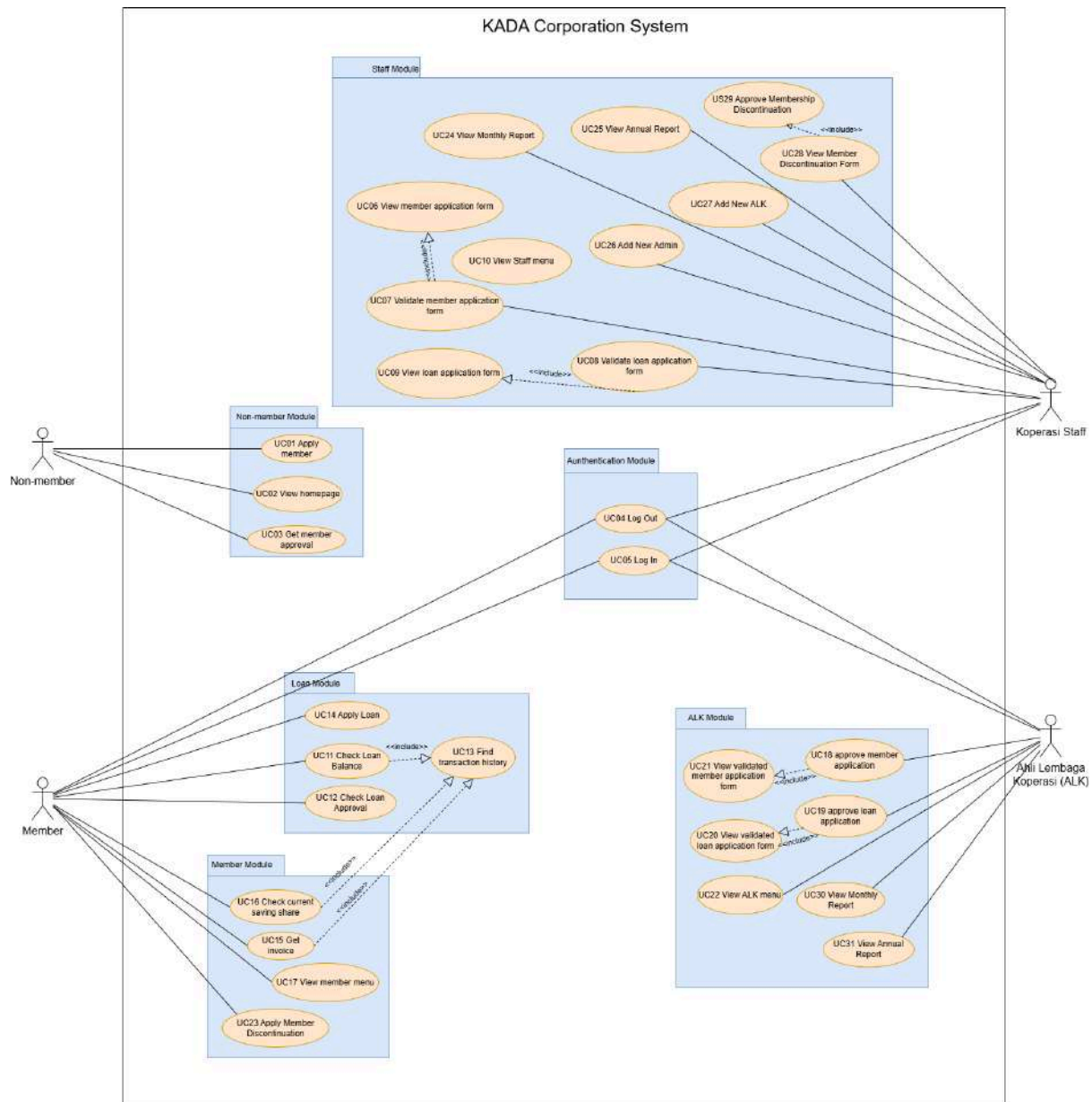


Figure 53: Use case diagram for Koperasi KADA System

4.2 Implementation View

The Component Diagram for the Koperasi KADA System represents the main functional units of the system and their interactions. The system follows the Model-View-Controller (MVC) architecture, which is divided into three layers: User Interface Layer (View), Business Logic Layer (Controller), and Data Access Layer (Model & Database).

The User Interface Layer (View) consists of the front-end components that allow users to interact with the system. These include the Login Page, where users (non-members, members, admin, and ALK) enter their credentials to access the system, and the different dashboards for each user type. The Member Dashboard allows members to apply for loans, check their savings, and view financial statements. The Admin Dashboard provides functionalities for reviewing and validating membership and loan applications. The ALK Dashboard enables high committee members (ALK) to approve or reject applications. This layer serves as the interface between users and the system, sending requests to the Controller Layer for processing.

The Business Logic Layer (Controller) is responsible for handling user requests, applying business rules, and processing data. The Authentication Controller manages user login, registration, and access control. The Membership Controller processes membership applications from non-members, while the Loan Controller handles loan applications, approvals, and loan balance checks. The Validation Controller ensures that verification documents for membership and loan applications are properly reviewed before approval. These controllers act as intermediaries, receiving user input from the View Layer, processing the request, and retrieving or updating data in the Model Layer (Database).

The Data Access Layer (Model & Database) stores and manages all system data using XAMPP. The Member Model maintains user details and account information, the Loan Model keeps track of loan applications and their statuses, and the Verification Model records document validation results. These models interact with the XAMPP Database, where all system records, including membership details, loan applications, and financial transactions, are securely stored. The Controller Layer communicates with these models to query, update, and retrieve data whenever needed.

Overall, the Component Diagram illustrates the modular structure of the system, where each layer functions independently yet cooperatively to ensure a secure, scalable, and efficient platform. The MVC architecture enhances code organization, making the system easier to maintain and extend in the future. The diagram ensures that all functionalities are well-defined, providing a clear separation of concerns between user interactions, business processes, and data management.

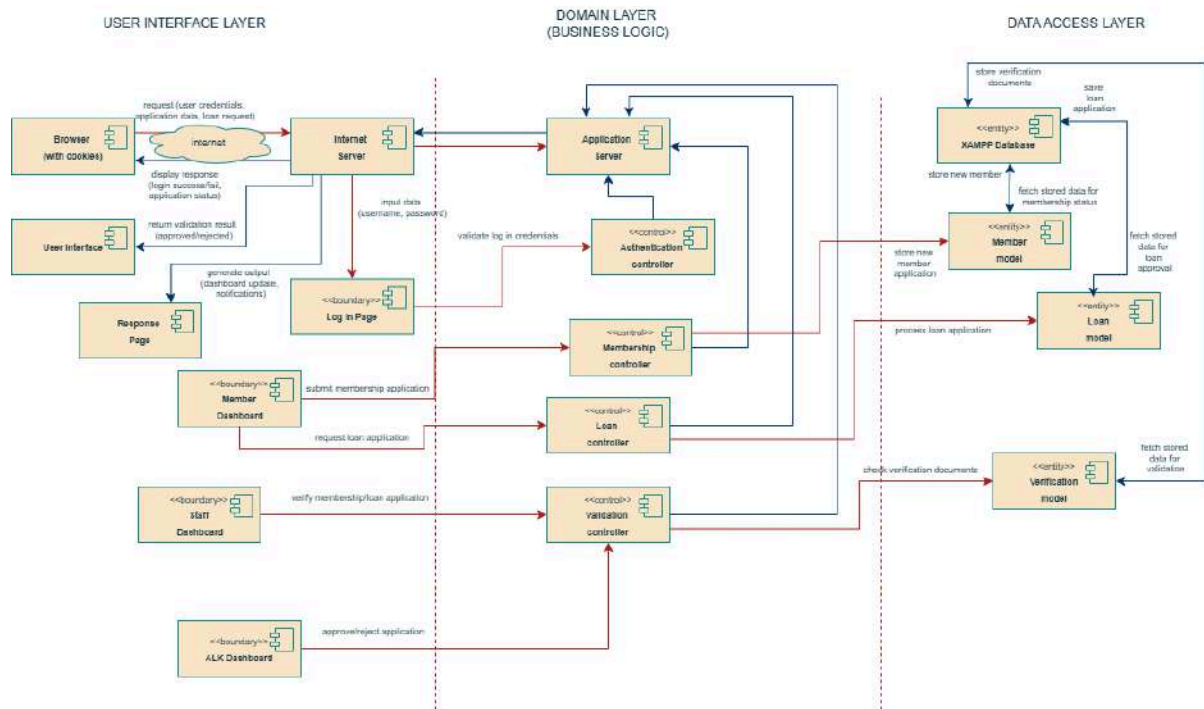


Figure 54: Component diagram of Koperasi KADA membership and loan application system

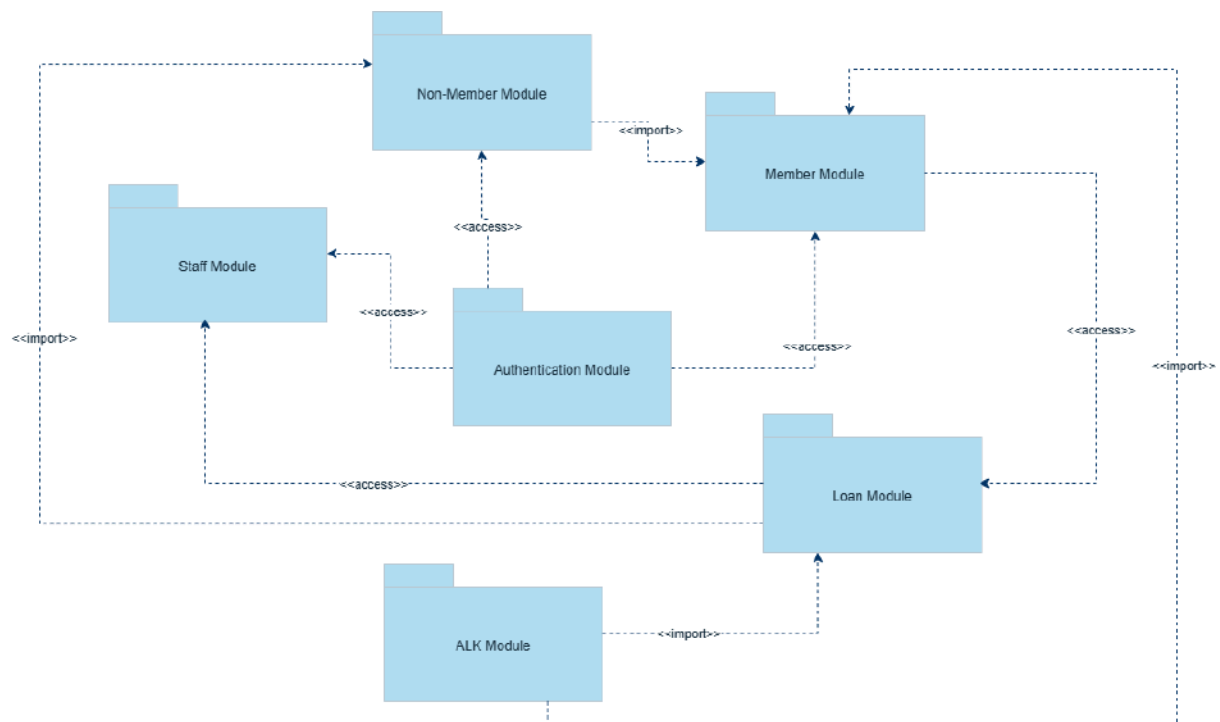


Figure 55: Package diagram for Koperasi KADA membership and loan application system

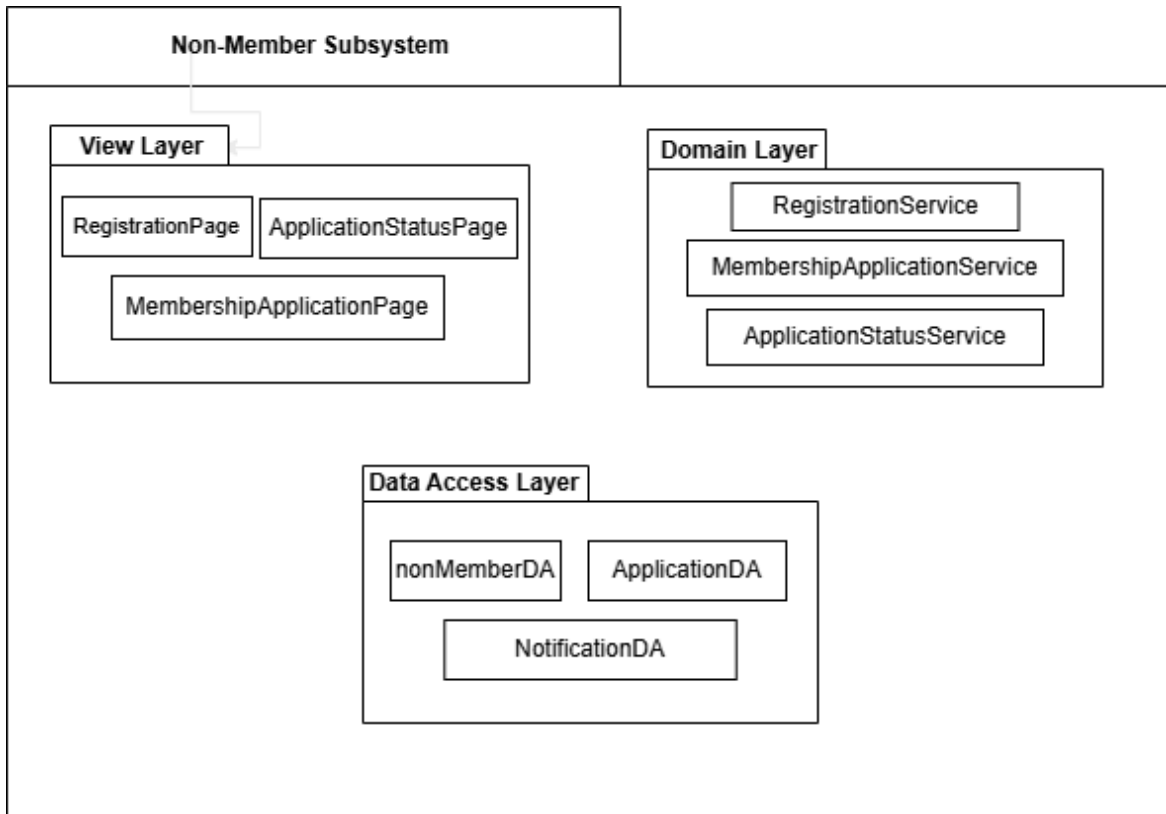


Figure 56: Package diagram for Member subsystem

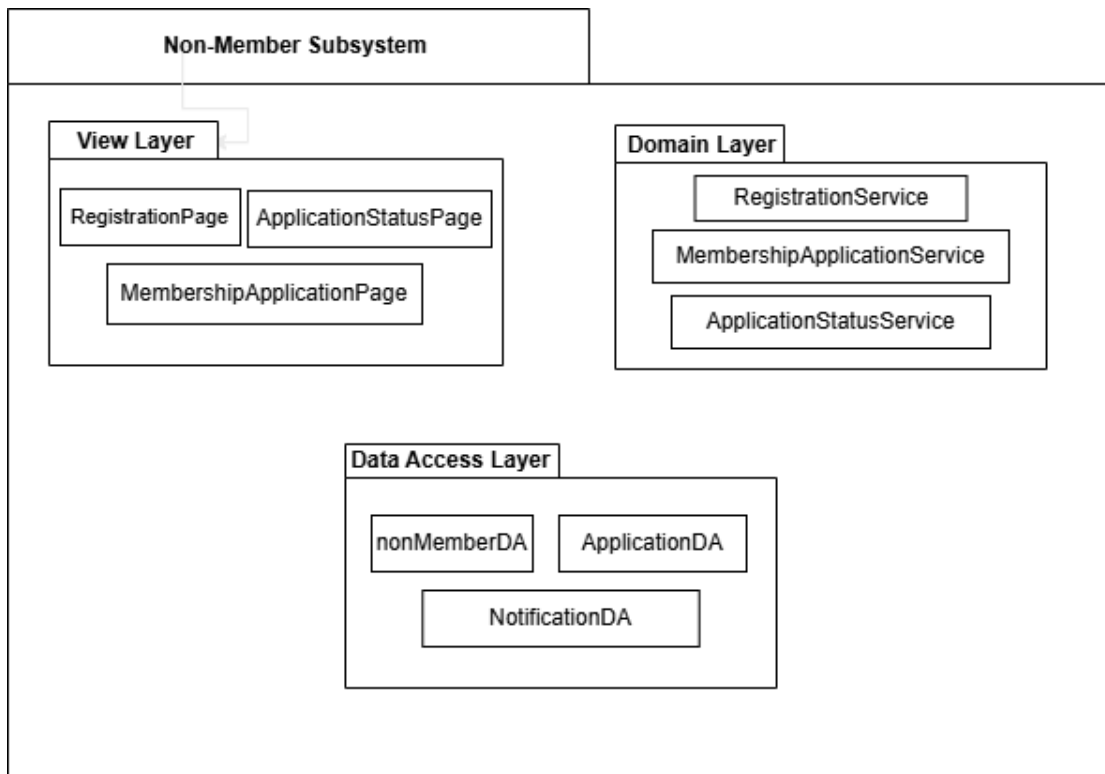


Figure 57: Package diagram for Non-Member subsystem

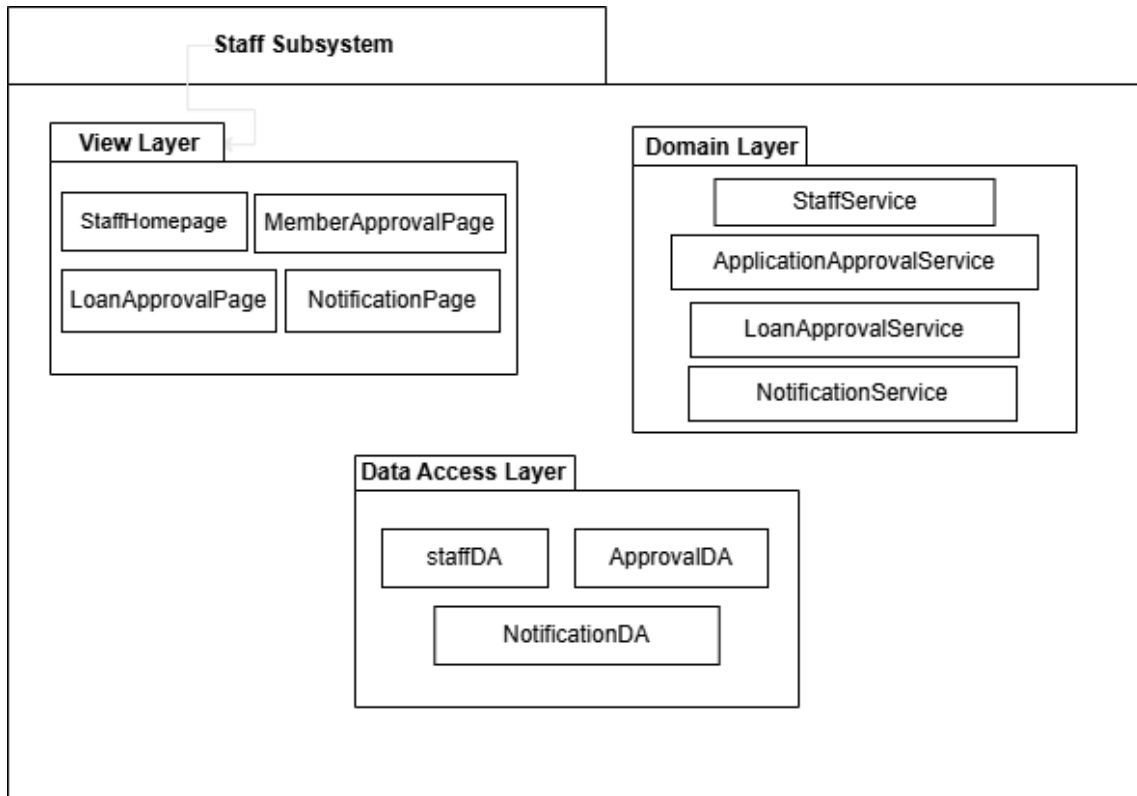


Figure 58: Package diagram for Staff subsystem

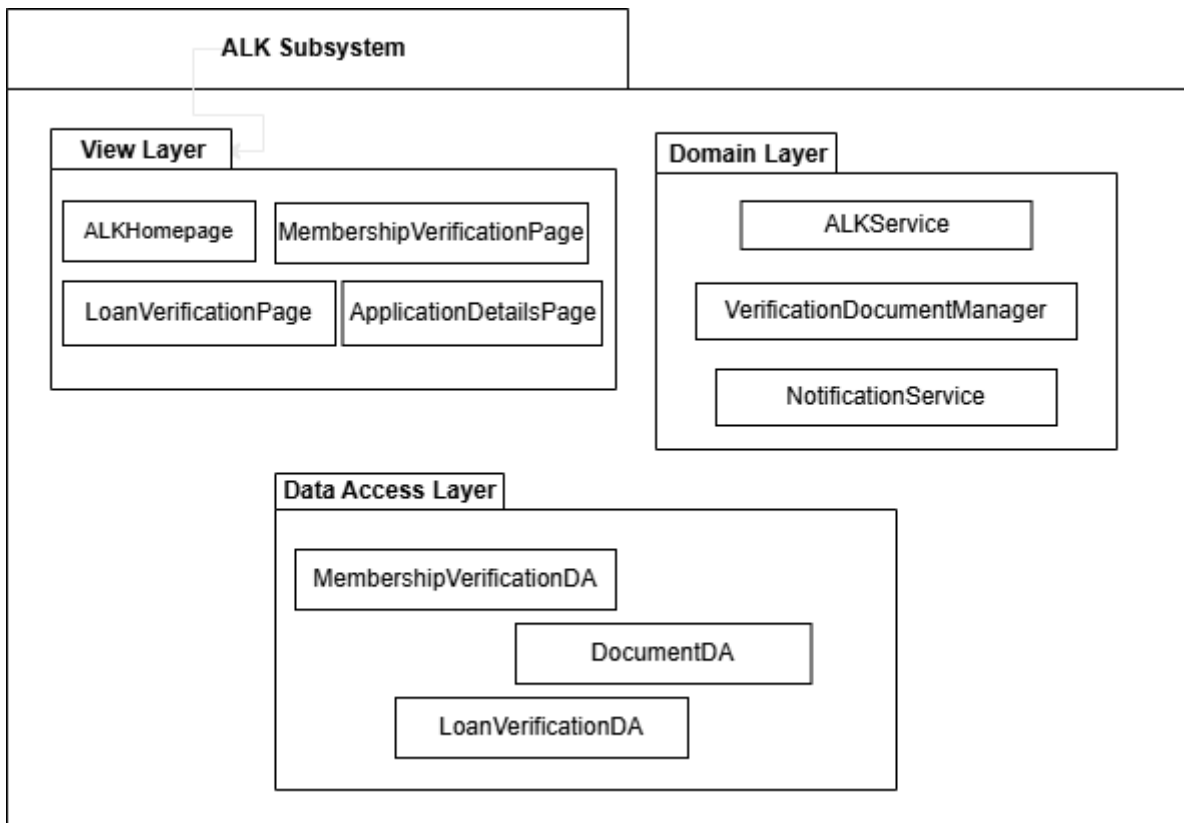


Figure 59: Package diagram for ALK subsystem

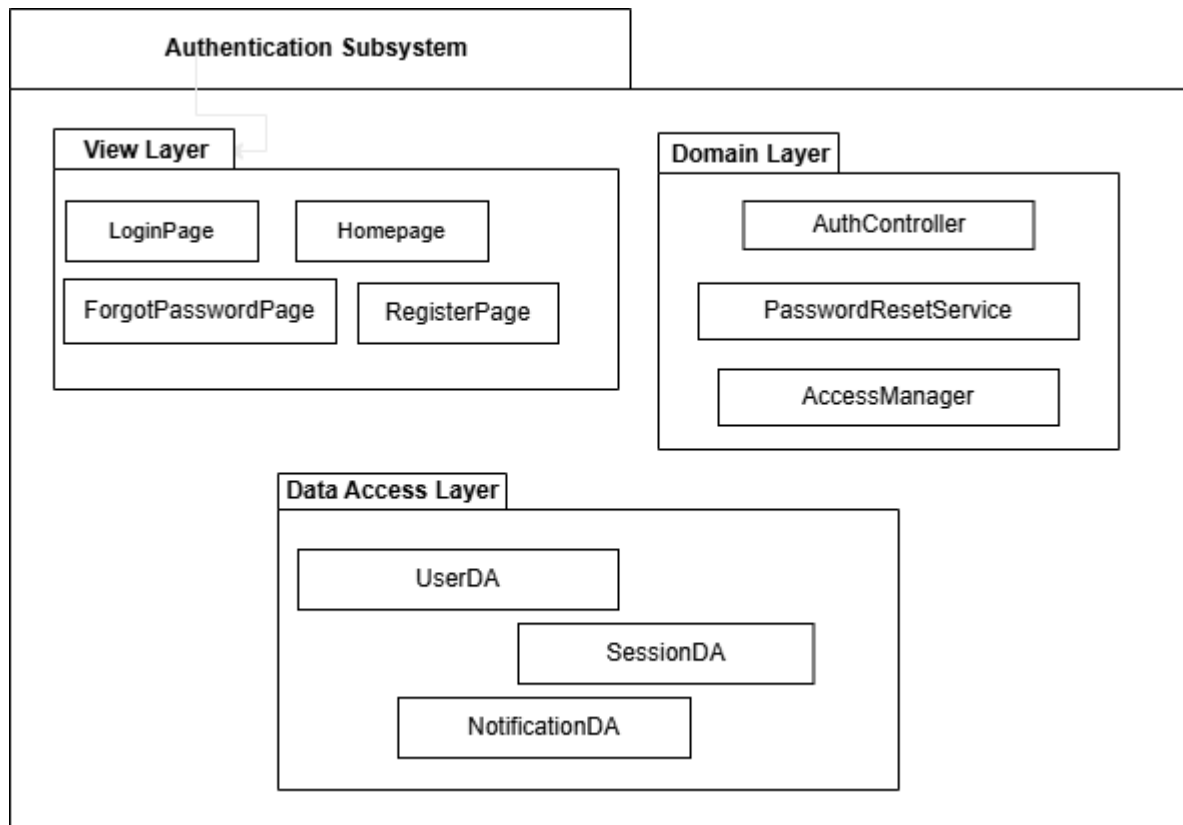


Figure 60: Package diagram for Authentication subsystem

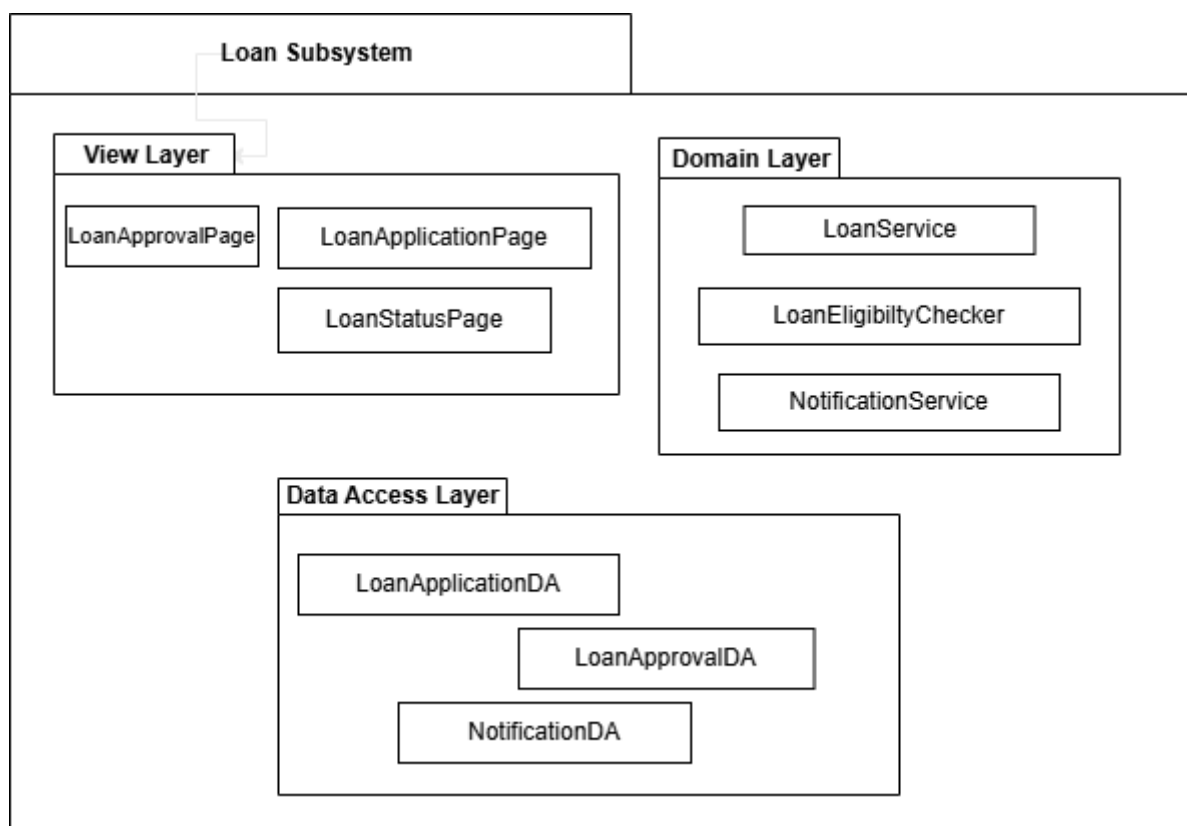


Figure 61: Package diagram for Loan subsystem

4.3 Logical View

In this subtopic, the class diagram for each module will be presented. There are a total of six modules in the system, which are Member Module, Non-Member Module, Admin Module, ALK Module, Authentication Module and Loan Module.

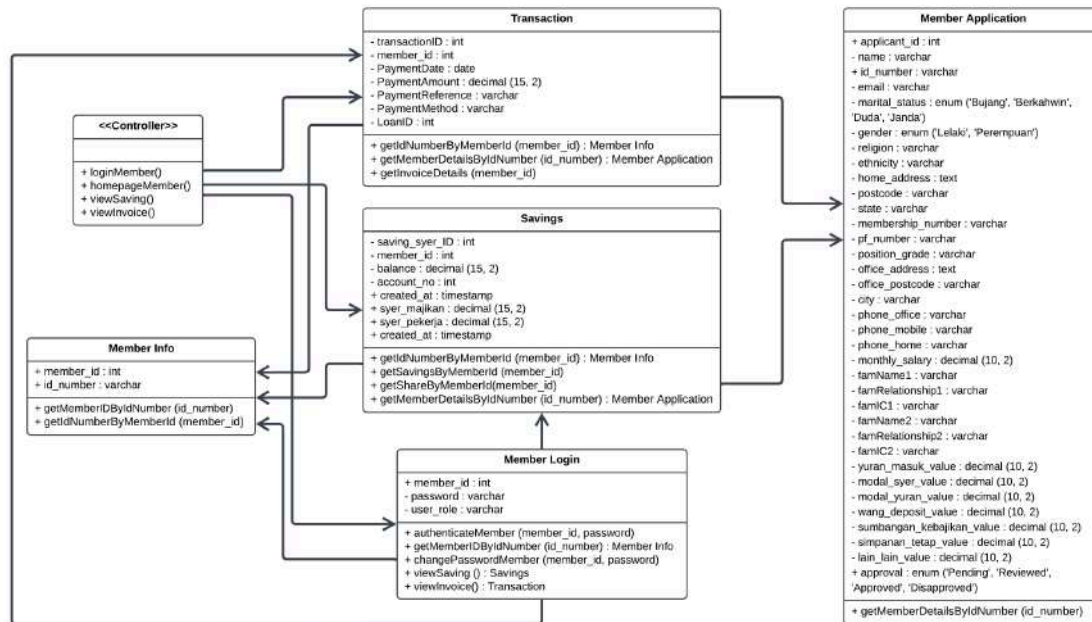


Figure 62: Class diagram for Member module

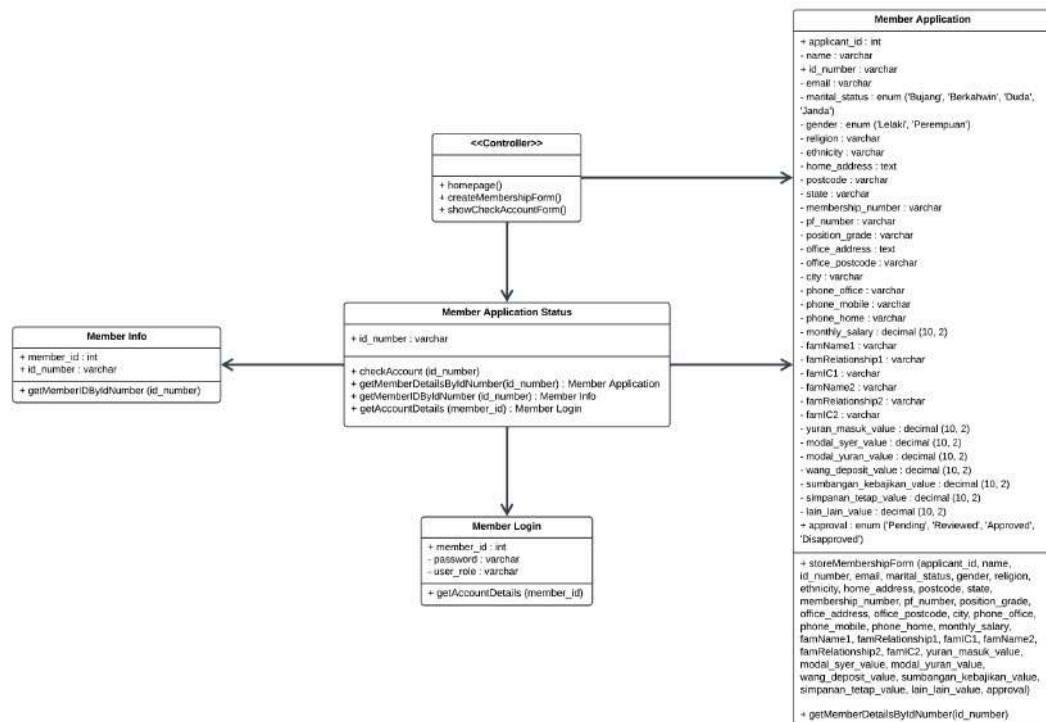


Figure 63: Class diagram for Non-Member module

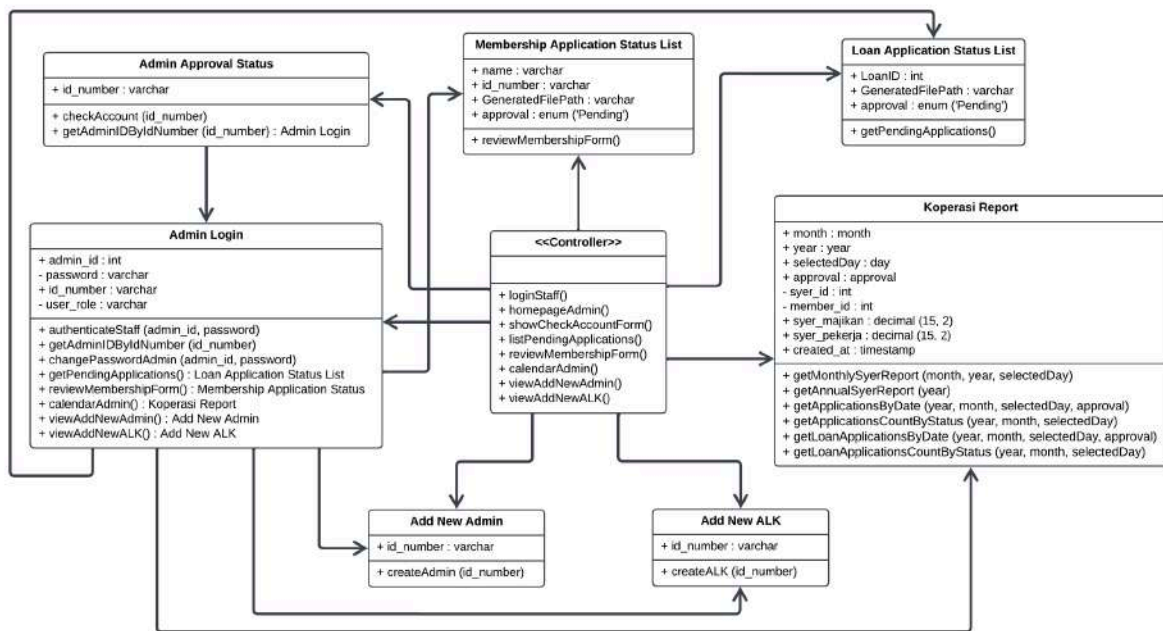


Figure 64: Class diagram for Admin module

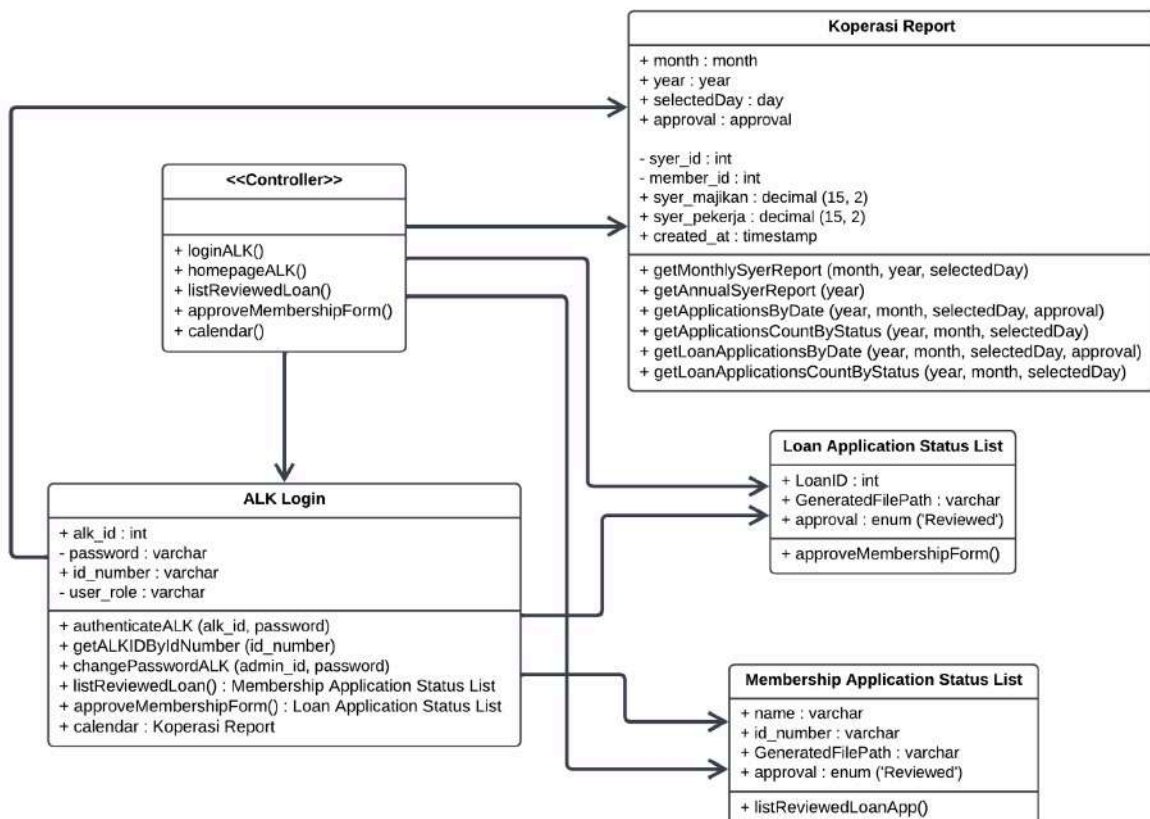


Figure 65: Class diagram for ALK module

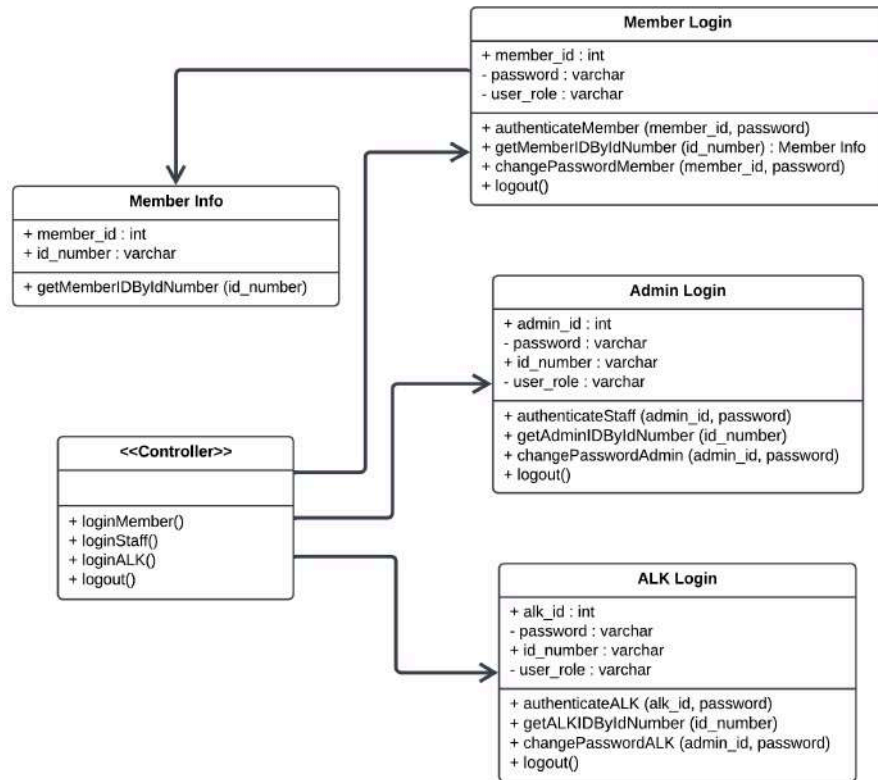


Figure 66: Class diagram for Authentication module

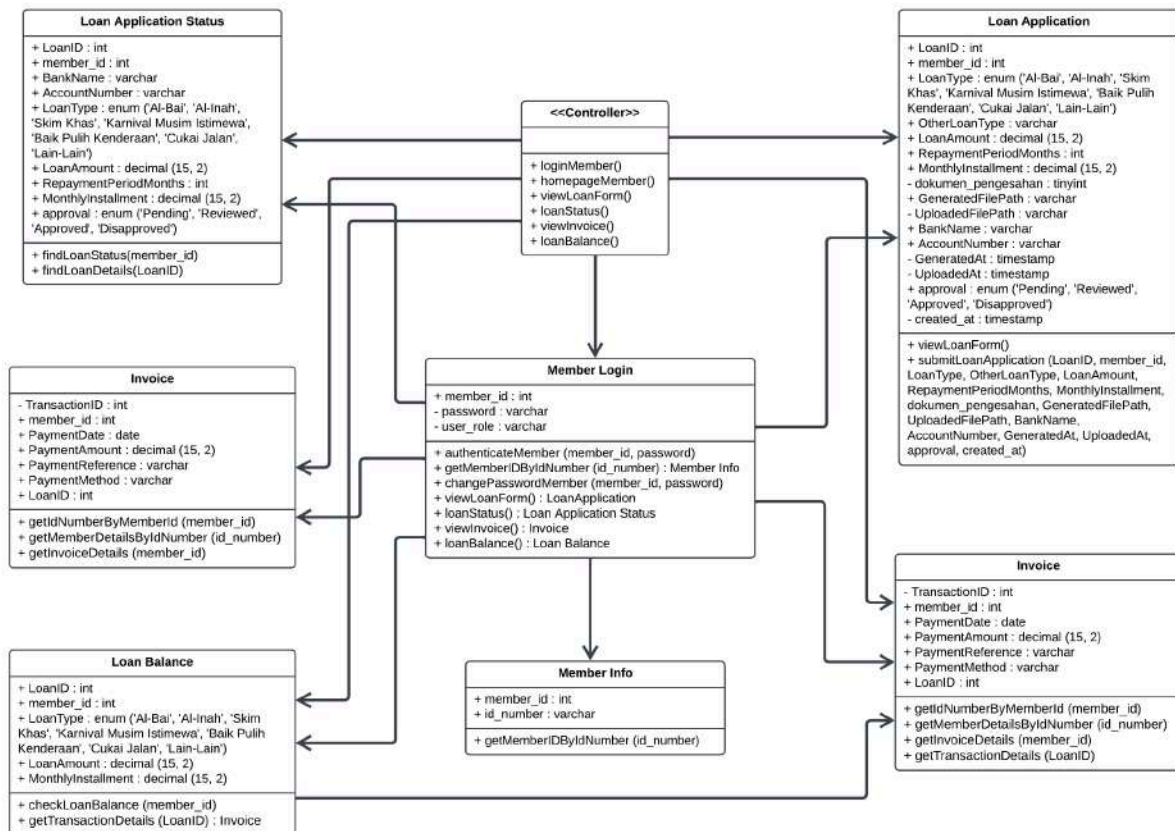


Figure 67: Class diagram for Loan module

4.4 Process View

In this subtopic, each process for each module which consists of Non-member, Authentication, Member, Loan, Admin and ALK modules will be explained and represented using a sequence diagram. Each module will be separated for better understanding with explanations provided for each diagram.

4.4.1 Authentication

In the authentication module, the user of Koperasi KADA will first select their role (Member, Admin, ALK) to access the system and they are required to fill in their username and password to ensure a secure access to the system by verifying user credentials. If the credentials are valid, users will be directed to their own homepage. However, if the credentials filled in is invalid, an error message will be displayed. For users that have gained access to the system, they can log out of the system which will invalidate their session. From this module, the security of the system will be maintained by allowing only authorized users to access it.

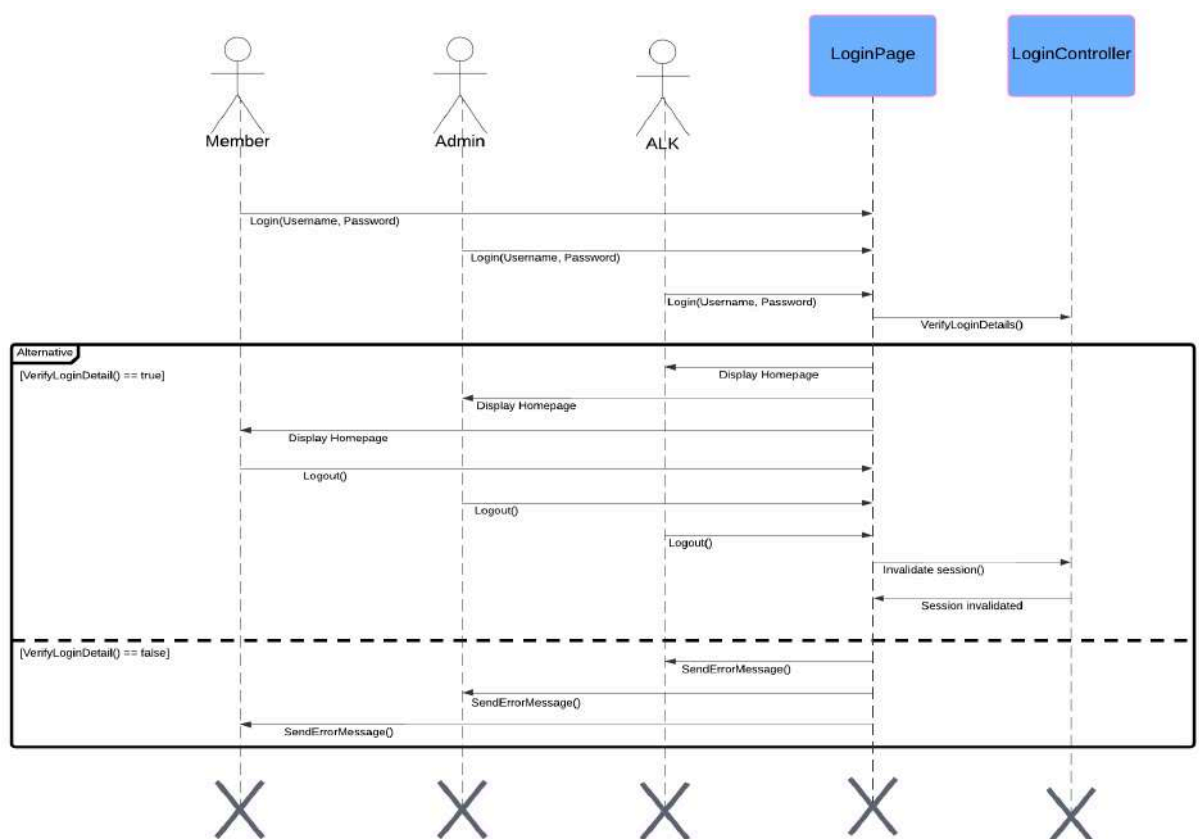


Figure 68: SD024 sequence diagram for Authentication module

4.4.2 Non-member

This module was an interaction of an employee in KADA who is not registered as a member of Koperasi KADA. This module allows the non-registered users to access a certain feature which is to apply for membership and check membership application status apart from viewing the homepage and accessing public information. Hence, non-member are required to fill in the membership application form by filling in the details through the membership form. The status for applied applications will be updated accordingly. The module allows any user to interact to access the system but with limited capacity in order to encourage users to apply for a membership in order to gain full access to the system.

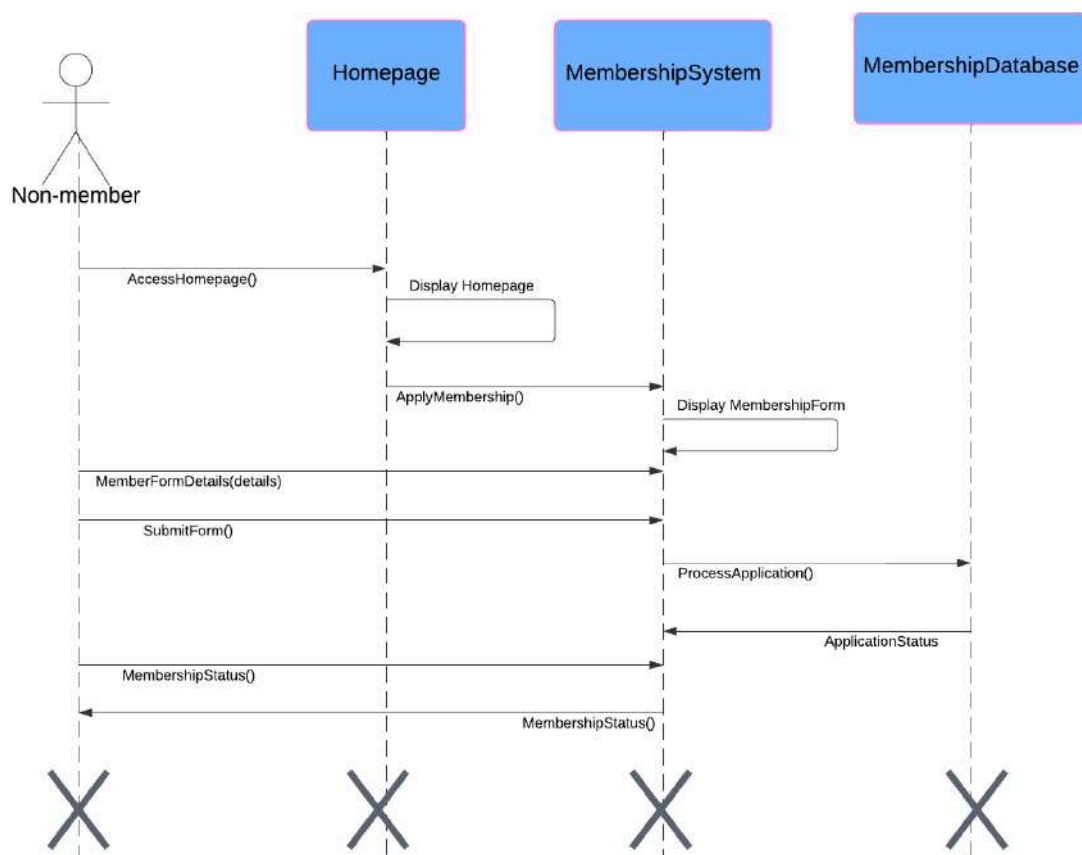


Figure 69: SD025 sequence diagram for Non-Member module

4.4.3 Member

The member module allows members to view their member-specific static homepage, where they can access a menu of options only after they successfully log in into the system. From the menu provided at the homepage, it allows members to check their current savings, check the invoice, apply for a loan, etc. However this module will focus on the process of member access to their financial activity information. By clicking on the menu button to check on their savings and invoice, the system will retrieve and display the detailed saving information and invoices.

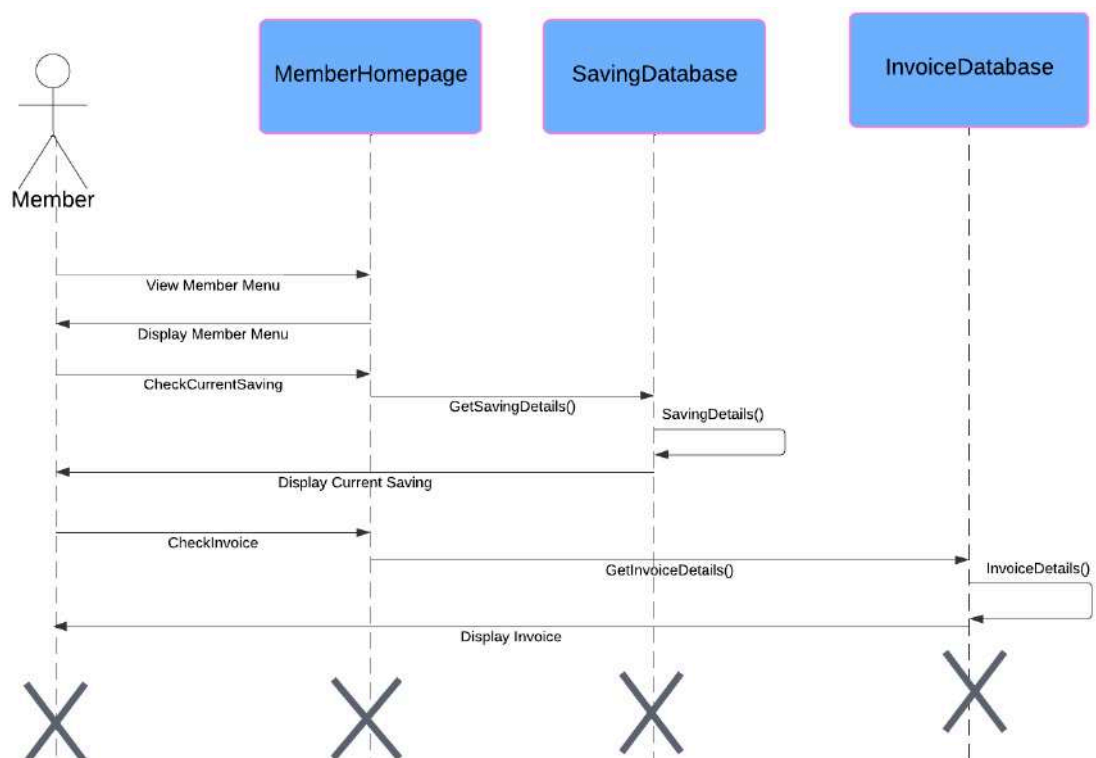


Figure 70: SD026 sequence diagram for Member module

4.4.4 Loan

In the loan module, it will focus the loan-related activities for members where these activities can only be accessed at the member-specific homepage after successfully login into the system. This module will integrate an efficient handling of every loan-related activity where it also allows accessible loan management services. Members can apply for a loan and submit the application. Every application submitted will be processed and they can check the status of the application applied to see whether the application has been approved or still under review. Moreover, members can check their loan balance and access their transaction history for every payment that has been made.

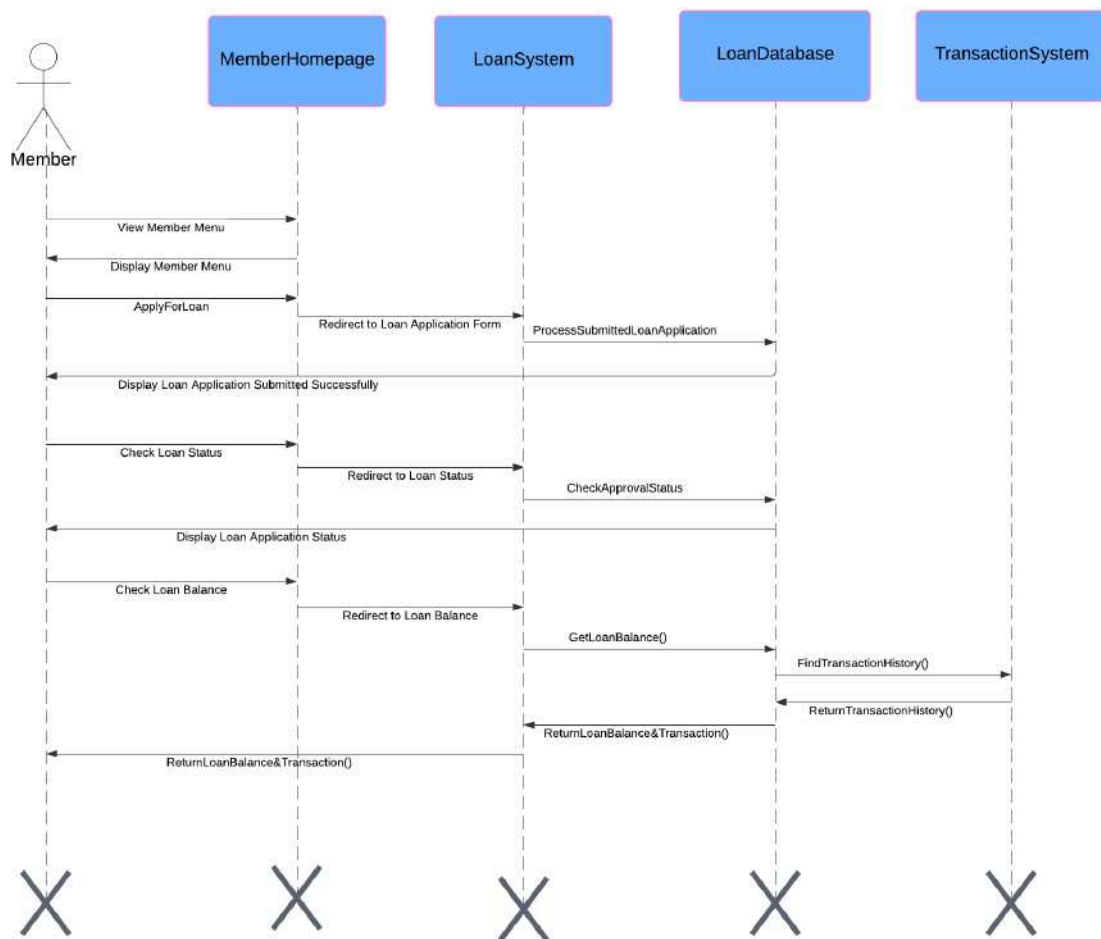


Figure 71: SD027 sequence diagram for Loan module

4.4.5 Admin

In the admin module, it will allow admins to view their staff-specific static homepage, where they can access a menu of options only after they successfully log in into the system. This module will focus on how admins will manage and validate applications submitted by both non-member and member of Koperasi KADA. As mentioned, admins can validate both membership applications applied by non-member and loan applications applied by members. Both will have the process where admins will review the application submitted and determine whether the application applied can be processed to the next step or not. This module will ensure all applications submitted will be thoroughly reviewed by admins before being reviewed by ALK.

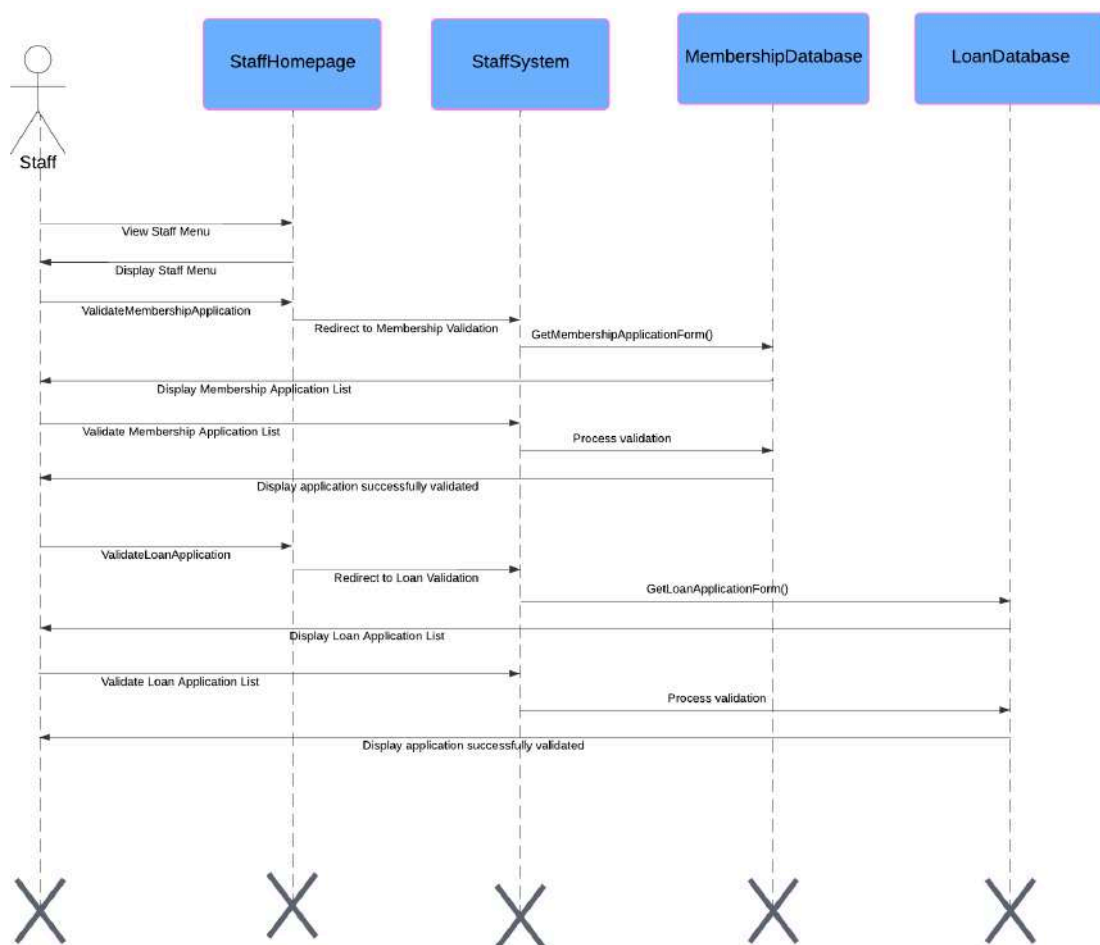


Figure 72: SD028 sequence diagram for Admin module

4.4.6 ALK

For the ALK module, it will have almost the same process with the staff module where ALK will view their staff-specific static homepage, where they can access a menu of options only after they successfully log in into the system. However, this module will focus more on approving the application that has been reviewed by staff. Similar with staff module, ALK will view the application that has been reviewed by staff and ALK will determine whether the application can be approved or not. Once the application is approved/disapproved, the system processes the applications and updates their status. This module will determine whether the non-member is qualified to be a member of Koperasi KADA and the member is qualified to apply for a loan where ALK will thoroughly check the application again after being reviewed by staff.

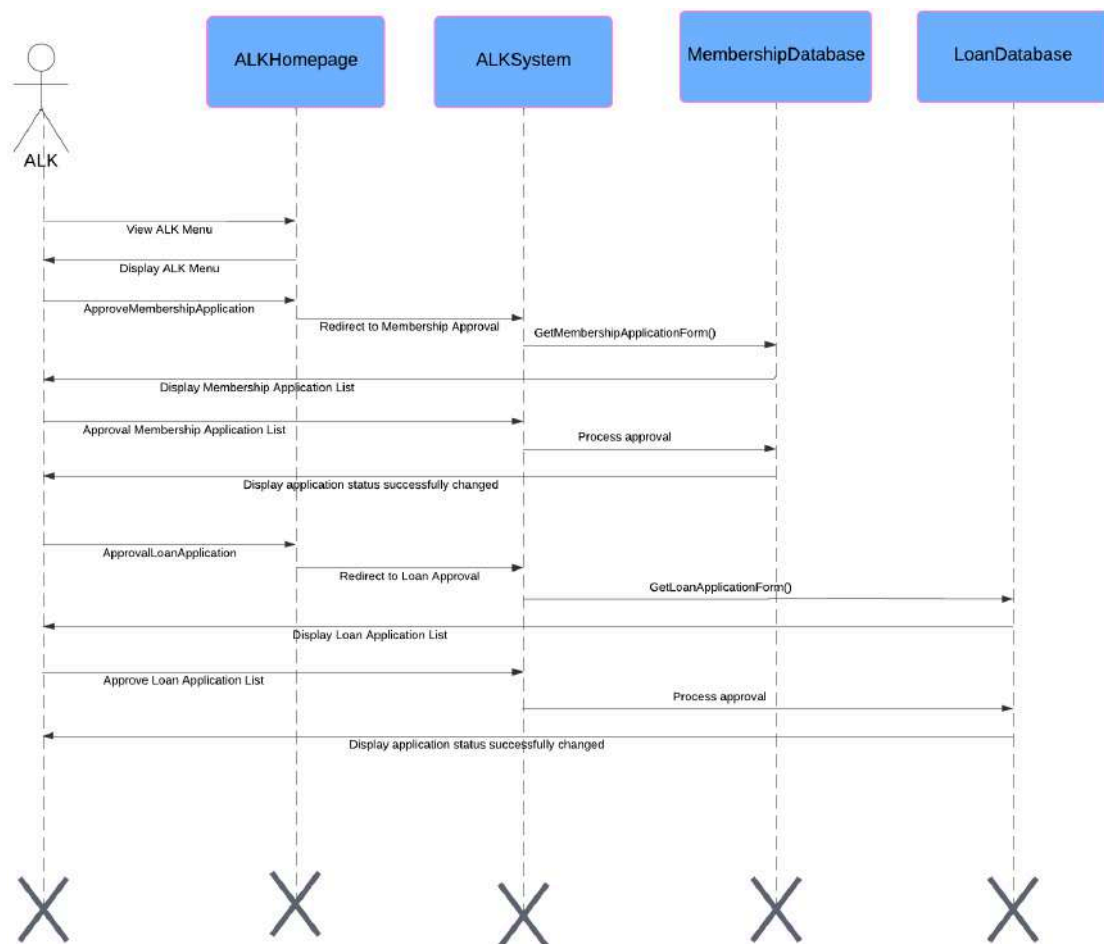


Figure 73: SD029 sequence diagram for ALK module

4.5 Physical/ Deployment View

This view describes how the system will be physically deployed on the hardware infrastructure. Clients will be connected to the system through the browser. From the browser, client's requests will be sent to the server and the application server will decide which module is involved in the request. Clients can view data stored in the database through the module in the application server. Deployment diagram below picture clearer view of how the system will be deployed.

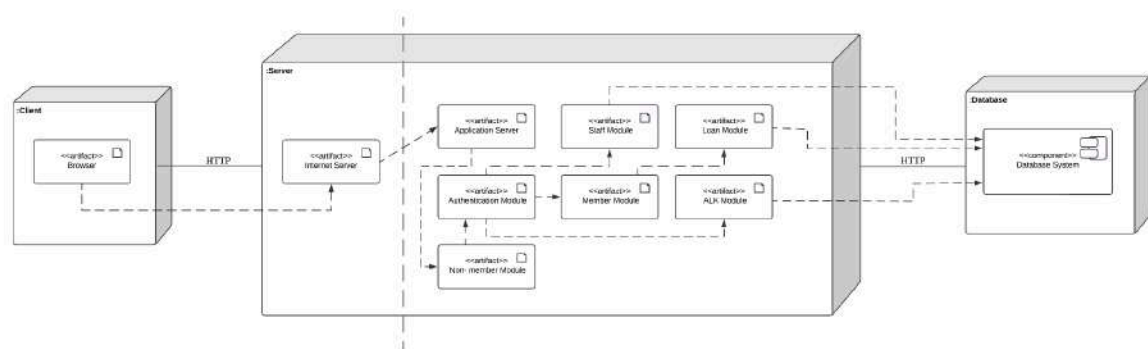


Figure 74: Deployment diagram for Koperasi KADA system

5. Data Design

The major data or systems entities are stored into a relational database named as Non-member, Member Application, Admin, ALK, Loan Application, Koperasi Member, Invoice, Loan Balance and Savings, processed and organized into 13 entities as listed in Table 31 describe in Figure 75.

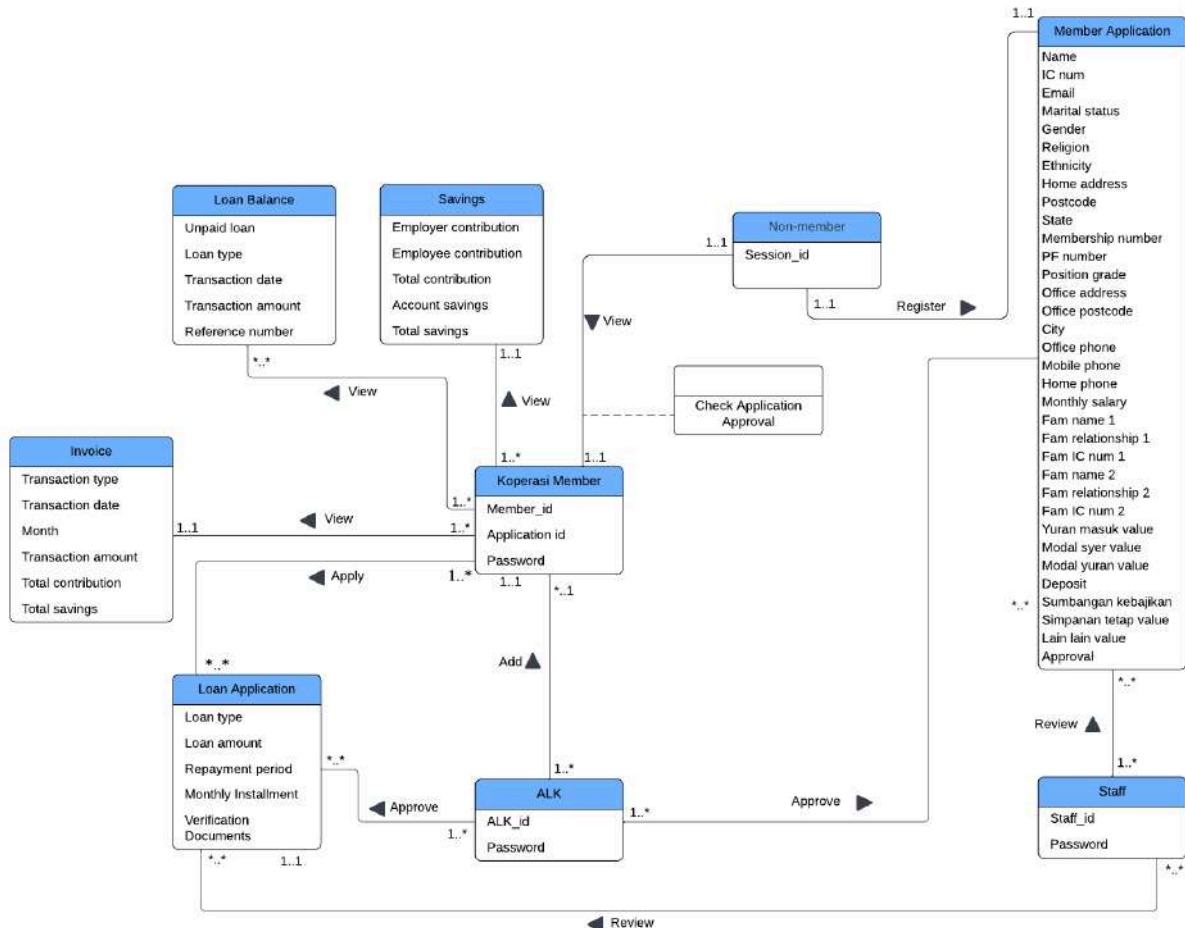


Figure 75: Global conceptual entity relationship diagram design

No.	Entity Name	Description
1	Non-member	General term to describe a person working in KADA who is not a member of Koperasi KADA.
2	Member Application	General term to describe the application filled in non-member that applying for a membership in Koperasi KADA
3	Admin	General term describing employee of Koperasi KADA that managed in review the application submitted
4	ALK	General term to describe an authorized personnel who is in charge to determine the approval of an application that has been reviewed by admin
5	Loan application	General term to describe the application filled in by Koperasi member that applying for a loan
6	Koperasi member	General term to describe the person in KADA who is a member of Koperasi KADA
7	Invoice	General term to describe the record of every transaction of every month
8	Loan balance	General term to describe the balance that need to be paid from the loan applied
9	Savings	General term to describe the funds saved by members from the fee paid every month

Table 31: Description of entities in the database

5.1 Data Dictionary

5.1.1 Entity: <ALK>

Attribute Name	Type	Description
ALK_id	INT	Unique identifier for ALK
Password	VARCHAR	Password for ALK

5.1.2 Entity: <Invoice>

Attribute Name	Type	Description
Transaction type	ENUM	Type of transaction
Transaction date	DATE	Date of transaction
Month	INT	Month of transaction
Transaction amount	DECIMAL	Total of amount of transaction
Total contribution	DECIMAL	Total contributions for the month
Total savings	DECIMAL	Cumulative savings till date

5.1.3 Entity: <Koperasi Member>

Attribute Name	Type	Description
Member_id	INT	Unique identifier for member
Application_id	INT	Unique identifier for the member's application
Password	VARCHAR	Password for the member account

5.1.4 Entity: <Loan Application>

Attribute Name	Type	Description
Loan type	ENUM	Type of loan applied for
Loan amount	DECIMAL	Total loan amount applied
Repayment period	INT	Period of loan repayment in months
Monthly installment	DECIMAL	Monthly payment amount
Verification document	STRING	Supporting document for verifications

5.1.5 Entity: <Loan Balance>

Attribute Name	Type	Description
Unpaid loan	DECIMAL	Remaining unpaid loan amount
Loan type	ENUM	Type of loan applied
Transaction date	DATE	Date of loan-related transaction
Transaction amount	DECIMAL	Amount transacted on the loan

Reference number	VARCHAR	Transaction reference number
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5.1.6 Entity: <Member Application>

Attribute Name	Type	Description
Nama	VARCHAR	Name of applicant
No. K/P	INT	Identification card number
Taraf perkahwinan	ENUM	Marital status of applicant
Alamat Rumah	STRING	Residential address
Poskod	INT	Postal code
Negeri	STRING	State of residence
No. Anggota	VARCHAR	Staff number
No. PF	VARCHAR	PF number
Jawatan & Gred	VARCHAR	Position and grade
Alamat Pejabat	STRING	Office address
Poskod	INT	Office postal code
Bandar	STRING	City of the office address
No. Tel/Fax	INT	Telephone or fax number
No. Tel Bimbit	INT	Mobile phone number
No. Tel Rumah	INT	Home phone number
Gaji Bulanan	DECIMAL	Monthly salary
Jantina	ENUM	Gender
Agama	STRING	Religion
Bangsa	STRING	Ethnicity
Maklumat Keluarga & Pewaris	STRING	Family and nominee details
Kuantiti Yuran	DECIMAL	Membership fee amount

5.1.7 Entity: <Non-member>

Attribute Name	Type	Description
Session_id	INT	Unique identifier for non-member session

5.1.8 Entity: <Savings>

Attribute Name	Type	Description
Employer contribution	DECIMAL	Contribution made by the employer
Employee contribution	DECIMAL	Contribution made by the employee

Total contribution	DECIMAL	Combined contribution from employer and employee
Account savings	DECIMAL	Savings in the account
Total savings	DECIMAL	Total cumulative savings

5.1.9 Entity: <Admin>

Attribute Name	Type	Description
Admin_id	INT	Unique identifier for admin
Password	VARCHAR	Password for admin

6. User Interface Design

The proposed system for Koperasi KADA has 4 user views, which are non-member, member, admin and Ahli Lembaga Koperasi (ALK). As for non-members, they can view the homepage where the details about Koperasi KADA is displayed. Non-members can also apply for membership by clicking on the 'Daftar Sekarang' button. Once they click on the 'Daftar Sekarang' button, the users will have to key-in their personal details in a form and click on 'Simpan Maklumat' button to submit their application. Once non-members have submitted their application, they can check the application status by clicking on the 'Semak Permohonan' button on the homepage to check their membership application status. If the membership application has been approved, the member identification number and default password will be displayed on the screen. Else, if the application is still being reviewed, is in pending status, disapproved or if the application doesn't exist, then appropriate messages will be displayed. Figure 76 to 82 in 6.1 shows the interfaces for non-members.

As for member view, members will have to click on the 'Daftar Masuk' button on the homepage to login to the system. After that, a button page with three user views will be displayed, which are 'Ahli Koperasi', 'Admin Koperasi' and 'Ahli Lembaga Koperasi'. Members have to click on the 'Ahli Koperasi' button and key-in their member identification number and password to login to the system. If members wish to change their password, they can click on the 'Tukar Kata Laluan' button to change their password. Members will have to enter their identification card number and enter their new password. Then, they have to click on the 'Simpan Maklumat' button to successfully update their password. Appropriate messages will be displayed based on successful or unsuccessful password reset. 'Kembali ke Halaman Utama' button will be displayed which brings it back to the homepage and 'Tukar Kata Laluan Lagi' button will be displayed if members wish to reset their password again. Once they have logged in, they'll view a page with 6 buttons, each representing a feature for members. The page will display 'Mohon Pinjaman' button to apply for loans, 'Semak Permohonan Pinjaman' button to check on loan application status, 'Semak Simpanan' button to view member's savings, 'Baki Pinjaman' button to view loan balance, 'Semak Penyata' button to get their invoice and last but not least, 'Mohon Berhenti Ahli' button to apply for membership termination. There will also be a small user icon on the navigation bar on the top of the screen to view member details. Members can click on the icon and click on the 'Kemaskini Maklumat' button to edit their information. The member details form with existing details will be displayed where members can make changes and have to click on the 'Simpan Maklumat' button to update the changes. If members click on

the 'Mohon Pinjaman' button, a form will be displayed for the members to fill in the details. Then, the members have to click on the 'Hantar Permohonan' button to submit their loan application form. If members click on the 'Semak Permohonan Pinjaman' button, the loan application status for all the loans applied will be displayed. If the loan is approved, then the loan details will be displayed. If the loan is being reviewed or is still pending, then appropriate messages will be displayed. If members click on the 'Semak Simpanan' button, members' savings along with the date will be displayed and if the members have loans with Koperasi KADA, then their employee share and employer share will be displayed along with the date. If the members click on 'Baki Pinjaman' button, the balance loan amount to be paid will be displayed in red along with the loan details such as loan type and monthly installment amount will be displayed. Apart from that, all the transactions that have been made previously to pay off the loan will also be displayed. If members click on the 'Semak Penyata' button, the members' details such as name, identification card number and membership number will be displayed along with the invoice. If the members click on 'Mohon Berhenti Ahli' button, they'll view a form to key in their details to apply for membership termination. Once they have filled in the form, they'll have to submit the form to successfully apply for membership termination. If the members click on the 'Pilihan' button on the navigation bar, they'll view other features so they can click on them to jump pages. To logout of the system, members have to click on the 'Daftar Keluar' button on the top right of the page. Figure 76 and Figure 83-97 in 6.1 shows the interfaces for members.

As for admin view, admins will have to click on the 'Daftar Masuk' button on the homepage to login to the system. After that, a button page with three user views will be displayed, which are 'Ahli Koperasi', 'Admin Koperasi' and 'Ahli Lembaga Koperasi'. Admins have to click on the 'Admin Koperasi' button and key-in their admin identification number and password to login to the system. If admins wish to change their password, they can click on the 'Tukar Kata Laluan' button to change their password. Admins will have to enter their identification card number and enter their new password. Then, they have to click on the 'Simpan Maklumat' button to successfully update their password. Appropriate messages will be displayed based on successful or unsuccessful password reset. 'Kembali ke Halaman Utama' button will be displayed which brings it back to the homepage and 'Tukar Kata Laluan Lagi' button will be displayed if admins wish to reset their password again. Once they have logged in, they'll view a page with 6 buttons, each representing a feature for admins. The page will display 'Senarai Permohonan Pinjaman' button to view loan application list, 'Senarai Permohonan Ahli' button to view membership application list, 'Laporan Koperasi' button to view Koperasi KADA report, 'Tambah Admin' button to add a new admin, 'Tambah

Ahli Lembaga Koperasi' button to add new ALK and lastly 'Senarai Permohonan Berhenti Ahli' button to view membership termination list. If admins click on the 'Senarai Permohonan Pinjaman' button, a list of loan applications with 'Pending' status will be displayed. Admins can view the verification documents submitted by applicants and then update the status to 'Reviewed' if the documents fit the requirements, or update the status to 'Disapproved' if it doesn't fit the requirements. Once the update has been successfully done, a message will be displayed to notify the successful update. The same goes to membership applications if admins click on the 'Senarai Permohonan Ahli' button, a list of membership applications with 'Pending' status will be displayed. Admins can view the verification documents submitted by applicants and then update the status to 'Reviewed' if the documents fit the requirements, or update the status to 'Disapproved' if it doesn't fit the requirements. Once the update has been successfully done, a message will be displayed to notify the successful update. If admins click on the 'Laporan Koperasi' button, Koperasi KADA's report will be displayed. The report has a monthly and annual summary for employee share displayed as 'Syer Pekerja' and employer share displayed as 'Syer Majikan' and has monthly report for membership and loan applications list. If admins click on any specific date on the calendar displayed on the report page, the report will change up to the specific date only. Admins can click on the 'Tambah Admin' button to add a new admin to the system. Admins have to key in the new admin's identification number and click on the 'Tambah Admin' button to complete the process. The same process applies for adding new ALK. Admins can click on the 'Tambah Ahli Lembaga Koperasi' button to add a new ALK to the system. Admins have to key in the new ALK's identification number and click on the 'Tambah ALK' button to complete the process. If admins click on the 'Senarai Permohonan Berhenti Ahli' button, a list of membership termination applications with 'Pending' status will be displayed. Admins can view the verification documents submitted by members and then update the status to 'Approved' if the documents fit the requirements, or update the status to 'Disapproved' if it doesn't fit the requirements. Once the update has been successfully done, a message will be displayed to notify the successful update. To check their admin approval status, admins can click on the 'Semak Permohonan' button on the homepage. If it is approved, the admin identification number and default password will be displayed on the screen. Else, if the admin approval process is still being reviewed, is in pending status, disapproved or if the application doesn't exist, then appropriate messages will be displayed. To logout of the system, members have to click on the 'Daftar Keluar' button on the top right of the page. Figure 76, Figure 78-83, Figure 98-108 in 6.1 shows the interfaces for admins.

As for ALK view, ALKs will have to click on the 'Daftar Masuk' button on the homepage to login to the system. After that, a button page with three user views will be displayed, which are 'Ahli Koperasi', 'Admin Koperasi' and 'Ahli Lembaga Koperasi'. Admins have to click on the 'Ahli Lembaga Koperasi' button and key-in their ALK identification number and password to login to the system. If ALKs wish to change their password, they can click on the 'Tukar Kata Laluan' button to change their password. ALKs will have to enter their identification card number and enter their new password. Then, they have to click on the 'Simpan Maklumat' button to successfully update their password. Appropriate messages will be displayed based on successful or unsuccessful password reset. 'Kembali ke Halaman Utama' button will be displayed which brings it back to the homepage and 'Tukar Kata Laluan Lagi' button will be displayed if ALKs wish to reset their password again. Once they have logged in, they'll view a page with 3 buttons, each representing a feature for ALKs. The page will display 'Pengesahan Permohonan Pinjaman' button to approve the loan applications, 'Pengesahan Permohonan Ahli' button to approve the membership applications and 'Laporan Tahunan' button to view Koperasi KADA's annual report. If ALKs click on the 'Pengesahan Permohonan Pinjaman' button, a list of loan applications with 'Reviewed' status will be displayed. ALKs can view the verification documents submitted by applicants and then update the status to 'Approved' if the documents fit the requirements, or update the status to 'Disapproved' if it doesn't fit the requirements. Once the update has been successfully done, a message will be displayed to notify the successful update. The same goes to membership applications if ALKs click on the 'Pengesahan Permohonan Ahli' button, a list of membership applications with 'Reviewed' status will be displayed. ALKs can view the verification documents submitted by applicants and then update the status to 'Approved' if the documents fit the requirements, or update the status to 'Disapproved' if it doesn't fit the requirements. Once the update has been successfully done, a message will be displayed to notify the successful update. If ALKs click on the 'Laporan Tahunan' button, Koperasi KADA's report will be displayed. The report has a monthly and annual summary for employee share displayed as 'Syer Pekerja' and employer share displayed as 'Syer Majikan' and has monthly report for membership and loan applications list. If admins click on any specific date on the calendar displayed on the report page, the report will change up to the specific date only. To check their ALK approval status, ALKs can click on the 'Semak Permohonan' button on the homepage. If it is approved, the ALK identification number and default password will be displayed on the screen. Else, if the ALK approval process is still being reviewed, is in pending status, disapproved or if the application doesn't exist, then appropriate messages will be displayed. There is a navigation bar on the top of the page. If the members click on the 'Pilihan' button, they'll view other features so they can click on

them to jump pages. To logout of the system, ALKs have to click on the 'Daftar Keluar' button on the top right of the page. Figure 76, Figure 78-83, Figure 109-115 in 6.1 shows the interfaces for ALKs.

6.1 Screen Images



Figure 76: Interface for homepage

Maklumat Pemohon

Name	Nombor K/P:	
Email		
Taraf Perkahwinan	Jantina	Agama
Bangsa	Alamat Rumah	
Poskod	Negeri	Bandar

Maklumat Pejabat

No. Anggota	No. PF
Jawatan & Gred	Alamat Pejabat
Poskod	No. Tel/Fax

Contact Details

No. Tel Bimbit

No. Tel Rumah

Maklumat Keluarga

Nama Ahli Keluarga 1

No. Kad Pengenalan

Hubungan

Nama Ahli Keluarga 2

No. Kad Pengenalan

Hubungan

Gaji Bulanan

Perkara	RM
Yuran Masuk	
Modal Syer	

Perkara	RM
Yuran Masuk	
Modal Syer	
Modal Yuran	
Wang Deposit Anggota	
Sumbangan Tabung Kebajikan (Al-Abrar)	
Simpanan Tetap	
Lain-lain	

Simpan Maklumat

Figure 77: Interface for membership application form

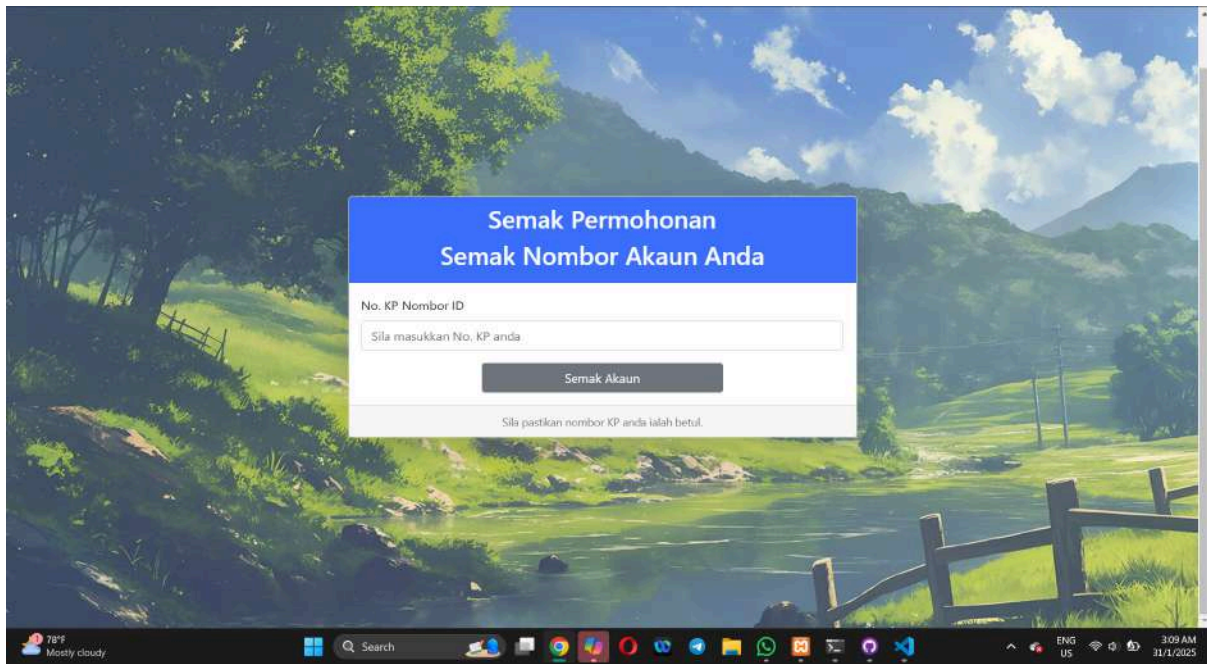


Figure 78: Interface to check membership application status

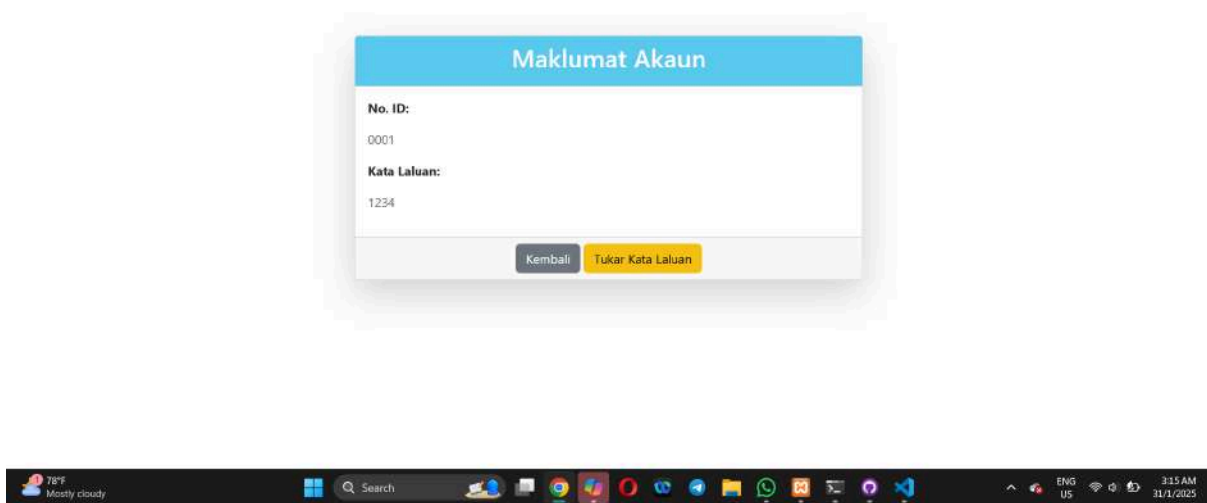


Figure 79: Interface to membership application status if approved



Figure 80: Interface to membership application status if pending



Figure 81: Interface to membership application status if reviewed



Figure 82: Interface to membership application status if disapproved

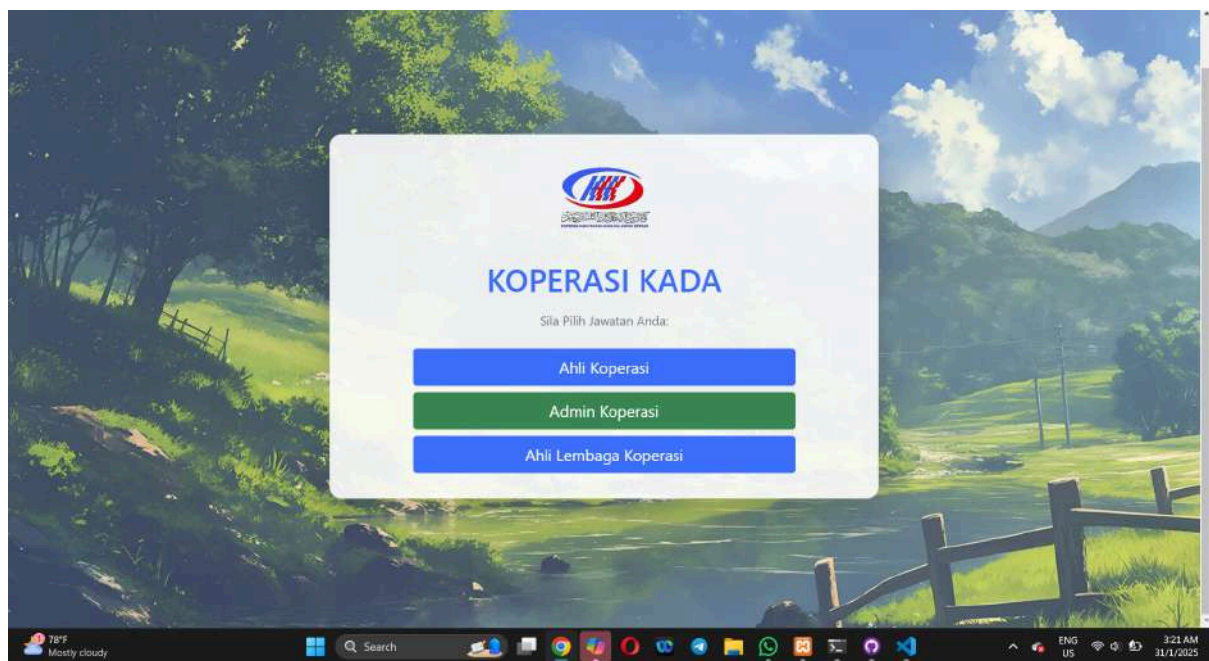


Figure 83: Interface for login page

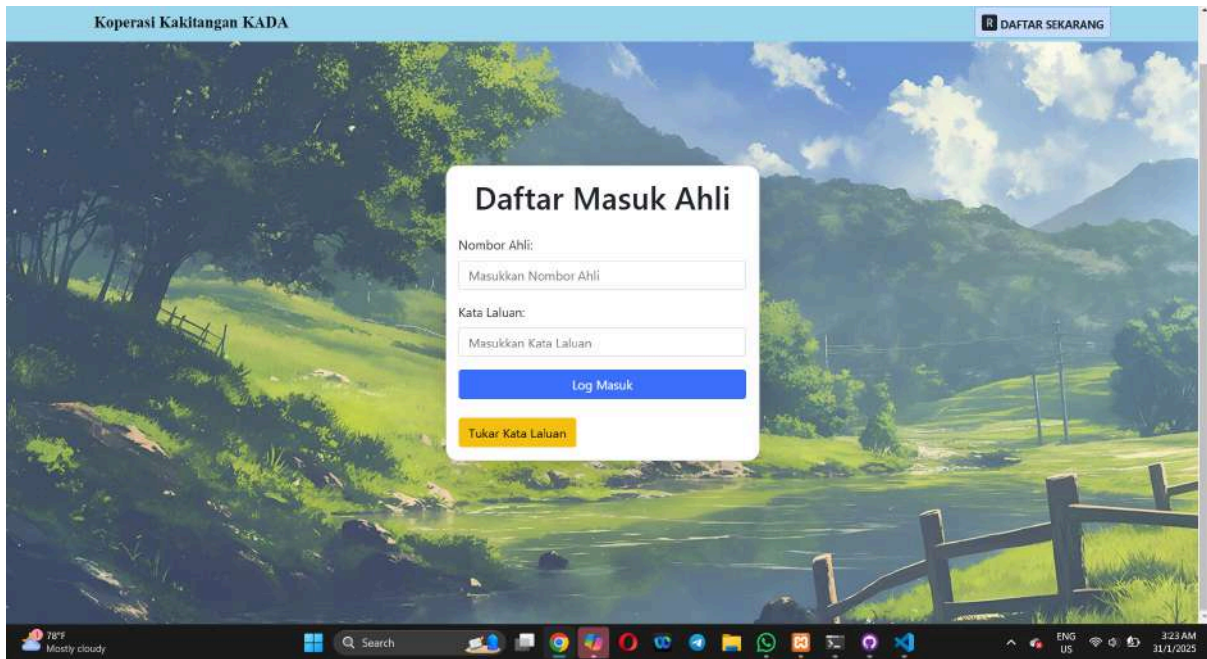


Figure 84: Interface for member login

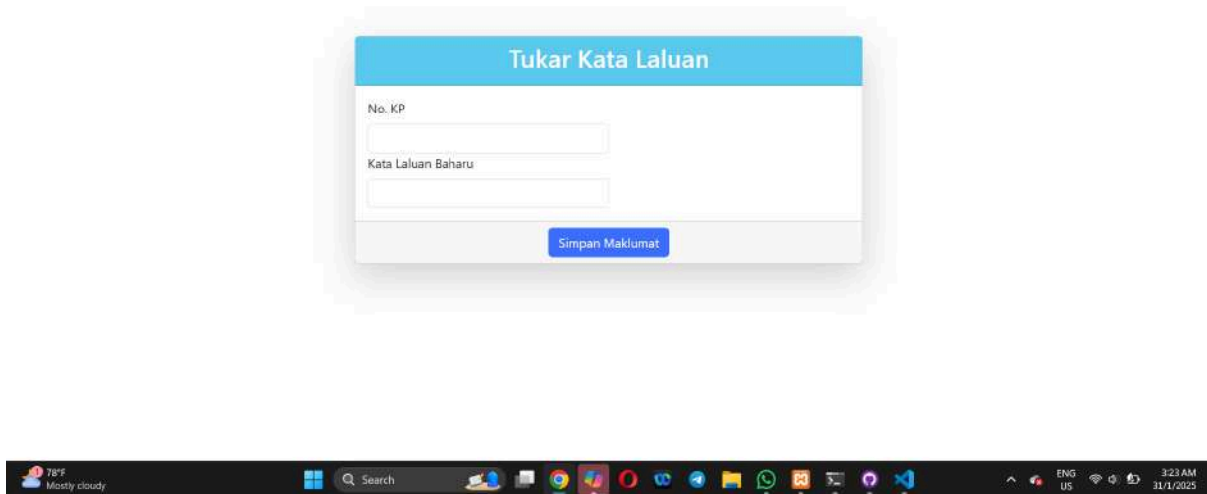


Figure 85: Interface for password reset



Figure 86: Interface for successful password reset



Figure 87: Interface for unsuccessful password reset



Figure 88: Interface for member homepage

Koperasi Kakitangan KADA
Pilihan
DAFTAR KELUAR

Gaji Bulanan

Gaji Bulanan

5000.00

Yuran & Sumbangan

Perkara	RM
Yuran Masuk	55.00
Modal Syer	0.00
Modal Yuran	0.00
Wang Deposit Anggota	0.00
Sumbangan Tabung Kobajikan (Al-Abrar)	0.00
Simpanan Tetap	0.00
Lain-lain	0.00

Kemaskini Maklumat

Figure 89: Interface for member details

Koperasi Kakitangan KADA
Pilihan
DAFTAR KELUAR

KEMASKINI MAKLUMAT PEMOHON

Maklumat Peribadi

Name

Harini A/P Sangaran

Nombor K/P:

041201070330

Email

harini.sangaran@gmail.com

Taraf Perkahwinan

Bujang

Jantina

Perempuan

Agama

Hindu

Bangsa

India

Alamat Rumah

21, Lorong Impian Indah 2, Taman Impian Indah

Poskod

14000

Negeri

Pulau Pinang

Bandar

Bukit Mertajam

Maklumat Pejabat

No. Anggota

0081

No. PF

5678

Jawatan & Grad

A

Alamat Pejabat

MA6 KTDI

Koperasi Kakitangan KADA

Pilihan [DAFTAR KELUAR](#)

No. Kad Pengenalan
750502025500

Gaji Bulanan
5000.00

Yuran & Sumbangan

Perkara	RM
Yuran Masuk	55.00
Modal Syer	0.00
Modal Yuran	0.00
Wang Deposit Anggota	0.00
Sumbangan Tabung Kebajikan (Al-Abrar)	0.00
Simpanan Tetap	0.00
Lain-lain	0.00

[SIMPAN MAKLUMAT](#)

Figure 90: Interface for update member details

Koperasi Kakitangan KADA

Pilihan [DAFTAR KELUAR](#)

JADUAL PEMBAYARAN BALIK PEMBIAYAAN

Tempoh (Tahun)	1	2	3	4	5	6
Tempoh (Bulan)	12	24	36	48	60	72
Jumlah Pembayaran	Ansuran Bulanan					
1,000.00	86.83	45.17	31.28	24.33	20.17	17.39
2,000.00	173.67	90.33	62.56	48.67	40.33	34.78
3,000.00	260.50	135.50	93.83	73.00	60.50	52.17
4,000.00	347.33	180.67	125.11	97.33	80.67	69.56
5,000.00	434.17	225.83	156.39	121.67	100.83	86.94
6,000.00	521.00	271.00	187.67	146.00	121.00	104.33
7,000.00	607.83	316.17	218.94	170.33	141.17	121.72

BORANG PERMOHONAN PINJAMAN

Jenis Pinjaman

Koperasi Kakitangan KADA Pilihan DAFTAR KELUAR

BORANG PERMOHONAN PINJAMAN

Jenis Pinjaman
 Sila Pilih Jenis Pinjaman
 Jika 'Lain-lain', Nyatakan
 Nyatakan Jenis Pinjaman Lain

Jumlah Pinjaman (RM)
 Masukkan Jumlah Pinjaman

Tempoh Bayaran Balik (bulan)
 Nyatakan Tempoh Bayaran Balik Bulanan

Ansuran Bulanan (RM)
 Masukkan Ansuran Bulanan

Nama Bank
 Nyatakan Nama Bank

Nombor Akaun
 Masukkan Nombor Akaun

Muat Turun Templat Dokumen Borang Permohonan Pinjaman

Muat Naik Dokumen Lengkap
 Choose file: No file chosen

HANTAR PERMOHONAN

Figure 91: Interface for loan application form

Koperasi Kakitangan KADA Pilihan DAFTAR KELUAR

SEMAK KELULUSAN PINJAMAN

Permohonan Anda Berjaya!

No. ID Member: 1
 Nama Bank: RHB Bank
 No. Akaun Bank: 1571003
 Jenis Pinjaman: Skim Khas
 Jumlah Pinjaman: 7000.00
 Tempoh Pembayaran Pinjaman: 48
 Ansuran Bulanan: 170.33

Permohonan Anda Sedang Diproses
 Sila tunggu keputusan kelulusan.

Permohonan Anda Ditolak
 Sila hubungi pihak kami untuk maklumat lanjut.

Permohonan Anda Dalam Penilaian
 Pihak kami sedang menilai permohonan anda. Sila semak semula kemudian.

Figure 92: Interface for loan application status



Figure 93: Interface for member savings



Figure 94: Interface for loan balance



Figure 95: Interface for invoice

Koperasi Kakitangan KADA Pilihan DAFTAR KELUAR

Borang Permohonan Berhenti Keahlian

Nama No KP
 Tarikh Lahir Umur
 Alamat Rumah
 Poskod Bandar Negeri
 No Ahli No PF
 Jawatan Alamat Pejabat
 Poskod Pejabat Bandar Pejabat No Tel/Fax

The screenshot shows a web form for membership termination. It includes several input fields and dropdown menus. At the bottom, there is a blue button labeled 'Hantar Permohonan'.

No Tel Bimbit	Jantina
	Lelaki
Agama	Nyatakan (Jika Lain-lain)
Islam	
Bangsa	Nyatakan (Jika Lain-lain)
Melayu	
Sebab Berhenti	

Hantar Permohonan

Figure 96: Interface for membership termination form



Figure 97: Interface for navigation bar for features choices and logout button

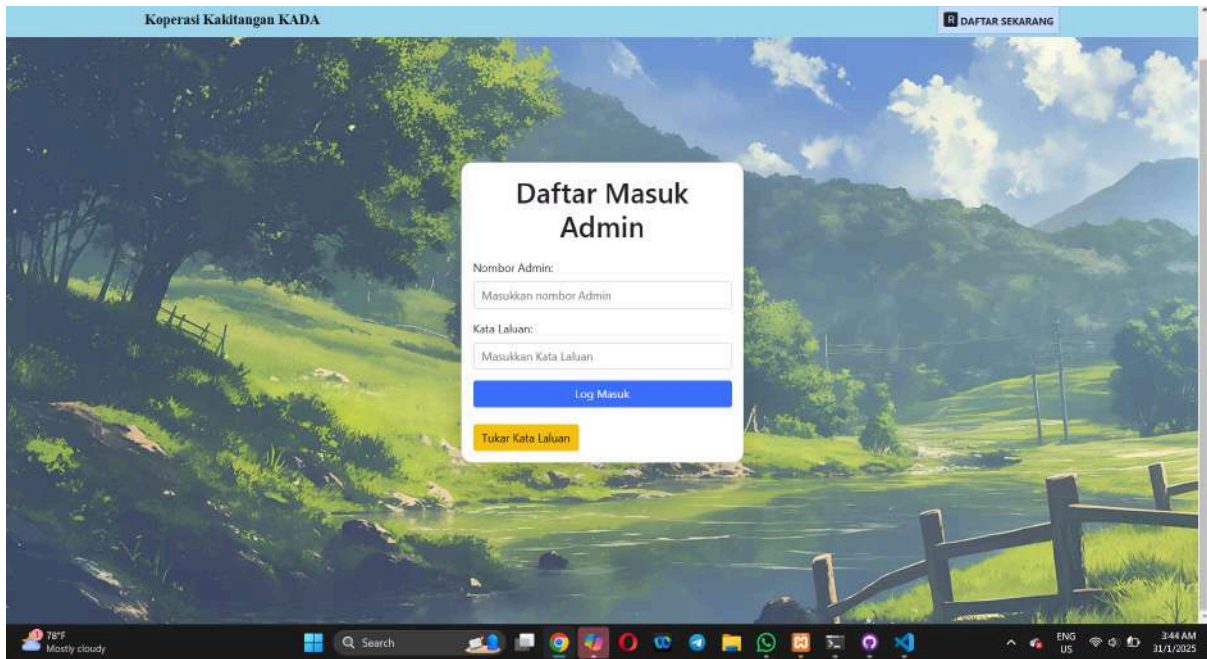


Figure 98: Interface for admin login

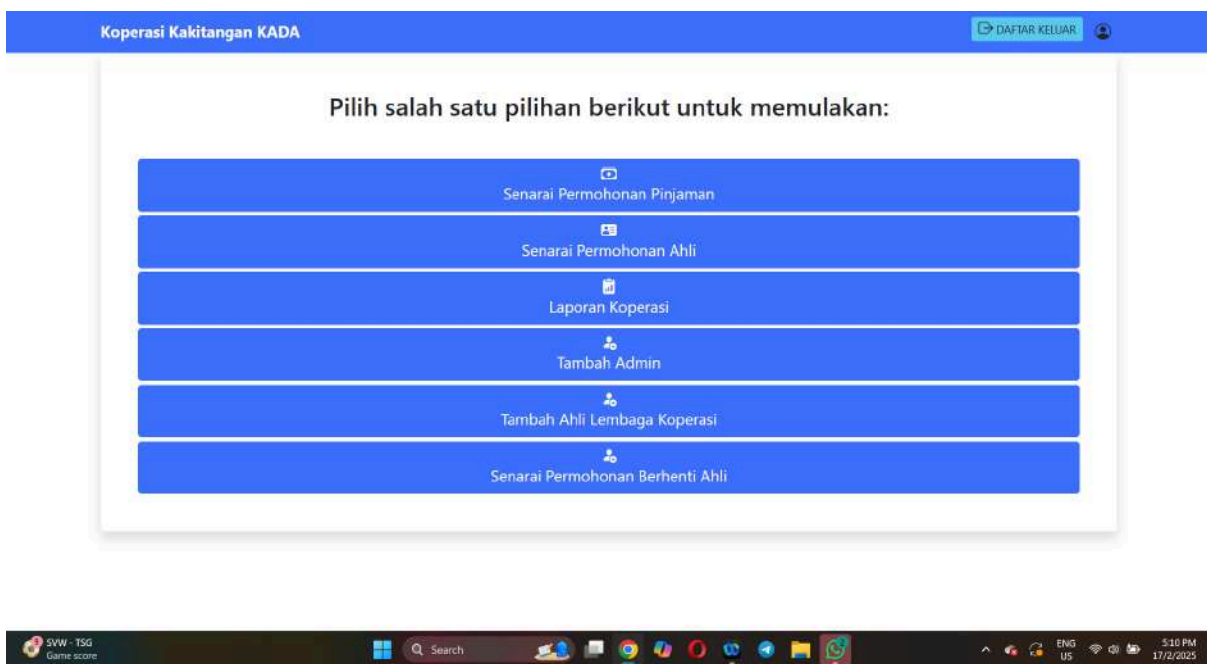


Figure 99: Interface for admin homepage

Senarai Permohonan Ahli

Nama	No. Kad Pengenalan	Dokumen Permohonan Ahli	Status Permohonan	Tindakan
Fatini Az-zahara Binti Baharom	040325080392	Dokumen Permohonan Ahli	Belum Selesai	<button>Kemas Kini</button>
Muhammad Filry Bin Mohd Nasir	670522070361	Dokumen Permohonan Ahli	Belum Selesai	<button>Kemas Kini</button>
Cheah Jo-Sie	001018070180	Dokumen Permohonan Ahli	Belum Selesai	<button>Kemas Kini</button>
	880623094769	Dokumen Permohonan Ahli	Belum Selesai	<button>Kemas Kini</button>

Kembali Semua



Figure 100: Interface for membership application list

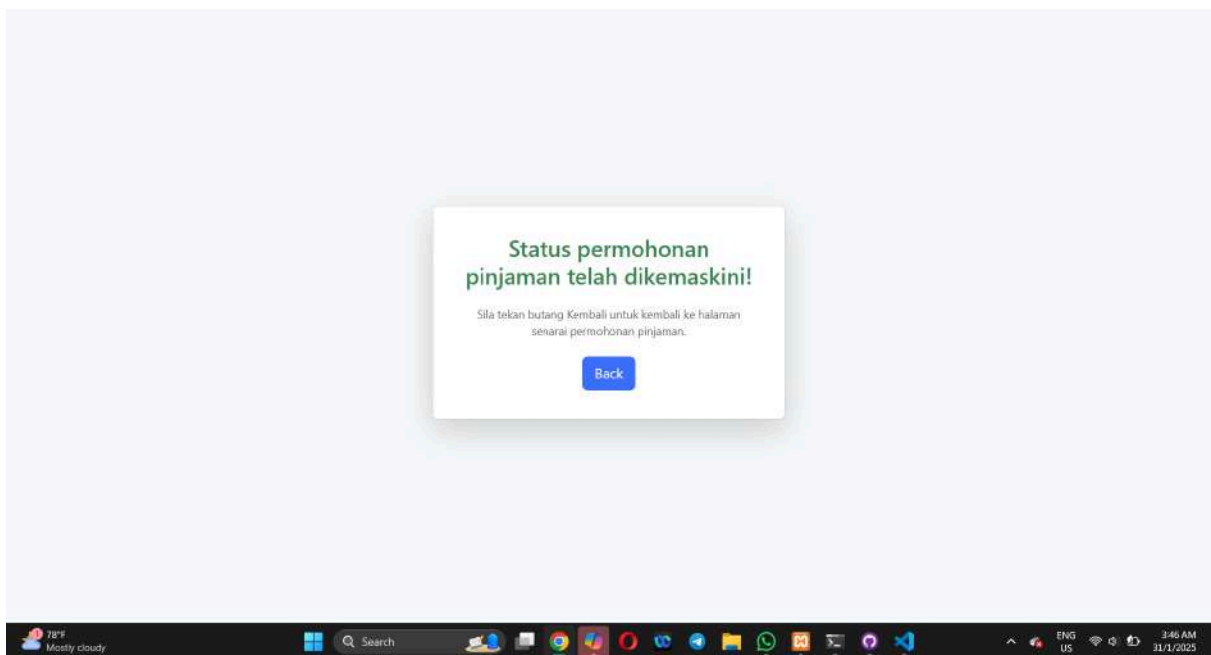


Figure 101: Interface for successful membership application status update

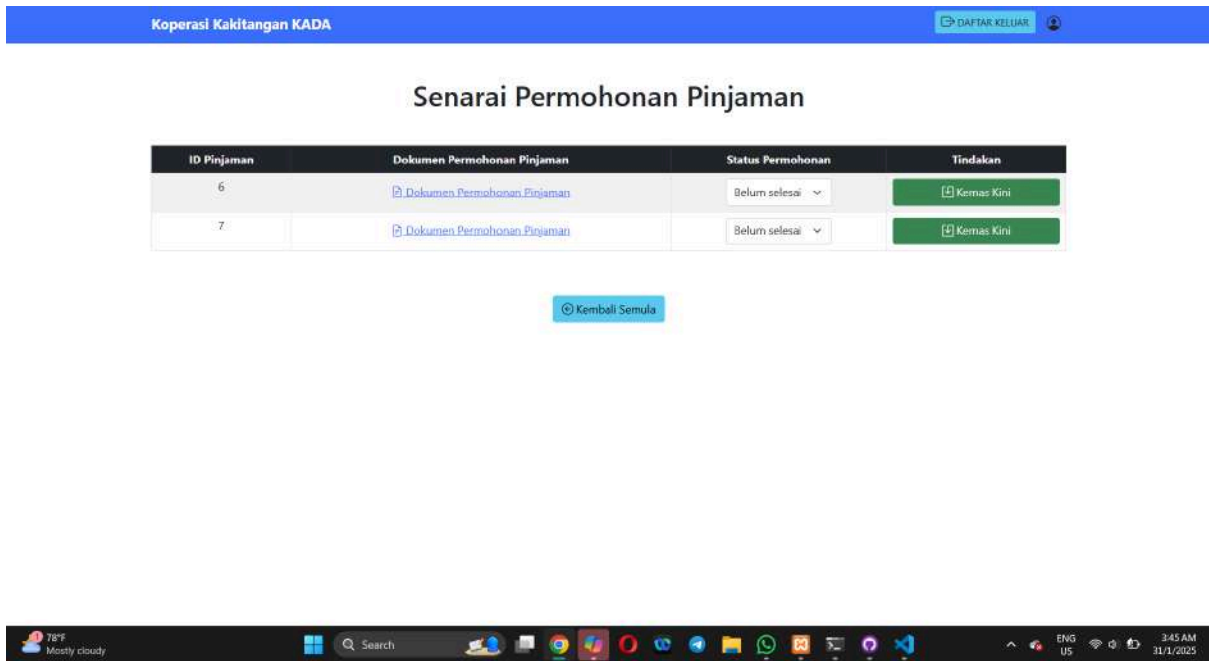


Figure 102: Interface for loan application list

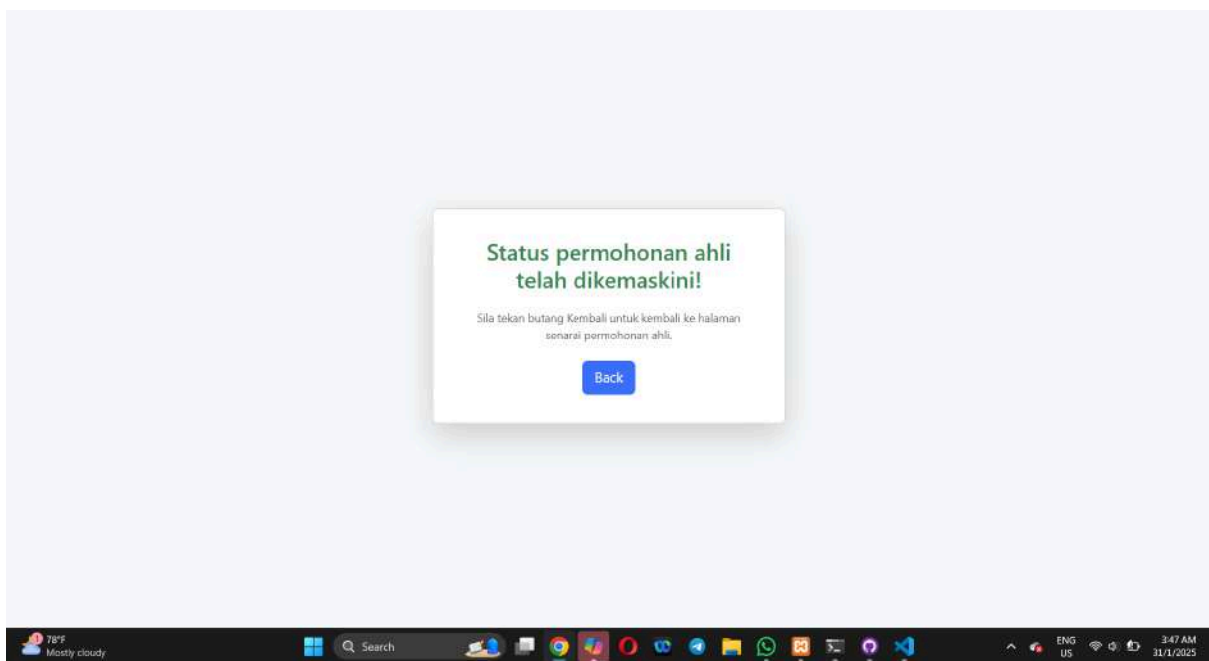


Figure 103: Interface for successful loan application status update

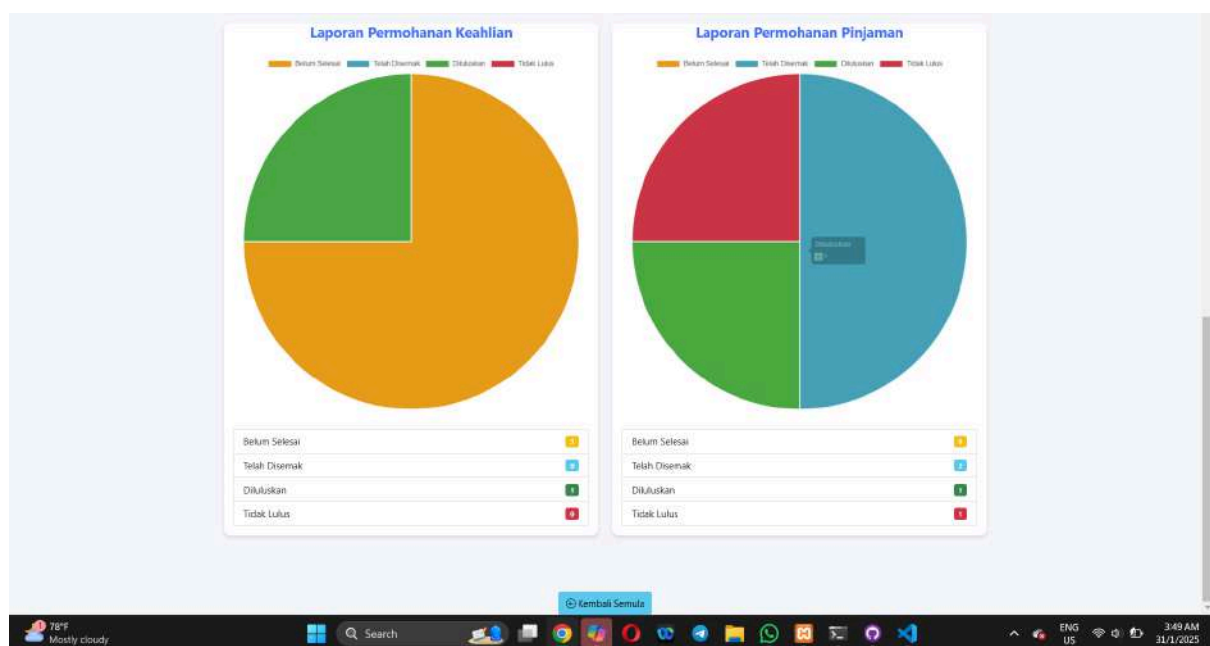
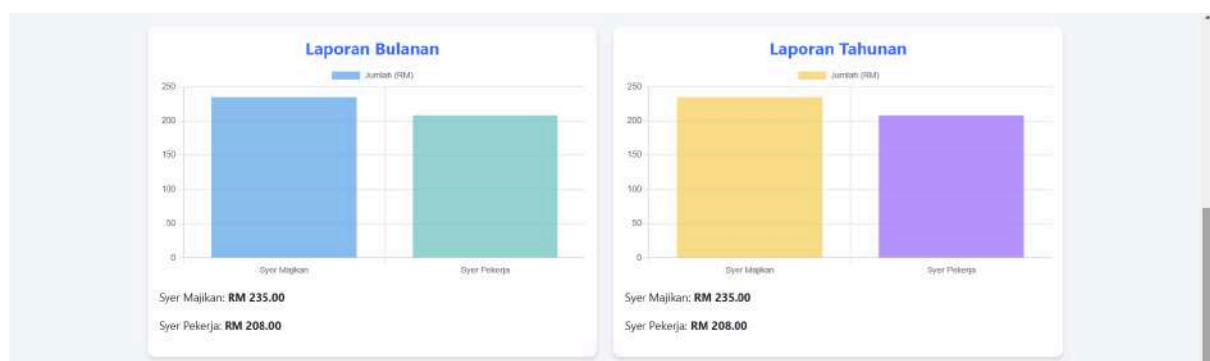
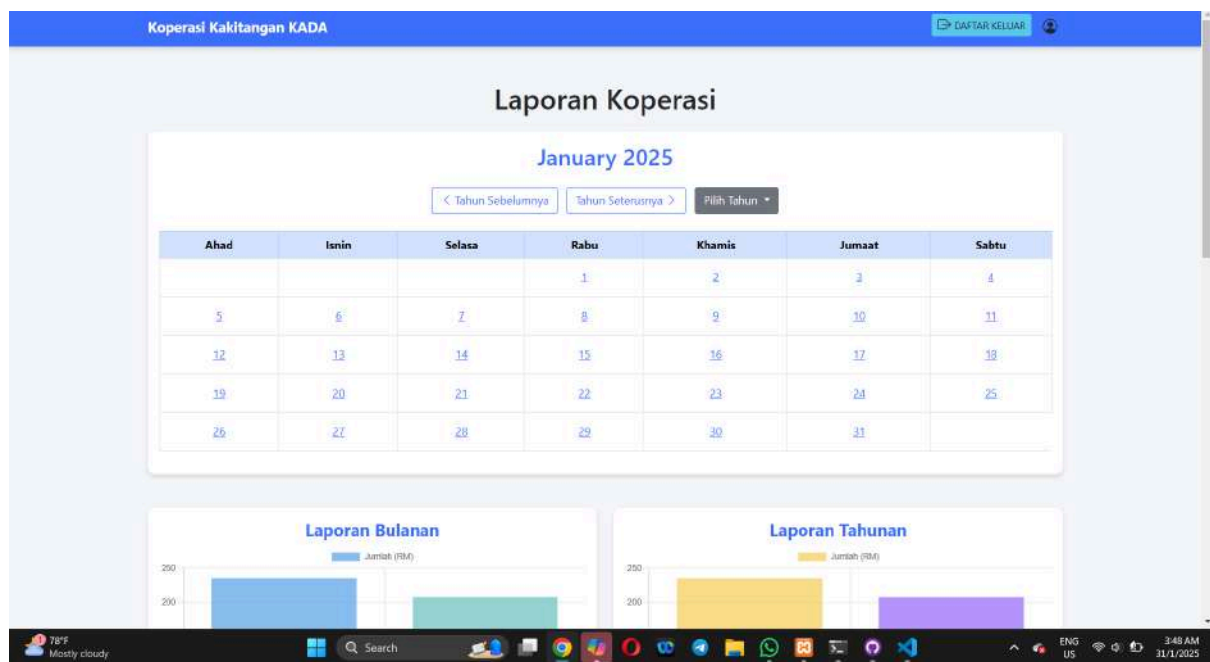


Figure 104: Interface for Koperasi KADA's report

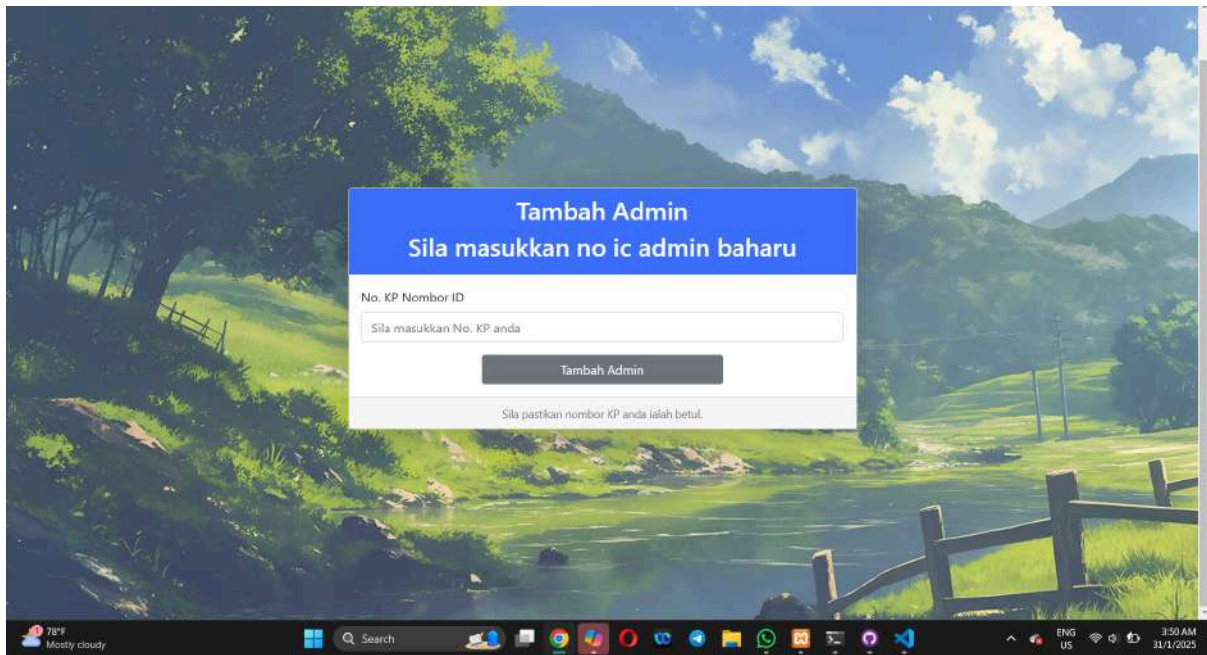


Figure 105: Interface for add new admin

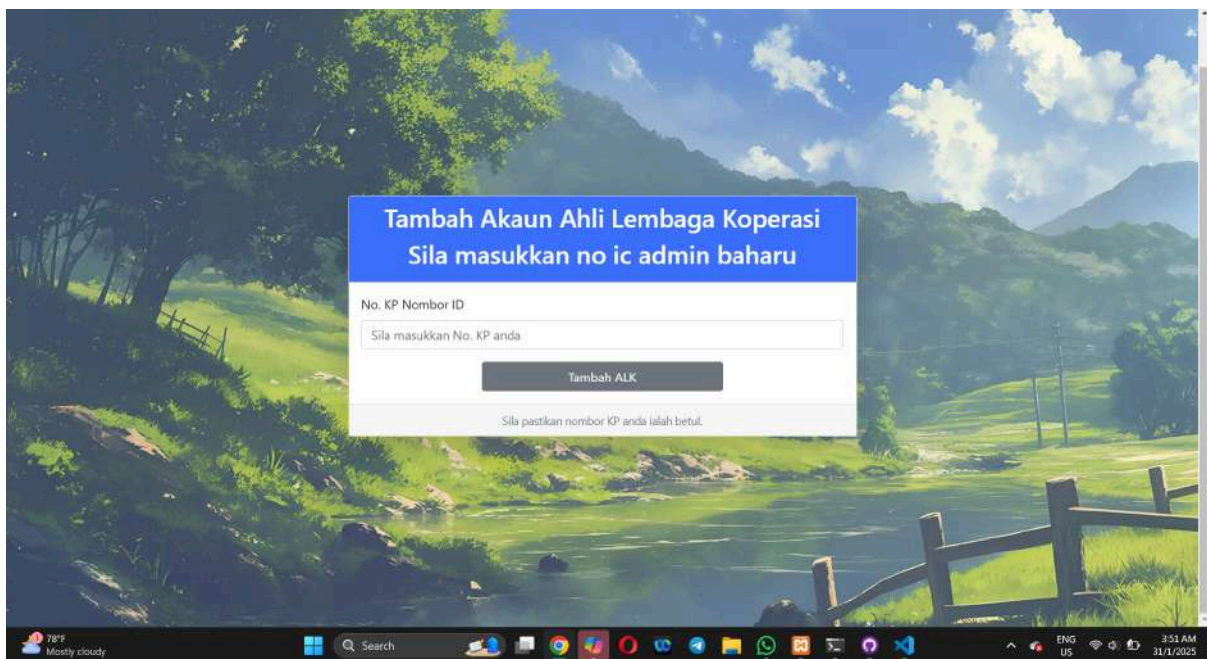


Figure 106: Interface for add new ALK

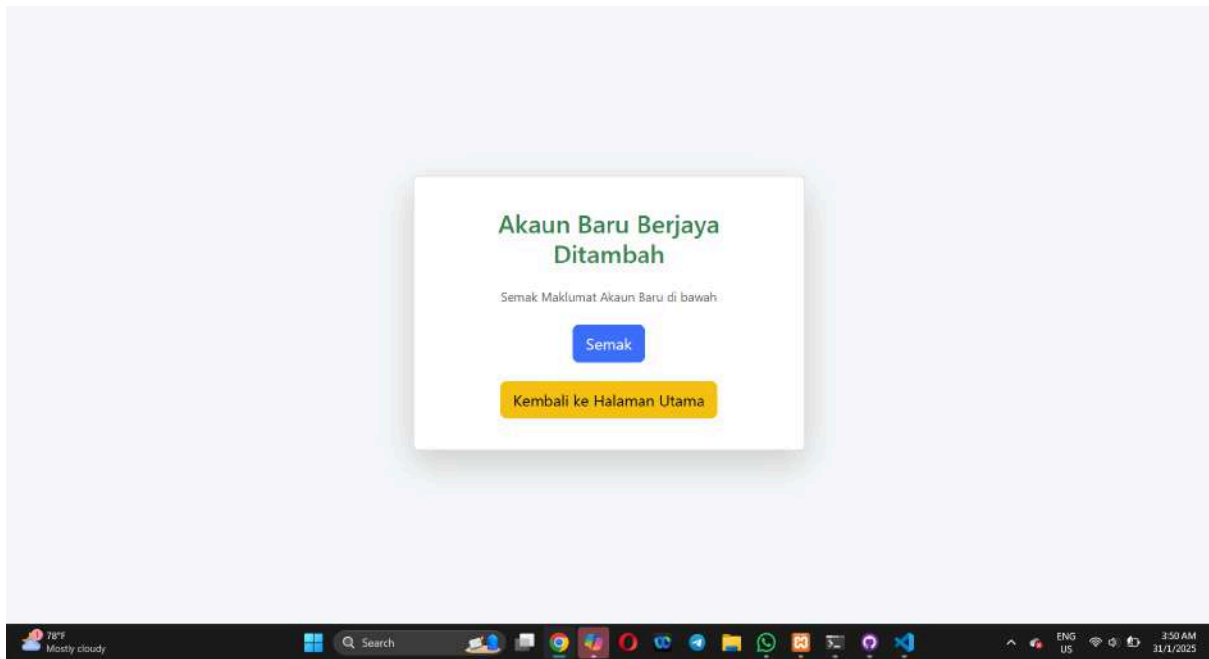


Figure 107: Interface for successful add new admin/ALK

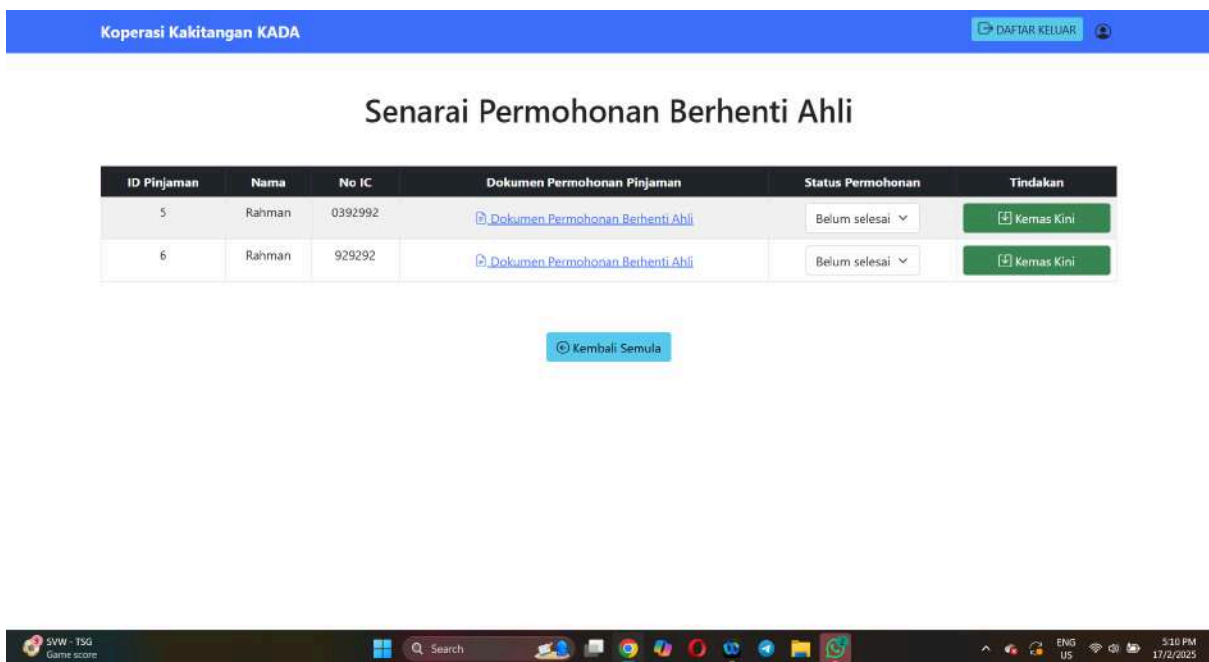


Figure 108: Interface for membership termination list

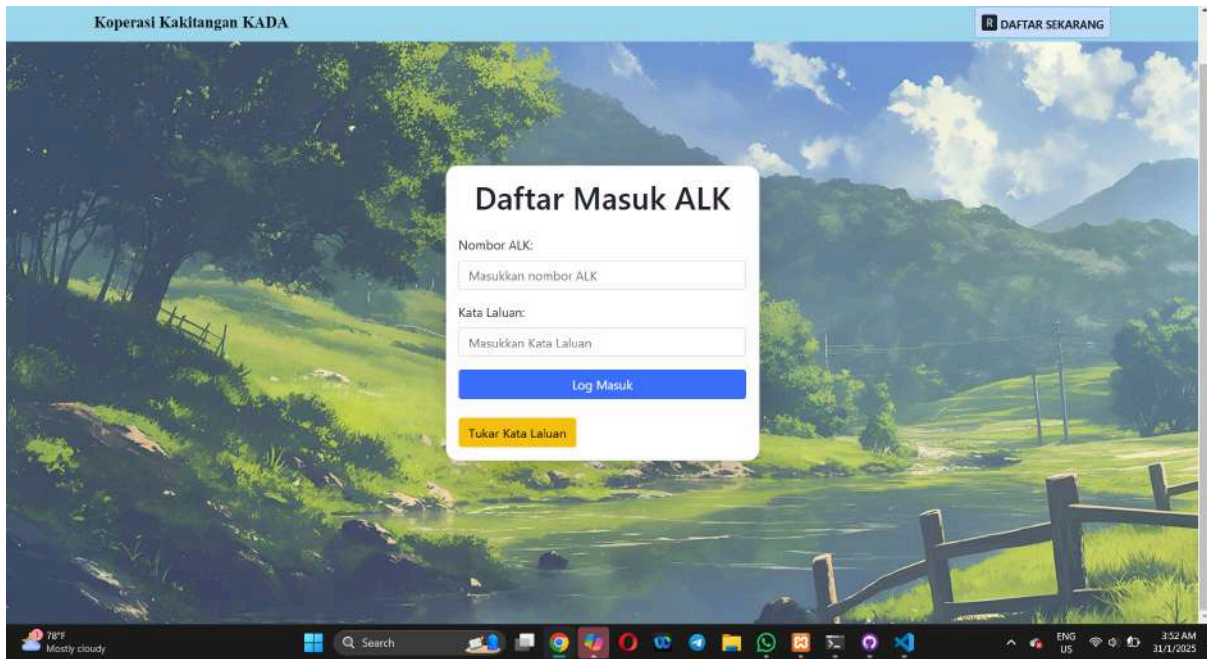


Figure 109: Interface for ALK login

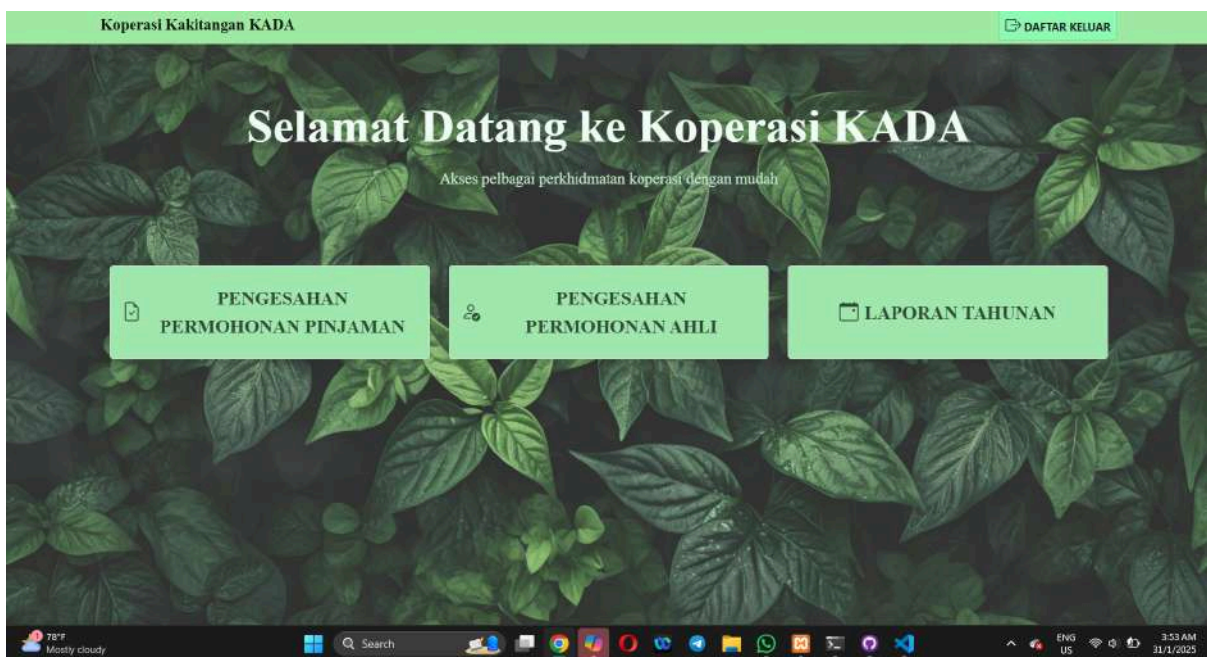


Figure 110: Interface for ALK homepage

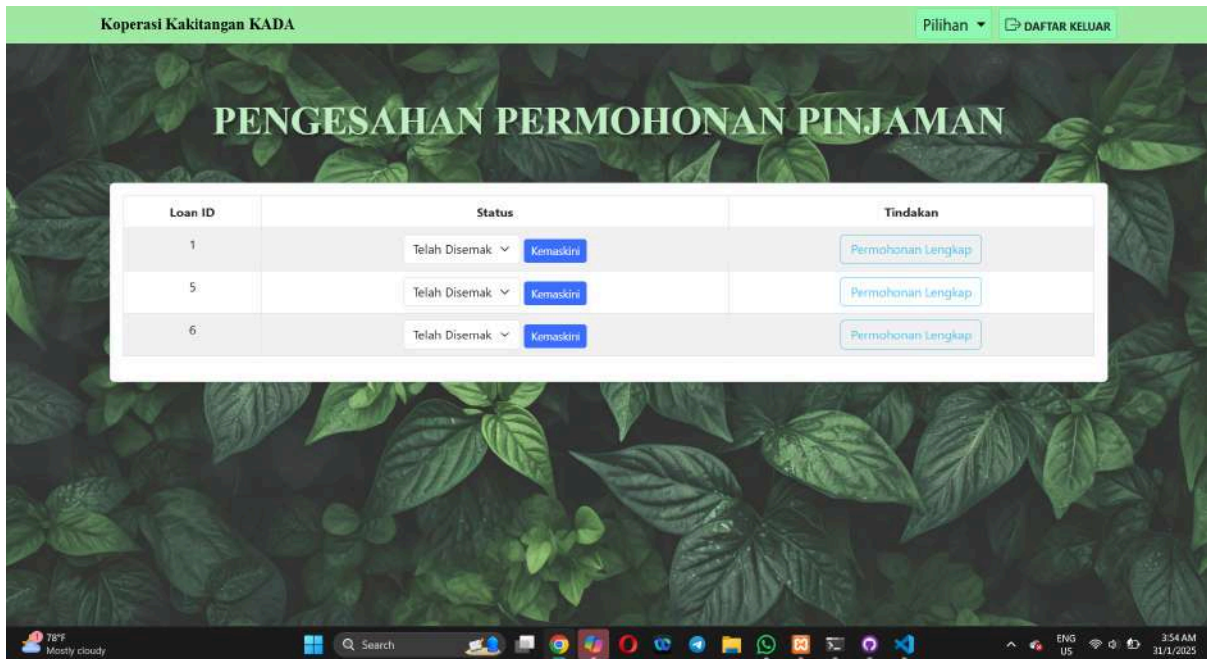


Figure 111: Interface for approve loan application

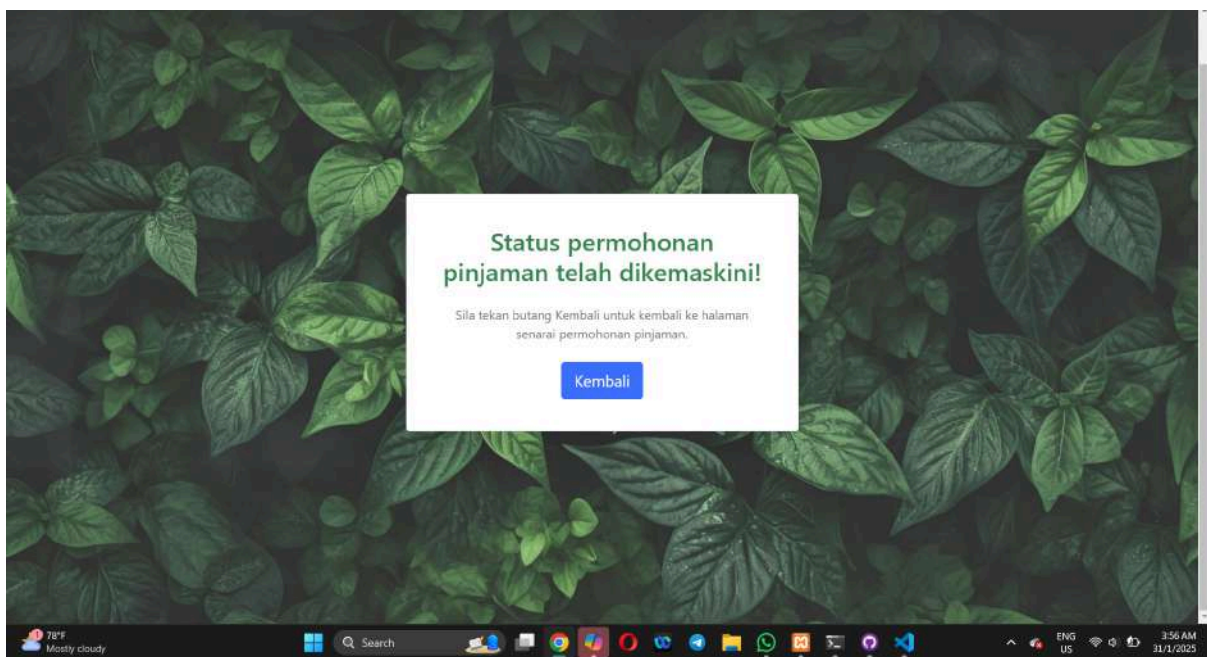


Figure 112: Interface for successful loan application status update

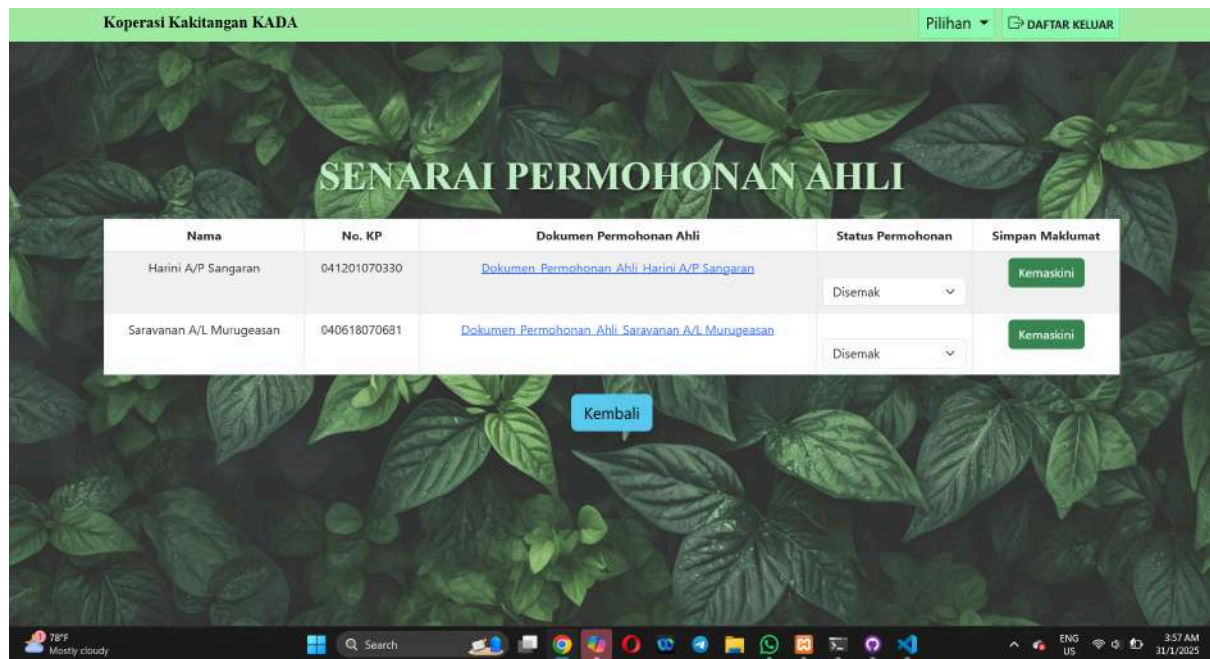


Figure 113: Interface for approve membership application

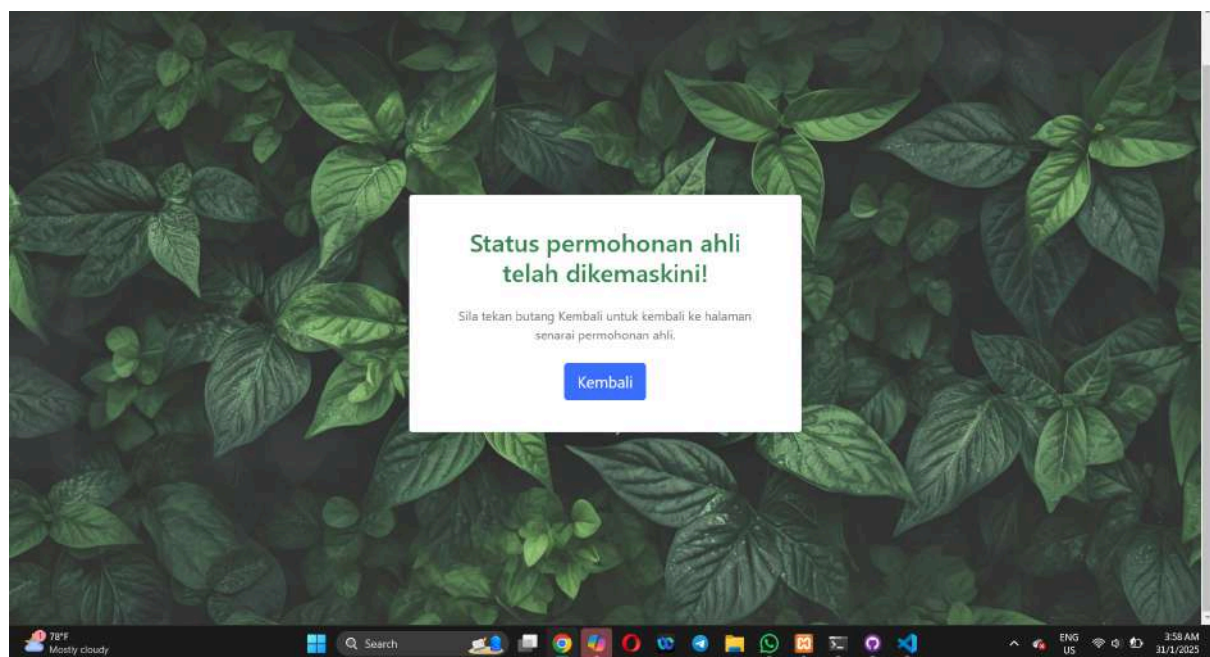


Figure 114: Interface for successful membership application status update

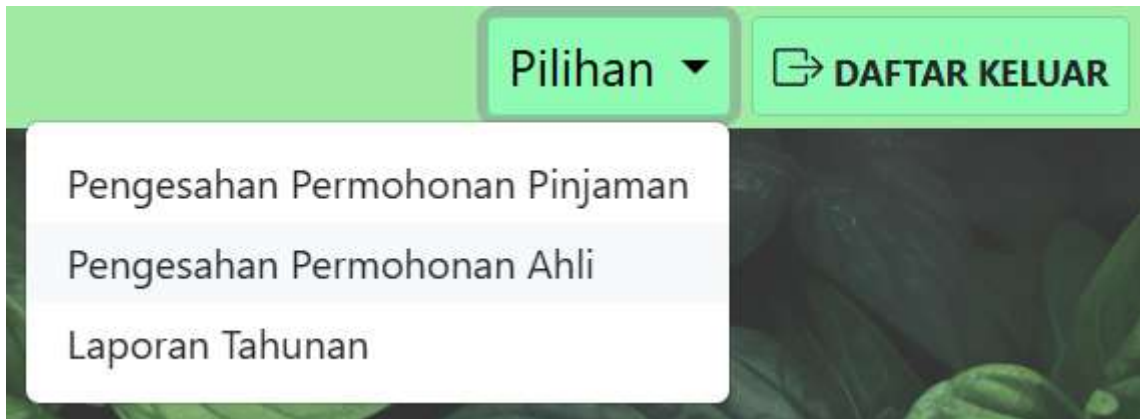


Figure 115: Interface for navigation bar for features choices and logout button

7. Traceability

Traceability in the Koperasi KADA System ensures that all user requirements are effectively linked to system components throughout the development lifecycle. It provides a structured approach to mapping user stories, system subsystems, and implementation details, ensuring that each functionality is properly developed, tested, and deployed.

By maintaining traceability, the project team can track dependencies, validate requirements, and ensure completeness from initial planning to final deployment. This approach helps in identifying gaps, improving system quality, and aligning development efforts with business objectives.

Sprint #	Subsystem/Package	User Story ID
sprint #1	Non-member module	US004
sprint #2	Member module	US001
sprint #3	Admin module	US002
sprint #4	ALK module	US003
sprint #5	Authentication module	US005
sprint #6	Loan module	US006

Table 32: Traceability